

Merged research report for the Surreal project

Omnific Groups and Lattices

Algebraic groups, elementary and Chevalley quotients, current Lie rings,
and missing lattice minima over the omnific integers

Built from five manuscripts dated September 23, 2026

Abstract

We study matrix groups and lattices over the omnific integers $\mathbf{Oz}$ and the Gaussian omnific integers $\mathbf{Oz}[i]$. Four phenomena are treated, each in its own part and under its own hypotheses.

Algebraic groups. A real algebraic group acquires no nonconstant points over the ring of surreal normal forms with nonnegative exponents exactly when it has no nonidentity real unipotent element, equivalently when its identity component is reductive with compact real derived group. For a closed subgroup scheme of GL_N over $\mathbb{Z}$ this gives an exact dichotomy: either every omnific point is an ordinary integral point, or the constant-term kernel contains an injective copy of the additive class of purely infinite normal forms. Over the Gaussian omnific integers the criterion is that the complex identity component be a torus. This answers, for such group schemes, a question on rigid affine varieties left open in the omnific Diophantine report.

Elementary groups. For $n \geq 3$ the kernel of constant-term reduction $E_n(\mathbf{Oz}) \rightarrow \mathrm{SL}_n(\mathbb{Z})$, and its Gaussian analogue, has no nontrivial homomorphism to any set-sized group, so $\mathrm{SL}_n(\mathbb{Z})$, respectively $\mathrm{SL}_n(\mathbb{Z}[i])$, is the universal set-sized quotient; this strengthens the universal ring-quotient theorem of the sibling report to arbitrary abstract group maps. Two independent proofs are given, and a third that rebuilds a ring from the root maps and so derives the group theorem from the ring theorem. The kernel is torsion-free and centerless, has no nontrivial set-sized normal subgroup, and its nonidentity conjugacy classes are proper classes; the Steinberg groups have the analogous universal quotient; and in the set-sized rings of supports of size below a regular uncountable κ every nonidentity kernel element has least detecting group of cardinality exactly $\kappa^{<\kappa}$. In rank two the answer reverses: E_2 is an amalgam, its kernel is a free product of copies of the purely infinite ideal, and one fixed countable group separates all elements, while finite groups see nothing of the kernel. Explicit non-elementary 2×2 unipotents become commutators after one stabilization.

Lattices. For finite sums of ordinary positive-semidefinite forms with successively infinitesimal weights we give exact criteria for shortest nonzero vectors, shortest unimodular vectors and closest points to every surreal target. When a shortest vector fails to exist the energy values have no surreal infimum, their lower bounds are an explicit cut, and one monomial vector improves every set-sized family at once; usual Gauss and LLL reduced bases then do not exist.

Chevalley groups and Lie rings. The result for $n \geq 3$ holds for the elementary and Steinberg groups of every root system without an A_1 component, including G_2 and the exceptional types; over $\mathbf{Oz}$ and $\mathbf{Oz}[i]$ a universal set-sized quotient exists exactly when there is no A_1 component. A current Lie ring $\mathfrak{l} \otimes \mathbf{Oz}$ has one exactly when $\mathfrak{l} \otimes \mathbb{Q}$ is perfect.

This report merges five independently written manuscripts; it states which source proved each result, prints shared results once, and keeps every source's limitations. It is an AI-assisted draft: it has not been refereed, and nothing in it has been formalized in Lean.

Sources: manuscripts 06 (the base text) and 10, pinned to repository commit 220784a, and 07, pinned to fb5c4b5; manuscript 11, added later and pinned to 9693b28; manuscript 12, added last and pinned to 7af8a30; their titles are in [subsection 1.2](#). Results already printed elsewhere in the collection are cited, not reprinted. The provenance of every section is recorded in [Appendix A](#); non-claims are collected in [Appendix B](#). Finite checks accompany the report and check finite identities and examples only.

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1 Scope, results, and conventions

An omnific integer has an ordinary integer constant coefficient and an arbitrary, possibly transfinite, purely infinite part whose coefficients are arbitrary real numbers. This mixture makes linear algebra over $\mathbf{Oz}$ quite different both from linear algebra over $\mathbb{Z}$ and from linear algebra over the surreal field. This report studies three linear questions over $\mathbf{Oz}$ and over the Gaussian omnific integers $\mathbf{Oz}[i]$.

- (I) Which algebraic groups acquire new omnific points ([Part I](#))?
- (II) Can the purely infinite directions of an elementary matrix group act on any ordinary set, when no algebraicity or continuity is imposed on the action ([Part II](#))?
- (III) Does a positive-definite form on $\mathbf{Oz}^n$ have a shortest vector, and what replaces it when it does not ([Part III](#))?
- (IV) Which root systems, and which Lie lattices, carry the obstruction of (II) over to elementary Chevalley groups and current Lie rings ([Part IV](#))?

The answers reveal three distinct mechanisms: geometric rigidity controlled by unipotent elements; an abstract representation obstruction controlled by the supply of surreal exponent scales and a third matrix index; and a competition between the constant-term retraction and monomial vectors lying on irrational kernels. Question (IV) locates the second mechanism: the third index is an instance of a root system without A_1 component, and for Lie rings the condition becomes rational perfection.

1.1 Principal results

Bracketed numbers name the source manuscripts ([subsection 1.2](#)). Here (D, k) is $(\mathbb{Z}, \mathbb{R})$ or $(\mathbb{Z}[i], \mathbb{C})$, Π_k is the class of normal forms with coefficients in k supported on strictly positive exponents, $\mathcal{B}_k = k \oplus \Pi_k$, and $\mathcal{A}_{D,k} = D \oplus \Pi_k$, so that $\mathcal{A}_{\mathbb{Z},\mathbb{R}} = \mathbf{Oz}$ and $\mathcal{A}_{\mathbb{Z}[i],\mathbb{C}} = \mathbf{Oz}[i]$ ([subsection 1.4](#)).

- (1) **Real omnific rigidity** [06]. For a real algebraic group G : $G(\mathcal{B}_{\mathbb{R}}) = G(\mathbb{R})$ iff $G(\mathbb{R})$ has no unipotent element other than 1 iff G has no subgroup isomorphic to $\mathbb{G}_a$ iff G^0 is reductive with $(G^0)^{\text{der}}(\mathbb{R})$ compact. For a closed subgroup scheme $\mathcal{G} \subseteq \text{GL}_{N,\mathbb{Z}}$ these are equivalent to $\mathcal{G}(\mathbf{Oz}) = \mathcal{G}(\mathbb{Z})$; otherwise $(\Pi, +)$ embeds in the kernel of $\text{ct} : \mathcal{G}(\mathbf{Oz}) \rightarrow \mathcal{G}(\mathbb{Z})$ ([theorem 4.4](#)).
- (2) **Gaussian omnific rigidity** [06]. For a complex algebraic group, $G(\mathcal{B}_{\mathbb{C}}) = G(\mathbb{C})$ iff G^0 is a torus; the same holds for $\mathbb{Z}[i]$ -models and $\mathbf{Oz}[i]$ ([theorem 4.5](#)). Thus $\text{SO}_3(\mathbf{Oz})$ has 24 elements while $\text{SO}_3(\mathbf{Oz}[i])$ contains injective unipotent families. Together, (1) and (2) answer the question **odg:q:affine** of the omnific Diophantine report for closed subgroup schemes of GL_N ([remark 4.6](#)).
- (3) **Bounded fibers, unipotent kernels, filtrations** [06]. Bounded real fibers over a torus give rigidity ([theorem 3.3](#)); the constant-term kernel of a unipotent group is $\exp(\mathfrak{u} \otimes \Pi_k)$ ([proposition 6.1](#)); $\text{UT}_n(\Pi_k)$ has the exact filtration $[\text{Fil}_r, \text{Fil}_s] = \text{Fil}_{r+s}$ and nontrivial characters ([theorem 6.2](#)).
- (4) **The elementary kernel for $n \geq 3$** [06, 07, 11]. Let $P_n = \ker(\text{ct} : E_n(\mathcal{A}_{D,k}) \rightarrow \text{SL}_n(D))$. It is generated by the elementary matrices with purely infinite parameters, it is perfect and nontrivial, $E_n(\mathcal{A}_{D,k}) = P_n \rtimes \text{SL}_n(D)$, and every homomorphism from P_n to a set-sized group is trivial. Hence composition with ct identifies the homomorphisms from

$E_n(\mathcal{A}_{D,k})$ and from $\mathrm{SL}_n(D)$ to every set-sized group (theorem 9.1). Two independent proofs are printed. The theorem implies, for $k \in \{\mathbb{R}, \mathbb{C}\}$, the universal ring-quotient theorem `osq:thm:universal` (remark 9.6); conversely a third proof (11) rebuilds a set-sized ring from the root maps of a group homomorphism and derives the theorem from the ring theorem (theorem 9.2).

- (5) **A congruence tower and the boundaries** [06, 07]. P_n is not simple: a proper-class chain of congruence subgroups separates its elements, though no set-sized subfamily does (theorem 10.1). The theorem fails in every fixed set-sized Hahn workspace and for finite supports (section 12); an abstract collision criterion isolates the size input (proposition 9.7). On the finite side, SL_n of the finite surreals has universal set-sized quotient $\mathrm{SL}_n(k)$ for $n \geq 3$ [merge] (proposition 11.1).
- (6) **Rank two** [07]. $E_2(\mathcal{A}_{D,k}) \cong \mathrm{SL}_2(D) *_{C_D} B_D$; the kernel P_2 is a free product of copies of $(\Pi_k, +)$ indexed by $\mathbb{P}^1(\mathrm{Frac} D)$; homomorphisms into the fixed countable group $\mathrm{SL}_2(\mathrm{Frac}(D)(t))$ separate every finite subset of E_2 ; neither E_2 nor P_2 has a universal set-sized quotient (theorems 13.2, 14.1 and 15.3).
- (7) **The detection trichotomy** [07]. A nonidentity $g \in E_n$ is detected by a finite group iff $\mathrm{ct}(g) \neq 1$; a nonidentity element of P_2 has least detecting cardinal $\aleph_0$; a nonidentity element of P_n , $n \geq 3$, has no set-sized detecting target (theorem 16.5).
- (8) **Characters and stabilization** [07]. $E_2(\mathbf{Oz})^{\mathrm{ab}} \cong \mathbb{Z}/12 \oplus \Pi$ (classical, Cohn; Mirzaii–Torres), while every abelian image of $E_2(\mathbf{Oz}[i])$ kills P_2 (propositions 17.1 and 17.2). A cusp-residue obstruction shows that $I + \omega^{2a} \begin{pmatrix} \alpha & 1 \\ -\alpha^2 & -\alpha \end{pmatrix}$, $\alpha \in k \setminus \mathrm{Frac} D$, lies in $\mathrm{SL}_2 \setminus E_2$; after one stabilization it is a commutator of two elements of P_3 (theorems 18.2 and 18.4).
- (9) **Structure of the kernel, Steinberg groups and set-sized fragments** [11]. For $n \geq 3$ the kernel P_n is torsion-free (so is the constant-term kernel of $\mathrm{GL}_n(\mathcal{A}_{D,k})$ for every n) and centerless, it has no nontrivial set-sized normal subgroup, every nonidentity element has a proper-class conjugacy class, and finite subgroups of $E_n(\mathcal{A}_{D,k})$ inject under ct (theorems 19.1 and 19.2, corollary 19.3, and proposition 19.4). The Steinberg group $\mathrm{St}_n(\mathcal{A}_{D,k})$ has universal set-sized quotient $\mathrm{St}_n(D)$ and a perfect kernel with no set-sized images (theorem 20.2). With the Bass–Milnor–Serre theorem the profinite completion of $E_n(\mathbf{Oz})$ is $\mathrm{SL}_n(\widehat{\mathbb{Z}})$ (corollary 19.7). In the set-sized ring of normal forms with fewer than κ exponents in $\bigoplus_{\alpha < \kappa} \mathbb{Q}$, κ regular uncountable, the kernel has no nontrivial image of cardinality below κ (theorem 21.3); combined with the sibling report’s field bound, each nonidentity kernel element has least detecting group of cardinality exactly $\kappa^{<\kappa}$ [merge] (proposition 21.12).
- (10) **Shortest vectors and missing infima** [10]. For a separated positive pencil $\mathcal{E} = \sum_j \varepsilon_j q_j$ on $\mathcal{A}_{D,k}^n$: a shortest nonzero vector exists iff the deepest kernel meets D^n nontrivially; otherwise the nonnegative lower bounds of the energy values are exactly $\varepsilon_m(\mathcal{O}_{\mathrm{fin}})_{\geq 0}$ and one monomial vector improves every set-sized family (theorems 24.1 and 24.2).
- (11) **Unimodular vectors and closest points** [10]. A rational-span criterion decides unimodular attainment, and unimodular energies always have a countable coinitial family (theorem 26.4); closest points exist for every surreal target iff every prefix kernel is rational, and then the closest points lie among at most 2^n (real) or 4^n (Gaussian) ordinary candidates after removing the target’s purely infinite part (theorems 27.2 and 27.5). The ordinary backbone of these statements is already in the Hahn–Tate report and is cited there (subsection 27.1).
- (12) **Reduction obstructions** [10]. In the nonattainment case no Gauss-reduced basis and no

δ -LLL basis with ordinary $\delta \in (1/4, 1]$ exists, and a Gaussian variant holds for $\delta \in (1/2, 1]$ (theorems 30.3 and 30.4).

- (13) **Chevalley groups and current Lie rings** [12]. For every unital $D \subseteq k$ and every finite reduced root system Φ without A_1 component, every homomorphism from $P_\Phi = \ker(\text{ct} : E_\Phi(\mathcal{A}_{D,k}) \rightarrow E_\Phi(D))$ to a set-sized group is trivial, so $E_\Phi(D)$ is the universal set-sized quotient, and likewise for Steinberg groups and other isogeny forms (theorems 38.2 and 40.1, corollary 38.3, and proposition 40.3); P_Φ is perfect, torsion-free, centerless and not simple (section 39). For $D = \mathbb{Z}, \mathbb{Z}[i]$, $E_\Phi(\mathcal{A}_{D,k})$ has a universal set-sized quotient exactly when Φ has no A_1 component (theorem 41.1). A current Lie ring $\mathfrak{l} \otimes \mathcal{A}_{D,k}$ has one exactly when $\mathfrak{l} \otimes \mathbb{Q}$ is perfect (theorems 42.3 and 42.6). Ring and group invisibility are equivalent for Dorroh unitizations (theorem 38.5), and detection thresholds transfer to every infinite cardinal; in the rings of section 21 the exact value $\kappa^{<\kappa}$ holds for every such Φ , and countably supported coefficients already suffice [merge] (corollaries 43.7 and 44.1). This answers question 34.3 for split elementary groups.

These statements are proved here, but proving a statement here is not a claim that nobody has previously proved it. Each source names its own most distinctive contributions and credits its classical inputs (Appendix B); no exhaustive historical-priority claim is made.

1.2 Provenance of this report

This report was first built from three manuscripts written independently and dated September 23, 2026. They are items 06, 07 and 10 of the batch placed in commit f4c9504; their shipped file prefixes are 06-algebraic-groups-, 07-matrix-dichotomy- and 10-shortest-vectors-. Sources 06 and 10 pin repository commit 220784a (19 commits before that placement), and source 07 pins fb5c4b5 (12 commits before it). Both pins precede the written omnific reports: at the pins the directories docs/surreal/omnific-diophantine-geometry and docs/surreal/set-sized-quotients-of-omnific-integers held only their placed base manuscripts, not the merged reports that now carry the labels cited below (Appendix A.3).

A fourth manuscript was added afterwards. It is item 05 of the batch placed in commit 66d7e55, shipped under the file prefix 11-matrix-shadows-, and is called source 11 here, after that prefix, as the first three sources are called after theirs. It is also dated September 23, 2026, and pins 9693b28 (24 commits before its placement), at which this report had already been written. Source 11 was written without knowledge of this report (Appendix A.3); its main theorem is theorem 9.1 again, and its other results are new to the collection.

A fifth manuscript was added last. It is item 07 of the batch placed in commit 7d04483, shipped under the file prefix 12-chevalley-quotients-, and is called source 12 here, after that prefix; it is not the source 07 above. Its title is *Arithmetic Quotients of Omnific Chevalley Groups and Lie Rings* (28 pages: a title page, a contents page and 26 numbered pages). It is dated September 23, 2026, and pins 7af8a30 (31 commits before its placement), at which this report already contained sources 06, 07, 10 and 11. It read this report's guide and source 11's proof audit, credits this report's type- A , rank-two, Steinberg and cardinal results, and extends them to all root systems and to current Lie rings (Part IV).

Source	Manuscript	Principal contribution to this report
06	<i>Omnific Points of Algebraic Groups</i> (22 pp.)	Base text and structure; the real and Gaussian dichotomies and the bounded-fiber principle (Part I); orthogonal examples; unipotent kernels and unitriangular filtrations; the $n \geq 3$ theorem with the proof by division by a difference of nearby monomials; normality of the root-generated subgroup for any constant ring; the congruence tower; fixed-workspace and finite-support boundaries.
07	<i>A Matrix-Size Dichotomy over the Omnific Integers</i> (25 pp.)	Base of Part II ; the $n \geq 3$ theorem with the proof by rescaling a large set field, including homomorphisms defined only on the kernel; rank two: amalgam, free-product kernel, countable detection; the finite/countable/no-set trichotomy; abelian characters; cusp residues, non-elementary unipotents and stabilization; the abstract collision criterion.
10	<i>Shortest Vectors and Missing Infima over the Omnific Integers</i> (26 pp.)	All of Part III : constant-term contraction, the shortest-vector dichotomy and exact lower cut, simultaneous improvement, the low-energy flag, the unimodular criterion, closest points for all surreal targets, the Pell comparison, the strict hierarchy, reduction obstructions, and the fixed-Hahn boundary.
11	<i>Arithmetic Shadows of Omnific Matrix Groups</i> (24 pp.)	A third proof of the $n \geq 3$ theorem, by reconstructing a ring from the root maps (section 9); torsion-freeness, centerlessness, set-sized normal subgroups and conjugacy classes, finite subgroups, a principal congruence chain and common congruence elements, the profinite completion (section 19); Steinberg groups (section 20); set-sized fragments, cardinal and action thresholds, and the exact-threshold criterion (section 21).
12	<i>Arithmetic Quotients of Omnific Chevalley Groups and Lie Rings</i> (28 pp.)	All of Part IV : the root-kernel ideal for every root system without A_1 component, the arithmetic quotient of elementary Chevalley and Steinberg groups, the ring-group equivalence, the structure of the relative kernel, isogeny forms, the componentwise classification, the Lie-ring classification, the cardinal transfer, explicit rank-two matrices, and a second proof of the rank-two amalgam theorem.

Where the merge had to choose.

- The theorem for $n \geq 3$ was proved by both 06 and 07, independently and by different mechanisms. It is printed once ([theorem 9.1](#)), in 06's generality for the constant ring and 07's intrinsic form for the kernel, with both proofs kept.
- Part II takes 07 as its base, because 07 contains the rank-two theory that 06 left open; 06's normality, splitting, tower and boundary results are inserted where they belong.
- Background on normal forms, units, gaps and divisibility, proved by all four sources and by the omnific reports, is printed once ([section 2](#)).
- Results already printed in other reports are cited rather than reprinted: $O_n(\mathbf{Oz})$ (`odg:cor:orthogonal`), the anisotropic orthogonal example (covered by `odg:cor:unipotent`), the lattice-projection lemma, the rational-flag minimum criterion, the 2^n parity bound and the Pell example of the Hahn–Tate report ([subsection 27.1](#)). Source proofs are kept only where they prove more or proceed differently.
- Part III was placed with Parts I–II rather than in a separate report: it studies modules and bases over the same rings, its reduction theory changes bases in $GL_n(\mathbf{Oz})$, and its ordinary backbone belongs to a surcomplex report about theta series, where its omnific content would be foreign.
- Source 11 proves [theorem 9.1](#) a third time. The theorem is not reprinted: 11 is added to its sources, and 11's proof, which rebuilds a ring from the root maps and invokes the ring theorem of the sibling report, is kept as a marked third route ([section 9](#)). 11's finite-residual theorem and finite-support counterexample are credited at [theorem 16.4](#) and [proposition 12.2](#), and its action and representation corollaries at [corollary 9.5](#). Its remaining results form three new sections at the end of Part II ([sections 19 to 21](#)), with the label sub-prefix `ogl:ker:.`
- Source 11 declines to identify the root-generated subgroup with the kernel; [proposition 7.3](#) (06) proves the identity, so 11's statements about both groups are printed as statements about P_n . 11's set-sized ring is the ring R_κ^{11} of the sibling report (`osq:rel:lem:embedding`; the superscript there names that report's manuscript 11, not source 11 here, and the ring is written $R_{<\kappa}$ below); its ring-level facts are cited there, and the merge combines them with 11's exact-threshold criterion ([proposition 21.12](#)).
- Source 12 is added as a new [Part IV](#) after the questions, with the label sub-prefix `ogl:ch:`, so that no earlier number changes. For $\Phi = A_{n-1}$ its theorems are those of [Part II](#) and are not reprinted; its general statements are printed with the type- A results credited at each ([subsection 35.4](#)). Its scalar theorem is the sibling report's `osq:thm:universal` and is cited. Its leading-term proof of the rank-two amalgam theorem is kept as a second route ([section 41](#)); its rank-two free product, additive functional and countable detection are 07's and are credited, not reprinted.
- Source 12's answer to [question 34.3](#) is recorded there, and the merge combines its cardinal transfer with [proposition 21.12](#) to extend the exact thresholds to every root system without A_1 component ([corollary 43.7](#)).

Results added by the merge carry the tag [merge] and have complete proofs. Appendix [A](#) records, section by section, which source proved what and every symbol renamed from a source.

1.3 Class size and foundations

The surreal field $\mathbf{No}$, the rings $\mathbf{Oz}$ and $\mathbf{Oz}[i]$, and most groups below are proper classes. Sources 06 and 07 work in Gödel–Bernays set theory with global choice, source 10 in NBG; 07 and 06 note that global choice is never used, and no proof printed here uses it. Source 11 allows Gödel–Bernays class theory with global choice and uses choice only for sets (a successor cardinal above a target, and the equivalence of reverse well ordering with the absence of increasing sequences); its set-sized fragments give ordinary ZFC statements. Source 12 works in Gödel–Bernays class theory with a ZFC set part, notes that global choice is harmless but unnecessary, and assumes no inaccessible cardinal. Each individual surreal, normal-form support, matrix, tuple and finite word is a set. A homomorphism from a class group is a class function satisfying the group identities; statements about all such homomorphisms are read one class function at a time, and no set of all such maps is formed. Residual properties are universally quantified kernel statements. A class quotient may be read as a kernel with its universal property, or through Scott’s trick (06).

Every cardinality contradiction is a contradiction about sets: for a target set H one produces, inside the source, a set of parameters that cannot inject into H . Sources 06 and 07 choose a cardinal $\lambda > |H|$; the Hartogs ordinal of H serves equally and needs no choice, as in the sibling report (`osq:thm:universal`). No cardinal is assigned to a proper class. Source 07 adds that the class statements concern class homomorphisms existing in the chosen foundational setting, not externally described maps in a larger metatheory.

1.4 Notation and conventions

Sign convention. Exponents are those of Conway normal forms: supports are reverse well ordered and ω^γ is *infinite* for $\gamma > 0$, even for infinitesimal $\gamma > 0$. *Purely infinite* refers to support: $-\omega$ and $\sqrt{2}\omega$ are purely infinite, $\omega - 1$ is infinite but not purely infinite, and ω^{-1} is not purely infinite. In the Hahn convention $t = \omega^{-1}$ all exponent signs change. This agrees with the omnific reports, whose letter Π we use for the purely infinite ideal (`found:eq:omnific` in the foundations report).

Two constant maps. `ct` extracts the coefficient of ω^0 from an element of $\mathcal{B}_k$; it is a ring homomorphism there, not on $\mathbf{No}$ (both ω and ω^{-1} have constant coefficient 0). `st` is the standard part of a *finite* surreal or surcomplex number. For example `ct`($\omega + 7$) = 7, although $\omega + 7$ is not finite. The two are never interchanged.

Two notions of finite. “Finite surreal” means bounded in modulus by an ordinary integer, not a normal form with finite support (10): the monomial $\omega^{1/\omega}$ has one term and is infinite. Group ranks, matrix sizes, polynomial degrees and lattice ranks are always ordinary finite integers.

Table 1: Notation used throughout.

Notation	Meaning in this report
(D, k)	$(\mathbb{Z}, \mathbb{R})$ or $(\mathbb{Z}[i], \mathbb{C})$; in some statements D is any unital subring of k .
F	The ambient field $\mathbf{No}$ or $\mathbf{No}[i]$.

Notation	Meaning in this report
Π_k, Π	Normal forms with coefficients in k supported on $\mathbf{No}^{>0}$, with 0 ; $\Pi = \Pi_{\mathbb{R}}, \Pi_{\mathbb{C}} = \Pi + i\Pi$.
$\mathcal{B}_k = k \oplus \Pi_k$	Nonnegative-support ring; $\mathcal{B}_{\mathbb{R}}, \mathcal{B}_{\mathbb{C}}$ as in the omnific Diophantine report.
$\mathcal{A}_{D,k} = D \oplus \Pi_k$	$\mathbf{Oz} = \mathcal{A}_{\mathbb{Z},\mathbb{R}}, \mathbf{Oz}[i] = \mathcal{A}_{\mathbb{Z}[i],\mathbb{C}}$. Direct sums of additive groups, not of rings.
ct, st	Constant coefficient on $\mathcal{B}_k$; standard part on finite elements.
$\deg_{\omega} f$	Largest exponent in $\text{supp } f$; not an ordinal and not a birthday.
$\mathcal{O}_{\text{fin}}, \mathfrak{m}$	Finite surreals (or surcomplex numbers) and infinitesimals; $\mathfrak{m}_k$ for coefficients in k .
$u \ll v$	$u/v \in \mathfrak{m}$, for $u \geq 0, v > 0$.
$K_D = \text{Frac } D$	$\mathbb{Q}$ or $\mathbb{Q}(i)$.
<i>Part I</i>	
$G, G^0, \mathcal{G}, T$	Algebraic group, its algebraic identity component, an integral model, a torus.
$\text{UT}_n(\Pi_k), \text{Fil}_r$	Unitriangular group with purely infinite entries and its superdiagonal filtration.
<i>Part II</i>	
$e_{ij}(a), E_n(A)$	$1_n + aE_{ij}$; the subgroup generated by these (finite words).
P_n	$\ker(\text{ct} : E_n(\mathcal{A}_{D,k}) \rightarrow E_n(D))$; generated by $e_{ij}(\Pi_k)$ for $n \geq 3$.
$P_n^{(a)}$	Congruence subgroup of P_n at scale ω^a .
$[x, y]$	$xyx^{-1}y^{-1}$.
G_D, C_D, B_D	$\text{SL}_2(D)$, its constant upper triangular subgroup, the upper triangular subgroup of $\text{SL}_2(\mathcal{A}_{D,k})$.
$U(x), \mathcal{T}$	$e_{12}(x)$; a transversal of G_D/C_D .
$P_{\text{fin}}, P_{\infty}$	Finite points of $\mathbb{P}^1(F)$; the infinite points and ∞ .
Φ_h	Exponent rescaling $\sum a_s \omega^s \mapsto \sum a_s \omega^{hs}$.
$\text{St}_n(B), x_{ij}(a), P_n^{\text{St}}$	Steinberg group, its generating symbols, and the kernel of $\text{ct} : \text{St}_n(\mathcal{A}_{D,k}) \rightarrow \text{St}_n(D)$ (section 20).
$P_n\langle a \rangle$	$\{g \in P_n : g - 1_n \in M_n(aA)\}$ for $0 \neq a \in A$; $P_n^{(a)} = P_n\langle \omega^a \rangle$.
$\Gamma_{\kappa}, e_{\alpha}$	$\bigoplus_{\alpha < \kappa} \mathbb{Q}e_{\alpha}$, least index dominant; its basis vectors (one index, no argument; root elements $e_{ij}(a)$ have two).
$R_{<\kappa}, \Pi_{<\kappa}, P_{n,<\kappa}$	The sibling report's R_{κ}^{11} and Π_{κ}^{11} (supports of size $< \kappa$ in Γ_{κ}), and the kernel of ct on $E_n(R_{<\kappa})$.
$\kappa^{<\kappa}, \mathfrak{c}$	$\sup_{\nu < \kappa} \kappa^{\nu}; 2^{\aleph_0}$.
<i>Part III</i>	
$q_j, \varepsilon_j, \mathcal{E}$	Ordinary positive-semidefinite forms, separated weights, the energy $\sum \varepsilon_j q_j$.
K_j, L_j, K_F	Prefix kernels, their integral points $K_j \cap D^n$, scalar extension of an ordinary subspace.

Notation	Meaning in this report
$\text{Um}(\mathcal{A}^n)$	Unimodular vectors (coordinates generate the unit ideal).
Low_j	Vectors of energy $\ll \varepsilon_j$.
Δ_j	Gram–Schmidt energies.
<i>Part IV</i> (reserved letters in subsection 35.2)	
Φ, α, β	A finite reduced root system (not the rescaling Φ_h); roots.
$G_\Phi, x_\alpha(a), E_\Phi(B)$	Split simply connected group scheme, its root elements, the elementary subgroup.
$P_\Phi(A), P_\Phi^{\text{St}}(A)$	Kernels of ct on $E_\Phi(\mathcal{A}_{D,k})$ and $\text{St}_\Phi(\mathcal{A}_{D,k})$; $P_{A_{n-1}} = P_n$.
$\mathfrak{l}, \mathfrak{l}_\mathbb{Q}, \mathfrak{l}(B)$	A Lie lattice, $\mathfrak{l} \otimes \mathbb{Q}$, the current Lie ring $\mathfrak{l} \otimes B$.

Renamed symbols. The sources’ names are changed where they collided with each other or with the omnific reports; the complete table is [table 3](#). The most important changes: the purely infinite ideal is Π_k (06 wrote $\mathcal{I}_k$, 07 $\mathcal{J}_k$, 10 $\mathcal{J}$); the constant term is ct (07 wrote c_0); the kernel of reduction is P_n (06’s $P_n(k)$, 07’s N_n); 06’s congruence subgroups N_a are $P_n^{(a)}$, so that no letter N carries two meanings; 10’s energy E is $\mathcal{E}$, because E_n is the elementary group; 10’s word *primitive* is *unimodular*, because the omnific Diophantine report uses *primitive* for “no common nonunit divisor” (`odg:thm:realray`), a weaker property. Source 11’s kernel $N_n(A)$ is P_n , its Steinberg kernel $K_n(A)$ is P_n^{St} (since K_D and K_j are taken), its congruence subgroups $N^{(m)}$ modulo $(1 + \omega)^m$ are $P_n \langle (1 + \omega)^m \rangle$, and its fragments $A_{\kappa,D}$, $\Pi_{\kappa,k}$ are the sibling report’s rings, written $R_{<\kappa}$, $\Pi_{<\kappa}$ (so that no Π_κ is confused with Π_k).

1.5 Relationship with the repository and the literature

The omnific integers are classical [5, 7]; L’Innocente and Mantova give a modern account and a factorization theory [10]. The collection now contains two omnific reports: the omnific Diophantine report `docs/surreal/omnific-diophantine-geometry` (prefix `odg:`) and the set-sized quotient report `docs/surreal/set-sized-quotients-of-omnific-integers` (prefix `osq:`). This report builds on both: Part I answers one of the former’s questions, and Part II strengthens the latter’s main theorem and uses its two collision lemmas. The euclidean-three-space report proves a parallel standard-part theorem for surreal rotation groups, and the Hahn–Tate report already contains the ordinary lattice backbone of Part III. The precise relations are in [section 33](#). Classical inputs (Tarski transfer, the structure of reductive groups, Cartan’s theorem, Cohn’s two-row theory, the abelianization of E_2 , the congruence subgroup theorem of Bass–Milnor–Serre, higher-rank Voronoi theory and LLL reduction) are attributed at their points of use, as is the precedent for recovering coefficient algebra from root commutators [18]. Part IV uses the integral Chevalley commutator identities [6, 8] and amalgam normal forms [17].

2 Normal forms, constant terms, and support gaps

This section collects the facts about normal forms used in all three parts. All four sources prove them (source 11 its units, gap and single-product facts), and so do the omnific reports (`osq:lem:ct`, `osq:lem:gap`, `osq:prop:common`, `odg:prop:units`); they are background, not new results.

2.1 Normal forms

Every surreal number has a unique Conway normal form

$$f = \sum_{s \in S} a_s \omega^s, \quad a_s \in \mathbb{R} \setminus \{0\}, \quad (2.1)$$

where $S \subseteq \mathbf{No}$ is a set and is reverse well ordered: every nonempty subset has a greatest element [5, 7, 3]. Addition is coefficientwise; multiplication is Hahn convolution, and the coefficient of a product at a fixed exponent is a finite sum. The sign of a nonzero normal form is the sign of its leading coefficient, and $\omega^{s+t} = \omega^s \omega^t$ is Conway's monomial law, not an analytic exponential. Complex coefficients give exactly $\mathbf{No}[i]$, by separating real and imaginary parts; a finite union of reverse-well-ordered supports is reverse well ordered, so no larger support convention is needed. For $f \in \mathbf{No}[i]$, $|f| = (f\bar{f})^{1/2}$; the leading coefficient of $f\bar{f}$ is the squared modulus of the leading coefficient of f (07).

The Hahn support lemma is used freely: products are defined, and a series supported on strictly negative exponents has a formal geometric inverse near 1, so that for $\gamma > 0$

$$(1 - \omega^{-\gamma})^{-1} = \sum_{m \geq 0} \omega^{-m\gamma}. \quad (2.2)$$

This is an identity of normal forms. It does not assert convergence of a sequence in the fine topology of $\mathbf{No}$ (06, 07, 10).

2.2 The rings and the constant term

For $k \in \{\mathbb{R}, \mathbb{C}\}$ and a unital subring $D \subseteq k$ put

$$\Pi_k = \{f : \text{supp } f \subseteq \mathbf{No}^{>0}\}, \quad \mathcal{B}_k = k \oplus \Pi_k, \quad \mathcal{A}_{D,k} = D \oplus \Pi_k, \quad (2.3)$$

with coefficients in k . The zero series belongs to Π_k . Then $\mathbf{Oz} = \mathcal{A}_{\mathbb{Z}, \mathbb{R}}$ and $\mathbf{Oz}[i] = \mathcal{A}_{\mathbb{Z}[i], \mathbb{C}}$; $\mathcal{B}_{\mathbb{R}} = \mathcal{A}_{\mathbb{R}, \mathbb{R}}$ and $\mathcal{B}_{\mathbb{C}} = \mathcal{A}_{\mathbb{C}, \mathbb{C}}$ are the nonnegative-support rings.

Lemma 2.1 (Constant term, units and discreteness [06, 07, 10]).

1. Π_k is an ideal of $\mathcal{B}_k$ and of $\mathcal{A}_{D,k}$, and $\text{ct} : \mathcal{A}_{D,k} \rightarrow D$ is a surjective ring homomorphism with kernel Π_k , split by the inclusion of constants. It commutes with conjugation and with every ordinary finite matrix.
2. $\deg_{\omega}(fg) = \deg_{\omega} f + \deg_{\omega} g$ for nonzero $f, g \in \mathcal{B}_k$, and $\mathcal{B}_k^{\times} = k^{\times}$, $\mathcal{A}_{D,k}^{\times} = D^{\times}$. In particular $\mathbf{Oz}^{\times} = \{\pm 1\}$ and $\mathbf{Oz}[i]^{\times} = \{\pm 1, \pm i\}$.
3. An element of $\mathcal{B}_k$ with a nonzero positive-exponent part has infinite modulus. Hence $\mathcal{B}_k$ has no nonzero infinitesimal, its finite elements are exactly k , the finite elements of $\mathcal{A}_{D,k}$ are exactly D , every nonzero element of $\mathcal{A}_{D,k}$ for $D \in \{\mathbb{Z}, \mathbb{Z}[i]\}$ has modulus at least 1, and every vector of $\mathcal{A}_{D,k}^n \setminus D^n$ has infinite Euclidean norm.

Proof. All exponents occurring are nonnegative, and a sum of two nonnegative exponents is zero only when both are zero; so the constant coefficient of a product is the product of the constant coefficients, and a product with a factor in Π_k has positive support. Additivity, the kernel and the section are immediate. Commutation with conjugation and ordinary matrices is coefficientwise.

Hahn multiplication has leading coefficient the product of the leading coefficients, which gives the degree formula. If $fg = 1$ with $f, g \in \mathcal{B}_k$, both degrees are nonnegative and add to

zero, so both are zero; a normal form with nonnegative support and leading exponent 0 is constant. Intersecting with $\mathcal{A}_{D,k}$ and applying the argument to the inverse gives $\mathcal{A}_{D,k}^\times = D^\times$.

If the positive-exponent part is nonzero, it has a greatest exponent $\gamma > 0$, and the leading term of the absolute value (real case) or of the squared modulus (complex case) is $|c|\omega^\gamma$ or $|c|^2\omega^{2\gamma}$, which dominates every ordinary constant; adding a constant does not change this. Nonzero integers and Gaussian integers have modulus at least one. For vectors apply this to one nonconstant coordinate. $\square$

Negative exponents are what matter, not size: $\omega + \omega^{-1}$ is infinite but does not lie in $\mathcal{B}_\mathbb{R}$, and ct is not order preserving on $\mathcal{B}_\mathbb{R}$, since $\omega - 1 > 0$ while $\text{ct}(\omega - 1) = -1$ (06, 10).

Lemma 2.2 (Bounded coordinates are constant [06]). *If $f \in \mathcal{B}_\mathbb{R}$ and $|f| \leq M$ for an ordinary real M , then $f \in \mathbb{R}$; hence $\mathbf{Oz} \cap \mathbb{R} = \mathbb{Z}$. If $z \in \mathcal{B}_\mathbb{C}$ has real and imaginary parts bounded by ordinary reals, then $z \in \mathbb{C}$.*

Proof. This is lemma 2.1(iii), applied coordinatewise. $\square$

2.3 Gaps below sets of exponents

Lemma 2.3 (Gaps [06, 07, 10]).

1. Every set of surreal numbers has a strict upper bound and a strict lower bound in $\mathbf{No}$.
2. For every set $S \subseteq \mathbf{No}^{>0}$ there is $\delta > 0$ with $m\delta < s$ for all $s \in S$ and all ordinary $m \geq 1$.
3. For every set-indexed family $(h_s)_{s \in S}$ of positive infinite surreals there is $\gamma > 0$ with $\omega^{2\gamma} \ll h_s$ for every s .

Proof. For a set T , the cuts $\{T \mid \emptyset\}$ and $\{\emptyset \mid T\}$ give (i). For (ii) apply the set cut $\{0 \mid T\}$ to $T = \{s/m : s \in S, m \geq 1\}$, and take $\delta = 1$ for empty S . For (iii) apply (ii) to the set $\{\deg_\omega h_s\}$ of positive exponents and take $\gamma = \delta/2$; the leading exponent of $\omega^{2\gamma}/h_s$ is $2\gamma - \deg_\omega h_s < 0$. $\square$

The full-class hypothesis is substantive (06, 07, 10). The support $\{1/m : m \geq 1\}$ of $\sum_{m \geq 1} \omega^{1/m} \in \Pi$ has no positive *real* lower bound, but it has a positive surreal one, for instance $1/\omega$. Confusing the two assertions would invalidate the main proofs of Parts II and III; in a fixed exponent group such as $\mathbb{R}$ they fail (sections 12 and 32).

Proposition 2.4 (Single-product divisibility [06, 07]). *For every set-sized family $\mathcal{F} \subseteq \Pi_k$ there is $\delta > 0$ with $\omega^{-\delta} f \in \Pi_k$ for all $f \in \mathcal{F}$. In particular every $c \in \Pi_k$ is a product $c = \omega^\delta (\omega^{-\delta} c)$ of two elements of Π_k , and $\Pi_k^r = \Pi_k$ for every ordinary $r \geq 1$.*

Proof. The union of the supports is a set of positive exponents; choose δ strictly below it by lemma 2.3. Translating supports by $-\delta$ keeps them positive. Induction gives the powers. $\square$

This is `osq:prop:common(i),(ii)` of the sibling report.

Part I

Omnific points of algebraic groups

This part is source 06's. An algebraic group is affine and of finite type over a field unless stated otherwise; for a real algebraic group G , G^0 is its *algebraic* identity component, not the identity component of the real Lie group $G(\mathbb{R})$. The main theorems are [theorem 4.4](#) and [theorem 4.5](#).

3 A bounded-fiber rigidity principle

3.1 Tori and finite schemes

A torus over k is an algebraic group that becomes a product of copies of $\mathbb{G}_m$ over an algebraic closure of k .

Lemma 3.1 (Torus freezing [06]). *For every real torus T , $T(\mathcal{B}_{\mathbb{R}}) = T(\mathbb{R})$. For every complex torus T , $T(\mathcal{B}_{\mathbb{C}}) = T(\mathbb{C})$.*

Proof. A complex torus is isomorphic over $\mathbb{C}$ to $\mathbb{G}_m^r$, and its $\mathcal{B}_{\mathbb{C}}$ -points are $(\mathcal{B}_{\mathbb{C}}^{\times})^r = (\mathbb{C}^{\times})^r$ by [lemma 2.1](#). For a real torus, extend scalars from $\mathcal{B}_{\mathbb{R}}$ to $\mathcal{B}_{\mathbb{R}} \otimes_{\mathbb{R}} \mathbb{C} = \mathcal{B}_{\mathbb{C}}$. After a complex splitting the resulting point is constant by the complex case; hence the original point is a constant complex point fixed by the real conjugation descent datum, and such points are exactly $T(\mathbb{R})$. This is ordinary Galois descent for points; $\mathcal{B}_{\mathbb{R}} \hookrightarrow \mathcal{B}_{\mathbb{C}}$ is injective. $\square$

The complex case is the special case, for the coordinate ring of a torus, of the remark in the omnific Diophantine report that an affine complex variety whose coordinate ring is generated by units has only constant $\mathcal{B}_{\mathbb{C}}$ -points (subsection “An algebraic-geometric formulation” of `odg:sec:rigidity`).

Lemma 3.2 (Finite schemes do not acquire points [06]). *If X is a finite scheme over $k = \mathbb{R}$ or $\mathbb{C}$, or more generally a zero-dimensional affine scheme of finite type, then $X(\mathcal{B}_k) = X(k)$.*

Proof. Every element of the finite-dimensional coordinate algebra $k[X]$ is algebraic over k , so its image under a $\mathcal{B}_k$ -point is algebraic over k in F . The only elements of $\mathbf{No}$ algebraic over $\mathbb{R}$ are real, because $\mathbb{R}$ is real closed and $\mathbf{No}$ is an ordered field extension; the complex case follows because $\mathbb{C}$ is algebraically closed. Nilpotents map to 0. The coordinate algebra of a zero-dimensional affine scheme of finite type is Artinian and finite-dimensional, so no reducedness is needed. $\square$

3.2 Bounded real fibers

Theorem 3.3 (Bounded fibers above a torus [06]). *Let $X \hookrightarrow \mathbb{A}_{\mathbb{R}}^m$ be an affine real scheme of finite type, T a real torus and $\pi : X \rightarrow T$ a morphism. Suppose that for every $t \in T(\mathbb{R})$ the real points of the fiber X_t are bounded in the given affine coordinates. Then $X(\mathcal{B}_{\mathbb{R}}) = X(\mathbb{R})$. No bound uniform in t is required.*

Proof. Let $x \in X(\mathcal{B}_{\mathbb{R}})$. By [lemma 3.1](#), $\pi(x) = t \in T(\mathbb{R})$. Applying `ct` gives a real point `ct`(x) of X_t , so that fiber is nonempty; choose an ordinary $M > 0$ bounding all coordinates of $X_t(\mathbb{R})$. The statement that every solution of the finitely many polynomial equations of X_t satisfies $-M \leq x_j \leq M$ is a first-order formula of ordered fields with real parameters. Every

embedding of real closed fields is elementary (Tarski's quantifier elimination; see [11]). Apply this to $\mathbb{R} \subset K$, where $K \subset \mathbf{No}$ is the real closure inside $\mathbf{No}$ of the field generated by $\mathbb{R}$ and the coordinates of x ; it is a set. So the bounds hold for x , and each coordinate of x is real by lemma 2.2. $\square$

The only transfer used is transfer between real closed *fields*. No elementary equivalence of $\mathbb{Z}$ and $\mathbf{Oz}$, and no transfer in a language with an omnific-integer predicate, is asserted (06). The same transfer proves the bounded semialgebraic rigidity theorem `odg:thm:bounded`; for algebraic sets, the case of the trivial torus of theorem 3.3 gives the constant-point form of that theorem, while `odg:thm:bounded` covers bounded semialgebraic sets and theorem 3.3 covers unbounded algebraic sets with bounded fibers over a torus. Neither contains the other.

Corollary 3.4 (Complex zero-dimensional fibers [06]). *If X is affine of finite type over $\mathbb{C}$ and $\pi : X \rightarrow T$ is a morphism to a complex torus whose fibers over complex points are zero-dimensional, then $X(\mathcal{B}_{\mathbb{C}}) = X(\mathbb{C})$.*

Proof. Freeze $\pi(x)$ by lemma 3.1, then apply lemma 3.2 to that fiber. $\square$

Corollary 3.5 (Compact-by-torus groups [06]). *Let G be a connected real reductive group whose derived group $H = G^{\text{der}}$ has compact real points. Then $G(\mathcal{B}_{\mathbb{R}}) = G(\mathbb{R})$.*

Proof. The quotient $T = G/H$ is a torus, by the structure theory of connected reductive groups [12]. A fiber with a real point g has real points $gH(\mathbb{R})$, a compact set, so its coordinates are bounded in any affine embedding; empty real fibers are bounded too. Apply theorem 3.3. $\square$

Provenance. The bounded-fiber formulation is 06's bridge between normal-form rigidity and algebraic groups. Its inputs are unit rigidity, classical torus splitting and real-closed-field transfer; it generalizes the bounded-coordinate and constant-product arguments of the omnific Diophantine report without treating them as new.

4 Unipotents and the classification

4.1 A polynomial subgroup from one unipotent

A matrix $u \in \text{GL}_N(k)$ is unipotent if $u - 1_N$ is nilpotent; the notion does not depend on the faithful representation.

Lemma 4.1 (Polynomial one-parameter subgroup [06]). *Let k have characteristic zero, $G \subseteq \text{GL}_{N,k}$ an algebraic group and $1_N \neq u \in G(k)$ unipotent. Then*

$$N_0 = \log u = \sum_{j=1}^{N-1} \frac{(-1)^{j+1}}{j} (u - 1_N)^j \neq 0, \quad v(t) = \exp(tN_0) = \sum_{j=0}^{N-1} \frac{t^j N_0^j}{j!}$$

define a closed immersion $v : \mathbb{G}_a \hookrightarrow G$, with $v(s+t) = v(s)v(t)$ and $v(-t) = v(t)^{-1}$ over every k -algebra.

Proof. The exponential and logarithm identities used are identities modulo a fixed nilpotence power, hence polynomial identities over $\mathbb{Q}$; so N_0 is nilpotent, $\exp N_0 = u$ and $N_0 \neq 0$. For each integer $m \geq 0$, $v(m) = u^m \in G(k)$, and $\det v(t) = 1$. Each defining equation of G pulls back to a polynomial in $k[t]$ vanishing at all nonnegative integers, hence to zero, so v lands in G . If $(N_0)_{pq} \neq 0$, the polynomial function $g \mapsto (\log g)_{pq}/(N_0)_{pq}$ recovers t on the image, so

$k[G] \rightarrow k[t]$ is surjective: v is a closed immersion and is injective on points. The group laws follow from the commuting nilpotent exponential identities. $\square$

Corollary 4.2 (Purely infinite parameters [06]). *Let $(D, k) = (\mathbb{Z}, \mathbb{R})$ or $(\mathbb{Z}[i], \mathbb{C})$. If the k -fiber of a closed subgroup scheme $\mathcal{G} \subseteq \mathrm{GL}_{N,D}$ contains a nonidentity unipotent, then the map of lemma 4.1 restricts to an injective homomorphism*

$$v : (\Pi_k, +) \hookrightarrow \mathcal{G}(\mathcal{A}_{D,k}), \quad \mathrm{ct}(v(a)) = 1_N.$$

Proof. For $a \in \Pi_k$ every positive-degree term of $v(a)$ has entries in Π_k , which is closed under multiplication and under multiplication by k ; so $v(a)$ and $v(-a)$ have entries in $D + \Pi_k$. The D -equations of $\mathcal{G}$ vanish, since their substitution into $v(t)$ is the zero polynomial. The constant term is 1_N , and $\log v(a) = aN_0$ gives injectivity. $\square$

Rationality of the unipotent is unnecessary: irrational coefficients of N_0 multiply purely infinite terms and do not affect the integral constant coefficient.

4.2 The real structural input

A connected linear algebraic group in characteristic zero has a unipotent radical $R_u(G)$ and is reductive when it is trivial; for connected reductive G the derived group is connected semisimple and G/G^{der} is a torus; a nontrivial unipotent group over $\mathbb{R}$ has nontrivial real points, and its exponential is an isomorphism of schemes from its Lie algebra [12, Proposition 14.32].

Lemma 4.3 (Real semisimple compactness criterion [06]). *A connected semisimple real algebraic group H has no nonidentity real unipotent element if and only if $H(\mathbb{R})$ is compact.*

Proof. If $H(\mathbb{R})$ is compact, average an inner product in a faithful representation; every element is then orthogonal, and an orthogonal unipotent matrix is the identity because orthogonal matrices are diagonalizable over $\mathbb{C}$.

Conversely, by Cartan's theorem a faithful representation has a basis in which H is stable under transpose [14, Theorem 1.16 and Example 1.17(c)]; so its real Lie algebra $\mathfrak{h}$ is stable under transpose. If every element of $\mathfrak{h}$ is skew-symmetric, the real Lie identity component of $H(\mathbb{R})$ is a closed subgroup of the orthogonal group, hence compact, and a real algebraic group has finitely many real connected components; so $H(\mathbb{R})$ is compact. Otherwise there is a nonzero symmetric $S \in \mathfrak{h}$. As $\mathfrak{h}$ is semisimple its center is zero, so $\mathrm{ad} S \neq 0$; for $\langle A, B \rangle = \mathrm{tr}(AB^T)$ the operator $A \mapsto [S, A]$ is self-adjoint (permute the factors cyclically in the trace), so it has an eigenvector $Y \neq 0$ with real eigenvalue $\lambda \neq 0$. Diagonalize S orthogonally; $[S, Y] = \lambda Y$ makes Y map the s -eigenspace of S into the $(s + \lambda)$ -eigenspace, so Y is nilpotent. The Lie exponential of tY , here a finite polynomial, lies in $H(\mathbb{R})$, and at $t = 1$ it is a nonidentity unipotent (the truncated logarithm recovers Y). $\square$

The proof uses ordinary real Lie theory only at the Cartan and exponential steps and no global exponential on $\mathrm{No}[i]$. One may also obtain the noncompact direction from real rank and relative root groups [13, Sections 22–23].

4.3 The two dichotomies

Theorem 4.4 (Omnific rigidity and real unipotents [06]). *For a real algebraic group G the following are equivalent:*

1. $G(\mathcal{B}_{\mathbb{R}}) = G(\mathbb{R})$;
2. $G(\mathbb{R})$ has no unipotent element other than the identity;
3. G has no real algebraic subgroup isomorphic to $\mathbb{G}_a$;
4. G^0 is reductive and $(G^0)^{\text{der}}(\mathbb{R})$ is compact.

Let $\mathcal{G} \hookrightarrow \text{GL}_{N,\mathbb{Z}}$ be a closed subgroup scheme and $G = \mathcal{G}_{\mathbb{R}}$. The conditions are also equivalent to $\mathcal{G}(\mathbf{O}_{\mathbf{z}}) = \mathcal{G}(\mathbb{Z})$. When they fail there is an injective homomorphism

$$(\Pi, +) \longrightarrow \ker(\text{ct} : \mathcal{G}(\mathbf{O}_{\mathbf{z}}) \rightarrow \mathcal{G}(\mathbb{Z})), \quad a \longmapsto \exp(aN),$$

for a fixed nonzero real nilpotent matrix N , the exponential being a finite polynomial; and every ordinary integral point q lies on the injectively parametrized family $a \mapsto q \exp(aN)$ of omnific points.

Proof. Assume (2). The unipotent radical of G^0 is trivial, so G^0 is reductive; its derived group has no real unipotents, hence compact real points by [lemma 4.3](#); so (4) holds and $G^0(\mathcal{B}_{\mathbb{R}}) = G^0(\mathbb{R})$ by [corollary 3.5](#). For $g \in G(\mathcal{B}_{\mathbb{R}})$ put $g_0 = \text{ct}(g) \in G(\mathbb{R})$. The component group $\pi_0(G) = G/G^0$ is finite, so its $\mathcal{B}_{\mathbb{R}}$ -points are constant ([lemma 3.2](#)); g and g_0 have the same image there, hence $g_0^{-1}g \in G^0(\mathcal{B}_{\mathbb{R}})$ is real, and applying ct shows it is 1. Thus (2) implies (1), and (4) implies (1) by the same component argument.

If $G(\mathbb{R})$ has a nonidentity unipotent, [lemma 4.1](#) (over $\mathcal{B}_{\mathbb{R}}$) gives nonconstant points and a real $\mathbb{G}_a$ subgroup; conversely a $\mathbb{G}_a$ subgroup gives nonconstant points by substituting a purely infinite parameter into its nonconstant polynomial coordinates, and its real points are unipotent. So (1), (2), (3) are equivalent, and (2) implies (4) was shown.

For the integral model: in the rigid case an omnific point is a constant point of G , whose matrix entries and inverse-determinant coordinate lie in $\mathbf{O}_{\mathbf{z}} \cap \mathbb{R} = \mathbb{Z}$ ([lemma 2.2](#)); so it is an ordinary integral point. In the nonrigid case [corollary 4.2](#) gives the injection, and left translation by $q \in \mathcal{G}(\mathbb{Z})$ gives the family through q . $\square$

Compactness of $G(\mathbb{R})$ is not the criterion: every real torus, including the noncompact $\mathbb{G}_m$, is rigid, while a noncompact semisimple group is not. Only the characteristic-zero fiber is used; smoothness of the integral model at the primes is not needed. Keeping the model fixed matters: ordinary integral points depend on the model, whereas the criterion for new omnific points depends only on the real fiber (06).

Theorem 4.5 (The Gaussian dichotomy [06]). *For a complex algebraic group G , $G(\mathcal{B}_{\mathbb{C}}) = G(\mathbb{C})$ if and only if G^0 is a torus. For a closed subgroup scheme $\mathcal{G} \hookrightarrow \text{GL}_{N,\mathbb{Z}[i]}$, $\mathcal{G}(\mathbf{O}_{\mathbf{z}}[i]) = \mathcal{G}(\mathbb{Z}[i])$ if and only if $(\mathcal{G}_{\mathbb{C}})^0$ is a torus. If rigidity fails, the constant-term kernel contains an injective image of $(\Pi_{\mathbb{C}}, +)$ under a finite matrix exponential.*

Proof. Over an algebraically closed field of characteristic zero, a connected group with no nontrivial connected unipotent subgroup is a torus [[12](#), Corollary 17.25]; so a connected group that is not a torus has a nonidentity unipotent. If G^0 is a torus, [lemma 3.1](#) freezes its points and the component argument of [theorem 4.4](#) gives $G(\mathcal{B}_{\mathbb{C}}) = G(\mathbb{C})$. Otherwise [lemma 4.1](#) and [corollary 4.2](#) supply injective purely infinite families, also on the integral model; constant points of $\mathbf{O}_{\mathbf{z}}[i]$ have entries in $\mathbb{Z}[i]$. $\square$

Remark 4.6 (The rigid-affine-variety question for group schemes [merge]). The omnific Diophantine report asks for which affine varieties X over $\mathbb{Z}$ one has $X(\mathbf{O}_{\mathbf{z}}) = X(\mathbb{Z})$ (odg:q:affine). For X a closed subgroup scheme of $\text{GL}_{N,\mathbb{Z}}$, [theorem 4.4](#) answers it completely: exactly when

$X_{\mathbb{R}}$ has no nonidentity real unipotent, equivalently when $X_{\mathbb{R}}^0$ is reductive with compact real derived group; in the Gaussian setting, $X(\mathbf{Oz}[i]) = X(\mathbb{Z}[i])$ exactly when $(X_{\mathbb{C}})^0$ is a torus (theorem 4.5). *Status* (batch 31): the question remains open for general affine varieties of dimension at least two. For curves that report now answers it, after this remark was written. Let $\mathbb{k}$ be a field of characteristic zero and call a point nonconstant if its coordinates lie in that report's Hahn ring over $\mathbb{k}$ of series with nonnegative support (for a nonzero exponent group), but not all in $\mathbb{k}$. A smooth geometrically integral affine curve over $\mathbb{k}$ has nonconstant points exactly when it is the affine line (odg:cr:thm:affine). A geometrically integral affine curve, singular or not, has them exactly when its normalization is the affine line (odg:sg:thm:main). Hence an affine model whose real fiber is geometrically integral with normalization other than $\mathbb{A}_{\mathbb{R}}^1$ satisfies $X(\mathbf{Oz}) = X(\mathbb{Z})$, and a $\mathbb{Z}[i]$ -model whose complex fiber is geometrically integral with normalization other than $\mathbb{A}_{\mathbb{C}}^1$ satisfies $X(\mathbf{Oz}[i]) = X(\mathbb{Z}[i])$ (odg:sg:thm:omnific). For such a $\mathbb{Z}[i]$ -model, nonordinary $\mathbf{Oz}[i]$ -points exist exactly when the complex normalization is $\mathbb{A}_{\mathbb{C}}^1$ and $X(\mathbb{Z}[i]) \neq \emptyset$ (odg:sg:cor:gaussian). Over $\mathbf{Oz}$ one case stays open: a real curve with normalization $\mathbb{A}_{\mathbb{R}}^1$ whose integer points all lack a real preimage under the normalization, such as the isolated real node of $Y^2 = X^2(X - 1)$ (odg:sg:q:real). In the terms of that report, a satisfactory criterion should separate polynomial parametrization, torus rigidity and genuinely algebraic branches; for group schemes the first two decide rigidity: nonconstant points exist exactly when a polynomial unipotent family exists, and otherwise the identity component is compact-by-torus. In the nonrigid case the theorem supplies families through every integral point but does not describe all omnific points. The theorem also contains several results of that report as special cases: the unipotent subgroup $(\Pi, +) \hookrightarrow \mathrm{SO}(q, \mathbf{Oz})$ for indefinite forms in at least three variables (odg:cor:unipotent; $\mathrm{SO}(q)_{\mathbb{R}}$ is then isotropic, hence has real unipotents), the finiteness of $\mathrm{SO}_n(\mathbf{Oz})$ (odg:cor:orthogonal; SO_n is compact), and the torus and unit remark cited after lemma 3.1. The explicit matrices of those results remain useful: theorem 4.4 gives existence of a family, while odg:cor:unipotent writes it down.

5 Orthogonal examples

5.1 Compact real groups and Gaussian escape

For the standard positive-definite form, $\mathrm{O}_n(\mathbf{Oz}) = \mathrm{O}_n(\mathbb{Z})$ is the group of signed permutation matrices, of order $2^n n!$, and $\mathrm{SO}_n(\mathbf{Oz})$ consists of those of determinant one (odg:cor:orthogonal, proved there by sources 01 and 02 of that report and reproved by 06 through the column bound $\sum_i g_{ij}^2 = 1$). In particular $|\mathrm{SO}_3(\mathbf{Oz})| = 24$.

Over $\mathbb{C}$ the same group scheme escapes (06). Take $v = (1, i, 0)^{\mathrm{T}}$, $w = (0, 0, 1)^{\mathrm{T}}$ and

$$N = vw^{\mathrm{T}} - wv^{\mathrm{T}} = \begin{pmatrix} 0 & 0 & 1 \\ 0 & 0 & i \\ -1 & -i & 0 \end{pmatrix}, \quad N^{\mathrm{T}} = -N, \quad N^3 = 0, \quad N^2 \neq 0.$$

The finite exponential is

$$g(a) = 1_3 + aN + \frac{a^2}{2}N^2 = \begin{pmatrix} 1 - a^2/2 & -ia^2/2 & a \\ -ia^2/2 & 1 + a^2/2 & ia \\ -a & -ia & 1 \end{pmatrix}. \quad (5.1)$$

For every $a \in \Pi_{\mathbb{C}}$ it lies in $\mathrm{SO}_3(\mathbf{Oz}[i])$ ($\exp(aN)^{\mathrm{T}} = \exp(-aN)$, and $\det = 1$ since N is nilpotent), has constant term 1_3 , and its $(1, 3)$ entry recovers a . Neither compactness of the real form nor a positive-definite defining form prevents new Gaussian points: the equation $z_1^2 + z_2^2 + z_3^2 = 1$ is not a positivity statement over $\mathbb{C}$. A conjugate-transpose condition, in

contrast, defines a real algebraic unitary group, which must be analyzed as a real group by [theorem 4.4](#), not as this complex orthogonal group (06).

5.2 Real, not rational, unipotents

The form $q = x^2 + y^2 - 3z^2$ is anisotropic over $\mathbb{Q}$, so $\mathrm{SO}(q)(\mathbb{Q})$ has no nonidentity unipotent; but $\mathrm{SO}(q)(\mathbf{Oz})$ has purely infinite unipotent families. The existence of such a family is a special case of `odg:cor:unipotent` (any nondegenerate indefinite integral form in at least three variables), whose construction already allows irrational real isotropic vectors; it is not reproved here. Source 06 adds two points that the earlier corollary leaves implicit.

No rational unipotent [06]. Anisotropy: a primitive integer solution of $x^2 + y^2 = 3z^2$ would have $3 \mid x, y$ (squares are 0, 1 modulo 3), hence $3 \mid z$. If $u \neq 1$ were a unipotent in $\mathrm{SO}(q)(\mathbb{Q})$ with $N = \log u$, then $\exp(tN)$ preserves q for all t ([lemma 4.1](#)), so $N^\top Q + QN = 0$ with $Q = \mathrm{diag}(1, 1, -3)$; for r maximal with $N^r \neq 0$ and rational z with $N^r z \neq 0$, $B(N^r z, N^r z) = (-1)^r B(z, N^{2r} z) = 0$, contradicting anisotropy.

An explicit family [06]. With $v = (\sqrt{3}, 0, 1)^\top$ and $w = (0, 1, 0)^\top$ one has $B(v, v) = B(v, w) = 0$, $B(w, w) = 1$, and

$$N = vw^\top Q - wv^\top Q = \begin{pmatrix} 0 & \sqrt{3} & 0 \\ -\sqrt{3} & 0 & 3 \\ 0 & 1 & 0 \end{pmatrix}, \quad N^\top Q + QN = 0, \quad N^3 = 0,$$

$$\exp(aN) = \begin{pmatrix} 1 - 3a^2/2 & \sqrt{3}a & 3\sqrt{3}a^2/2 \\ -\sqrt{3}a & 1 & 3a \\ -\sqrt{3}a^2/2 & a & 1 + 3a^2/2 \end{pmatrix}. \quad (5.2)$$

For $a \in \Pi$ this lies in $\mathrm{SO}(q)(\mathbf{Oz})$ with constant term 1_3 , and its $(3, 2)$ entry is a ; $a = \omega$ gives an explicit nonconstant omnific matrix. The polynomial subgroup exists over $\mathbb{R}$, not over $\mathbb{Q}$, and that suffices for omnific integrality. There is no conflict with the rigidity of nonzero binary norm fibers in the omnific Diophantine report: the ambient group here is three-dimensional with noncompact real derived group.

5.3 A comparison table

Group or model	Real omnific behavior	Reason
A real torus	All points constant	Units freeze after complex splitting
Compact semisimple group	All points constant	Bounded real coordinates
$\mathrm{SO}(x^2 - y^2)$	All points constant	A torus, although noncompact
SL_n , $n \geq 2$	Purely infinite families	Elementary $\mathbb{G}_a$ subgroups
$\mathrm{SO}(x^2 + y^2 - 3z^2)$	Purely infinite families	Real, not rational, unipotents

The entries concern constant-term rigidity, not finite generation of an arithmetic subgroup or elementary generation of all omnific matrices (06). The table agrees with the quadratic-level dichotomy `odg:thm:quadclassification`; for instance $\mathrm{SO}(x^2 - y^2)$ is a torus, matching its case of binary forms.

6 Unipotent kernels and unitriangular filtrations

Let U be a unipotent algebraic group over $k \in \{\mathbb{R}, \mathbb{C}\}$; in a faithful representation it is conjugate over k to a group of upper unitriangular matrices, its Lie algebra $\mathfrak{u}$ consists of nilpotent matrices with a common nilpotence bound, and exponential and logarithm are mutually inverse polynomial maps [12, Proposition 14.32].

Proposition 6.1 (The constant-term kernel of a unipotent group [06]). *There is a polynomial-coordinate identification*

$$\ker(U(\mathcal{B}_k) \rightarrow U(k)) = \exp(\mathfrak{u}(k) \otimes_k \Pi_k),$$

where the right side means finite combinations of a fixed k -basis of $\mathfrak{u}$ with coefficients in Π_k , and multiplication corresponds to the truncated Baker–Campbell–Hausdorff polynomial. For an integral model with unipotent k -fiber, its constant-term identity fiber is the same group inside the faithful representation.

Proof. Exponential and logarithm are polynomials over k , so they commute with ct. A Lie-algebra point has zero constant term exactly when its coordinates lie in Π_k , and then its exponential has constant term the identity. The Baker–Campbell–Hausdorff expression truncates, and its brackets have coefficients in Π_k . For an integral model these points and their inverses satisfy the defining equations as in corollary 4.2; the converse follows by embedding into the k -fiber and taking logarithms. $\square$

No convergence of an exponential series at an infinite argument is asserted; every displayed matrix function is a finite polynomial.

Put $U_n = \text{UT}_n(\Pi_k)$, the upper unitriangular matrices with entries above the diagonal in Π_k , and for $r \geq 1$

$$\text{Fil}_r = \{g \in U_n : g_{ij} = 0 \text{ whenever } 0 < j - i < r\}, \quad \text{Fil}_1 = U_n, \quad \text{Fil}_n = 1.$$

Theorem 6.2 (Exact commutator filtration [06]). *For all $r, s \geq 1$, $[\text{Fil}_r, \text{Fil}_s] = \text{Fil}_{r+s}$. Consequently the lower central series is $\gamma_r(U_n) = \text{Fil}_r$, the derived series is $U_n^{(j)} = \text{Fil}_{2^j}$, and for $n \geq 2$ the nilpotency class is $n-1$ and the derived length is $\lceil \log_2 n \rceil$. The first superdiagonal induces $U_n/[U_n, U_n] \cong (\Pi_k, +)^{n-1}$.*

Proof. Write $\text{Fil}_r = 1 + J_r$ with J_r the corresponding strictly upper triangular matrices; $J_r J_s \subseteq J_{r+s}$ and finite geometric inversion of $1 + X$ give $[\text{Fil}_r, \text{Fil}_s] \subseteq \text{Fil}_{r+s}$. Elimination of successive superdiagonals writes every member of Fil_t as a finite product of $e_{ij}(a)$ with $j - i \geq t$ and $a \in \Pi_k$. For a generator $e_{ij}(a)$ of Fil_{r+s} with $r + s < n$ choose $h = i + r$, so $h - i \geq r$ and $j - h \geq s$, write $a = bc$ with $b, c \in \Pi_k$ (proposition 2.4), and use $[e_{ih}(b), e_{hj}(c)] = e_{ij}(bc)$. If $r + s \geq n$ both sides are trivial. Induction gives the two series; $\text{Fil}_t \neq 1$ exactly when $t < n$, witnessed by $e_{1n}(a)$, $a \neq 0$. The first superdiagonal is additive, onto $(\Pi_k)^{n-1}$, with kernel Fil_2 . $\square$

For every positive surreal exponent ρ , coefficient extraction $f \mapsto [\omega^\rho]f$ is a nonzero additive map $\Pi_k \rightarrow k$; composed with a first-superdiagonal coordinate it is a nontrivial homomorphism $U_n \rightarrow (k, +)$ for $n \geq 2$. A purely infinite unipotent group can therefore be visible to small groups. The opposite-root commutators of Part II destroy all such visibility; the invisibility theorem there is not a statement about all purely infinite matrix groups (06).

Provenance. The exponential correspondence and the unitriangular commutator calculus are classical specializations; the exact formulas use the single-product property of Π_k . They describe the nonrigid alternative and are not presented as independent discoveries (06).

Part II

Elementary groups and their set-sized images

This part asks how much of an omnific matrix a set-sized group can detect. A group representation of matrices need not come from a homomorphism of their coefficient rings, so the question is genuinely stronger than the ring question answered in the sibling report (`osq:thm:universal`). The answer depends sharply on whether a third row and column are available: two rows admit nonmultiplicative coefficient specializations, and three rows provide the commutator relation that recovers multiplication, which the supply of surreal scales turns into a cardinality obstruction (07). The part follows source 07, with source 06's normality, splitting, tower and boundary results inserted. Source 11's third proof of the main theorem is in [section 9](#); its structural, Steinberg and cardinal results form the last three sections of the part ([sections 19 to 21](#)). [Part IV](#) (source 12) replaces the third index by an arbitrary root system without A_1 component.

7 Setting

For a commutative unital ring A , $E_n(A)$ is the subgroup of $\mathrm{SL}_n(A)$ generated by the elementary matrices $e_{ij}(a) = 1_n + aE_{ij}$, $a \in A$, $i \neq j$. Generation always means finite words. *We never identify $E_n(A)$ with $\mathrm{SL}_n(A)$ for an omnific ring*; for $n = 2$ they differ ([theorem 18.2](#)). Direct multiplication of matrix units gives, for distinct i, h, j ,

$$e_{ij}(a)e_{ij}(b) = e_{ij}(a + b), \quad [e_{ih}(a), e_{hj}(b)] = e_{ij}(ab), \quad (7.1)$$

finite polynomial identities over any commutative ring [[19](#), Chapter III, Section 1].

Throughout, $k \in \{\mathbb{R}, \mathbb{C}\}$, $D \subseteq k$ is a unital subring (for most statements $D = \mathbb{Z}$ or $\mathbb{Z}[i]$), $A = \mathcal{A}_{D,k}$, and

$$P_n = \ker(\mathrm{ct} : E_n(A) \rightarrow E_n(D)), \quad (7.2)$$

where ct is applied entrywise; the inclusion of constant matrices is a section. For $D = \mathbb{Z}, \mathbb{Z}[i]$ Euclidean row reduction gives $E_n(D) = \mathrm{SL}_n(D)$ (the entries of the first column of a determinant-one matrix generate the unit ideal; reduce it to a unit times e_1 , remove the unit by an elementary diagonal block, and induct; over $\mathbb{Z}[i]$ use the Gaussian norm). These are finite ordinary matrix arguments; no Euclidean algorithm in **Oz** is assumed (06, 07).

Lemma 7.1 (The split relative kernel [07, 11]). *For every $n \geq 2$, P_n is the normal closure in $E_n(A)$ of the matrices $e_{ij}(x)$, $x \in \Pi_k$.*

Proof. Let N' be that normal closure; $N' \subseteq P_n$. Modulo N' , $e_{ij}(d + x) = e_{ij}(d)e_{ij}(x)$ is congruent to the constant $e_{ij}(d)$, so every word is congruent to a matrix of $E_n(D)$; if the word lies in P_n , its constant representative is the identity. So $P_n = N'$. Only the split inclusion of D is used, not a relative exactness theorem. $\square$

Source 11 proves the same lemma for every split unital ring $D \oplus I$ with ideal I (its Lemma 5.1), by the same finite-word replacement, and stresses that it uses neither normality of the root-generated subgroup, nor a presentation of E_n , nor $E_n = \text{SL}_n$.

For $n \geq 3$ the normal closure is not needed: the root subgroups already generate a normal subgroup (06).

Proposition 7.2 (An abstract idempotent-ideal lemma [06]). *Let A be a commutative unital ring, $n \geq 3$, and I an ideal in which every element is a product of two elements of I . Then $P = \langle e_{ij}(a) : a \in I, i \neq j \rangle$ is perfect and normal in $E_n(A)$.*

Proof. For a generator $e_{ij}(a)$ choose a third index h and $a = bc$ with $b, c \in I$; (7.1) writes it as a commutator of two elements of P , so $P = [P, P]$. Conjugating a generator by $z = e_{kl}(r)$, $r \in A$, gives, except in the opposite-root case $(k, l) = (j, i)$, either the generator or a product of root elements with parameters a and $\pm ra$, all in I . For the opposite root write $e_{ij}(a) = [e_{ih}(b), e_{hj}(c)]$ and use

$$e_{ji}(r)e_{ih}(b)e_{ji}(-r) = e_{jh}(rb)e_{ih}(b), \quad (7.3)$$

$$e_{ji}(r)e_{hj}(c)e_{ji}(-r) = e_{hi}(-cr)e_{hj}(c); \quad (7.4)$$

both conjugated factors lie in P , hence so does their commutator. $\square$

A version with $I^2 = I$ and finite sums of products follows by splitting $e_{ij}(a)$ along (7.1); the single-product hypothesis holds for Π_k by proposition 2.4. The perfectness half, in this $I^2 = I$ form and for any group generated by root maps, is also source 11's (corollary 9.3(iii)).

Proposition 7.3 (Exact split kernel [06]). *For $n \geq 3$ and every unital subring $D \subseteq k$,*

$$E_n(A) = P_n \rtimes E_n(D), \quad P_n = \langle e_{ij}(a) : a \in \Pi_k, i \neq j \rangle.$$

Proof. Let P' be the subgroup on the right; it is normal by propositions 2.4 and 7.2. Writing $e_{ij}(d + b) = e_{ij}(b)e_{ij}(d)$ and moving purely infinite factors to the left using normality writes every $g \in E_n(A)$ as $g = pq$ with $p \in P'$, $q \in E_n(D)$; applying ct gives $\text{ct}(g) = q$. Hence $P_n = P'$, $P' \cap E_n(D) = 1$, and the constant inclusion is a section. $\square$

8 Two collision mechanisms

A set-sized target identifies two of any sufficiently large set of root elements. To turn such a collision into the death of an arbitrary root element one needs a divisor: a purely infinite quotient of the prescribed parameter by the difference of the two colliding parameters. The two sources find the divisor in different ways, and both ways are already lemmas of the sibling report, which is where the sources found them.

8.1 Division by a difference of nearby monomials

Lemma 8.1 (Absorption [06]). *Fix $0 \neq c \in \Pi_k$. There is $\varepsilon > 0$ such that for all $0 < \beta < \alpha < \varepsilon$,*

$$\frac{c}{\omega^\alpha - \omega^\beta} = \sum_{m \geq 0} c \omega^{-\alpha - m(\alpha - \beta)} \in \Pi_k. \quad (8.1)$$

For every ordinal Λ the interval $(0, \varepsilon)$ contains Λ distinct exponents, for instance $\alpha_\xi = \varepsilon/(2 + \xi)$, $\xi < \Lambda$, with surreal arithmetic on the embedded ordinals.

Proof. Choose $\delta > 0$ strictly below $\text{supp } c$ (lemma 2.3) and put $\varepsilon = \delta/(4\omega)$ in the exponent field, so that $m\varepsilon < \delta$ for every ordinary $m \geq 1$. With $\gamma = \alpha - \beta > 0$, $(\omega^\alpha - \omega^\beta)^{-1} = \omega^{-\alpha}(1 - \omega^{-\gamma})^{-1} = \sum_{m \geq 0} \omega^{-\alpha - m\gamma}$ by (2.2). Every exponent of the product with c has the form $s - \alpha - m\gamma$ with $s \in \text{supp } c$, and $s - \alpha - m\gamma > \delta - (m+1)\varepsilon > 0$; cancellation cannot create a nonpositive exponent. The α_ξ are distinct and lie in $(0, \varepsilon)$. $\square$

This is the subordinate division lemma `osq:lem:division` of the sibling report (sources 09, 10, 11 there), with the admissible scales $(0, \varepsilon)$ inside its class $U_{\mathbf{No}}(\text{supp } c)$; source 06 cites it as the geometric collision construction of the quotient report's base manuscript, adapted to produce an arbitrary prescribed root parameter.

Remark 8.2 (No series is mapped term by term [06]). If $d = \omega^\alpha - \omega^\beta$ and b is the whole series in (8.1), the only identity used is $db = c$. A target homomorphism is applied to a finite matrix commutator involving b , never to the summands. For the truncations b_M one has

$$db_M = c(1 - \omega^{-(M+1)(\alpha-\beta)}), \quad (8.2)$$

with nonzero remainder; replacing b by a truncation would not prove the theorem (compare `osq:rem:nolimit`).

8.2 Rescaling a large set field

Lemma 8.3 (Exponent scaling [07]). *For every $h \in \mathbf{No}^{>0}$, $\Phi_h(\sum_s a_s \omega^s) = \sum_s a_s \omega^{hs}$ is an ordered field automorphism of $\mathbf{No}$ fixing $\mathbb{R}$, with inverse $\Phi_{1/h}$; it extends coefficientwise to $\mathbf{No}[i]$.*

Proof. Multiplication by h is a strictly increasing additive bijection of the exponent group; it preserves reverse well ordering, set size, exponent sums and the finite convolution formulas, and the leading coefficient. $\square$

This is the dilation S_h of the surcomplex automorphism report (`saut:eq:dilation`, with $t = \omega^{-1}$) and of `odg:def:ex:dilation`. No compatibility with the surreal exponential, simplicity order or birthday is asserted (07).

Lemma 8.4 (Set-field compression [07]). *For every $\delta > 0$ and every infinite set cardinal λ there is a set subfield $L \subset \mathbf{No}$ with $|L| \geq \lambda$ and $\text{supp } a \subset (-\delta, \delta)$ for every $a \in L$; in particular $(a - b)^{-1}$ has the same support bound for distinct $a, b \in L$.*

Proof. Let L_0 be the subfield generated by λ distinct surreals (for instance embedded ordinals); it is a set with at least λ elements. The union T of the supports of its members is a set; choose $\beta > 0$ with $|s| < \beta$ for $s \in T$ (lemma 2.3), put $h = \delta/\beta$ and $L = \Phi_h(L_0)$. Then $|hs| < \delta$ for every exponent of every member of L , and $(a - b)^{-1} \in L$. $\square$

The quantifiers matter: L is chosen after δ and λ , and the supports of the whole field, not only of its generators, are compressed (07).

Lemma 8.5 (A commutator-ready family [07]). *Let $0 \neq x \in \Pi_k$ and H a set. There are $\delta > 0$ and a set subfield $L \subset \mathbf{No}$ with $|L| > |H|$ such that, with $\tau = \omega^{2\delta}$,*

$$\tau a \in \Pi_k \quad (a \in L), \quad \frac{x}{\tau(a-b)} \in \Pi_k \quad (a, b \in L, a \neq b).$$

Proof. Choose $\delta > 0$ with $4\delta < s$ for all $s \in \text{supp } x$ and apply [lemma 8.4](#) with a cardinal larger than $|H|$. Every exponent of τa lies in $(\delta, 3\delta)$, and every exponent of $x\omega^{-2\delta}(a-b)^{-1}$ exceeds $4\delta - 2\delta - \delta = \delta$. The auxiliary field may stay real when x has complex coefficients. $\square$

This is the scaled-field collision lemma `osq:lem:collision` of the sibling report (source 06 there, with 03) in the form needed for root parameters: there $m = X^{g/2}$ and $mL \subseteq \Pi_K$ give $(mx - my) \cdot m/(x - y) = m^2$; here τa plays the role of mx and the prescribed x that of m^2 . Source 07 credits the scaled-field collision to the quotient report's base manuscript.

9 Three or more rows: no set-sized images

Theorem 9.1 (The purely infinite elementary kernel is invisible to sets [06, 07, 11]). *Let $k \in \{\mathbb{R}, \mathbb{C}\}$, $D \subseteq k$ a unital subring, $A = \mathcal{A}_{D,k}$ and $n \geq 3$.*

1. $P_n = \langle e_{ij}(a) : a \in \Pi_k, i \neq j \rangle$, and $E_n(A) = P_n \rtimes E_n(D)$.
2. P_n is perfect and nontrivial.
3. Every class homomorphism from P_n to a set-sized group is trivial.
4. For every set-sized group H , composition with ct is a bijection $\text{Hom}(E_n(D), H) \rightarrow \text{Hom}(E_n(A), H)$. Thus $E_n(D)$ is the universal set-sized group quotient of $E_n(A)$; it is $\text{SL}_n(D)$ for $D \in \{\mathbb{Z}, \mathbb{Z}[i]\}$, so for $E_n(\mathbf{Oz})$ and $E_n(\mathbf{Oz}[i])$, and $\text{SL}_n(k)$ for $D = k$.

No continuity, preservation of infinite sums, definability or algebraicity of a homomorphism is assumed.

Assertion (iii) is stronger than the factorization (iv): a homomorphism defined only on the kernel need not extend to $E_n(A)$, and all three proofs below treat such maps directly (06, 07, 11). Source 06 states (i)–(iv) for $D \in \{\mathbb{Z}, \mathbb{Z}[i]\}$ and, in its corollary on other constant rings, for every set-sized unital $D \subseteq k$; source 07 states them for $D \in \{\mathbb{Z}, \mathbb{Z}[i]\}$, and its proof uses D only through the split inclusion of [lemma 7.1](#), so it applies verbatim to every unital $D \subseteq k$. Source 11 states them for $D \in \{\mathbb{Z}, \mathbb{Z}[i]\}$ (its Theorem A, proved as its Theorem 5.2 with Lemma 5.3), also for maps defined only on the kernel; its proof uses D only through the split retraction, so it too applies to every unital $D \subseteq k$. Every $D \subseteq k$ is a set.

Proof of (i), (ii) and (iv) given (iii). (i) is [proposition 7.3](#). (ii): perfectness is [proposition 7.2](#), and $e_{12}(\omega) \neq 1$. (iv): a homomorphism $\psi : E_n(A) \rightarrow H$ restricts to the trivial map on P_n by (iii), so it factors uniquely through ct in the split decomposition; restriction to the constant subgroup gives the factor, and for every g the matrix $g \text{ct}(g)^{-1}$ lies in P_n . Conversely every homomorphism $E_n(D) \rightarrow H$ composes with ct . Over the field k ordinary elimination gives $E_n(k) = \text{SL}_n(k)$. $\square$

First proof of (iii): rescaled set fields [07]. By [lemma 7.1](#) it suffices to kill each conjugate $g e_{ih}(x)g^{-1}$ with $g \in E_n(A)$, $0 \neq x \in \Pi_k$, $i \neq h$. Let $\psi : P_n \rightarrow H$. Choose an index $j \notin \{i, h\}$ and take τ, L from [lemma 8.5](#). Since P_n is normal, $a \mapsto \psi(g e_{ij}(\tau a)g^{-1})$ is defined on L , and $|L| > |H|$ gives distinct $a, b \in L$ with equal images; by (7.1) $\psi(g e_{ij}(\tau(a-b))g^{-1}) = 1$. Put $d = x/(\tau(a-b)) \in \Pi_k$. The conjugated commutator identity

$$[g e_{ij}(\tau(a-b))g^{-1}, g e_{jh}(d)g^{-1}] = g e_{ih}(\tau(a-b)d)g^{-1} = g e_{ih}(x)g^{-1}$$

is an identity between elements of P_n , and applying ψ kills the normal generator. The conjugator g need not lie in P_n : only the two conjugated root elements and their commutator, all in the domain, are fed to ψ . $\square$

Second proof of (iii): division by nearby monomials [06]. By (i), P_n is generated by the $e_{ij}(c)$, $c \in \Pi_k$, without conjugates. Let $\varphi : P_n \rightarrow H$, $c \neq 0$, and fix a third index h . Take ε from [lemma 8.1](#) and a set $T \subset (0, \varepsilon)$ that does not inject into H (its Hartogs ordinal, or a cardinal $> |H|$, indexes such a set). The elements $e_{ih}(\omega^\alpha)$, $\alpha \in T$, lie in P_n and cannot have distinct images, so for some $\alpha > \beta$ in T , $\varphi(e_{ih}(\omega^\alpha - \omega^\beta)) = 1$. With $d = \omega^\alpha - \omega^\beta$ and $b = c/d \in \Pi_k$,

$$\varphi(e_{ij}(c)) = \varphi([e_{ih}(d), e_{hj}(b)]) = [1_H, \varphi(e_{hj}(b))] = 1_H.$$

Every generator, hence every finite word, is killed. $\square$

The third route turns a group homomorphism into a ring homomorphism. It rests on an abstract lemma about groups generated by root maps. For a commutative ring I , possibly without unit, write x_{ij} ($1 \leq i \neq j \leq n$) for maps from I into a group; the letters x_{ij} always carry two indices and are not the elements $x \in \Pi_k$ of the proofs above.

Theorem 9.2 (Root-kernel reconstruction [11]). *Let $n \geq 3$, let I be a commutative ring, possibly without unit and possibly a proper class, and let Σ be a group generated by additive homomorphisms $x_{ij} : (I, +) \rightarrow \Sigma$, $i \neq j$, with*

$$[x_{ih}(a), x_{hj}(b)] = x_{ij}(ab) \quad (i, h, j \text{ distinct}, a, b \in I). \quad (9.1)$$

For a homomorphism $\varphi : \Sigma \rightarrow H$ to a set-sized group put $J_\varphi = \{a \in I : \varphi(x_{ij}(a)) = 1 \text{ for all } i \neq j\}$. Then J_φ is an ideal of I , and there are a set-sized ring R_φ , possibly without unit, and a surjective additive and multiplicative map $\vartheta_\varphi : I \rightarrow R_\varphi$ with kernel J_φ and

$$|R_\varphi| \leq |H|^{n(n-1)}.$$

Moreover $R_\varphi = 0$ if and only if φ is trivial.

Proof. Each condition defining J_φ is the kernel of an additive homomorphism, so J_φ is an additive subgroup. For $a \in J_\varphi$, $b \in I$ and $i \neq j$ choose $h \notin \{i, j\}$; then $\varphi(x_{ij}(ab)) = [\varphi(x_{ih}(a)), \varphi(x_{hj}(b))] = [1, \varphi(x_{hj}(b))] = 1$. This holds for every ordered pair, so $ab \in J_\varphi$, and commutativity gives a two-sided ideal. Put $\vartheta_\varphi(a) = (\varphi(x_{ij}(a)))_{i \neq j} \in H^{n(n-1)}$ and $R_\varphi = \vartheta_\varphi(I)$, a subclass of a set and hence a set. By additivity, $\vartheta_\varphi(a) = \vartheta_\varphi(b)$ if and only if $a - b \in J_\varphi$. Define $\vartheta_\varphi(a) + \vartheta_\varphi(b) = \vartheta_\varphi(a + b)$ and $\vartheta_\varphi(a)\vartheta_\varphi(b) = \vartheta_\varphi(ab)$; these are well defined because J_φ is a subgroup and an ideal, and the ring axioms are inherited from I . If $R_\varphi = 0$, every generator of Σ is killed, so φ is trivial; the converse is clear. $\square$

The ring structure on R_φ is transported from I/J_φ ; it is not the componentwise product in the group $H^{n(n-1)}$, which carries no ring structure. No set of class-sized cosets is formed: the tuple image is the carrier (11).

Corollary 9.3 (Transfer from rings to groups [11]). *In the setting of [theorem 9.2](#):*

1. *if every additive multiplicative map from I to a set-sized ring is zero, every homomorphism from Σ to a set-sized group is trivial;*
2. *if κ is an uncountable cardinal and every additive multiplicative map from I to a ring of cardinality $< \kappa$ is zero, every homomorphism from Σ to a group of cardinality $< \kappa$ is trivial;*
3. *if $I^2 = I$ then Σ is perfect, and if every element of I is a single product bc , each $x_{ij}(a)$ is a single commutator of root elements.*

Proof. (i) Apply the hypothesis to ϑ_φ . (ii) If $|H| < \kappa$ then $|H|^{n(n-1)} < \kappa$: it is finite for finite H and equals $|H|$ for infinite H . (iii) For $a = \sum_{\ell=1}^r b_\ell c_\ell$ and $h \notin \{i, j\}$, $x_{ij}(a) = \prod_\ell [x_{ih}(b_\ell), x_{hj}(c_\ell)]$. $\square$

Lemma 9.4 (Normal generation [11]). *Let U be a subgroup of a group G and N its normal closure. If every homomorphism $U \rightarrow H$ is trivial for a given group H , so is every homomorphism $N \rightarrow H$. If U is perfect, so is N .*

Proof. N is generated by the conjugates gUg^{-1} , $g \in G$. For $f : N \rightarrow H$, the map $u \mapsto f(gug^{-1})$ is a homomorphism $U \rightarrow H$, hence trivial. Each gUg^{-1} is perfect when U is, so it lies in $[N, N]$. $\square$

Third proof of (iii): a ring rebuilt from the root maps [11]. Let $U = \langle e_{ij}(a) : a \in \Pi_k, i \neq j \rangle$. The restrictions $x_{ij} = e_{ij}|_{\Pi_k}$ are additive and satisfy (9.1) by (7.1), so theorem 9.2 applies with $I = \Pi_k$ and $\Sigma = U$. Every additive multiplicative map from Π_k to a set-sized ring is zero: this is part (iii) of the ring theorem `osq:thm:universal` of the sibling report, which source 11 reproves (its Theorem 3.3) with the division identity of lemma 8.1 in the form $c/(\omega^\alpha - \omega^\beta) \in \Pi_k$ for $0 < \beta < \alpha < \delta$, where $m\delta < s$ for all $s \in \text{supp } c$ and ordinary $m \geq 1$, proving strong summability of the quotient as well as positivity, and with the exponents $\delta/(\xi + 2)$, $\xi < |R|^+$, to force a collision in the target ring R . By corollary 9.3(i) every homomorphism from U to a set-sized group is trivial. P_n is the normal closure of U (lemma 7.1), and lemma 9.4 kills every homomorphism from P_n to H . By (i) $U = P_n$, so the last step is not needed here; source 11 does not use (i), and kills a map on the kernel separately on each conjugate of U . $\square$

The first two proofs do not construct a ring map from A to H or to an endomorphism ring; both apply to arbitrary abstract group maps on P_n , and for $k = \mathbb{C}$ neither needs an order on coefficients, only positivity of exponents (06, 07). The third proof does build a ring, but from the root maps of the given group homomorphism, not from any ring map on A , so it too applies to arbitrary abstract group maps defined only on the kernel (11). Neither of the first two proofs uses the other's stronger ingredients: 06's needs only single-product divisibility, gaps and the geometric inverse, and 07's needs the field automorphism Φ_h and a set field closed under inverses. The third is not independent of them in mechanism: it imports the ring theorem, whose proofs use the same collision identities (section 8); what it adds is the general transfer corollary 9.3, which also gives the Steinberg and cardinal versions below (theorems 20.2 and 21.3). The recovery of coefficient algebra from root commutators has substantial precedent [18]; source 11 claims neither that idea nor the normal-generation machinery, only its elementary finite-tuple form for cardinality and nonunital parameter rings. Perfectness alone would kill only abelian images; the cardinal collision kills images in finite simple groups and in arbitrary nonabelian groups as well. The theorem does not make P_n trivial: its tautological representation on the proper-class space F^n is faithful and has no set-sized target (06, 07).

Corollary 9.5 (Actions and linear representations [06, 07, 11]). *For $n \geq 3$, every action of P_n on a set is trivial, and every action of $E_n(A)$ on a set factors uniquely through ct . Every linear representation of P_n on a set-sized vector space over a set-sized field, of any dimension, is trivial; the same holds for representations on set-sized abelian groups or on any structure whose automorphism group is a set.*

Proof. An action on a set X is a homomorphism to the set-sized group $\text{Sym}(X)$; for linear representations use the invertible linear maps of the space. $\square$

Source 11 proves the same (its Theorem 7.1 and Corollary 7.2) and adds two points. P_n has no nontrivial transitive action on a set with more than one point. Restriction to constant matrices and pullback by `ct` identify the representations of $E_n(A)$ on set-sized vector spaces with those of $E_n(D)$, intertwiners included: a linear map intertwines two pulled-back representations exactly when it intertwines their restrictions to the constant subgroup, since every matrix acts through its constant term. Here “arithmetic” means only this factorization; the representations of $SL_n(D)$ are not classified, nor asserted to be algebraic (11).

Remark 9.6 (The ring theorem as a consequence [merge]). [Theorem 9.1](#) implies the universal set-sized quotient theorem `osq:thm:universal` of the sibling report for coefficient fields $K \in \{\mathbb{R}, \mathbb{C}\}$ and every unital $D \subseteq K$. *Proof.* Let S be a set-sized unital ring, possibly noncommutative, and $\psi : A \rightarrow S$ a unital ring homomorphism. Entrywise application of ψ is multiplicative on matrices and sends $e_{ij}(a)$ to $1 + \psi(a)E_{ij} \in GL_n(S)$, so it restricts to a homomorphism $E_n(A) \rightarrow GL_n(S)$ into a set-sized group, with $n = 3$. By [theorem 9.1\(iii\)](#) it kills $e_{12}(x)$ for $x \in \Pi_k$, so $\psi(x) = 0$ and $\psi = \psi|_D \circ \text{ct}$; this is part (ii) of the ring theorem. For an additive multiplicative map $\varphi : A \rightarrow S$ into a possibly nonunital set-sized ring, let $S^+ = \mathbb{Z} \times S$ be the unitalization, with $(m, s)(m', s') = (mm', ms' + m's + ss')$; then $m + x \mapsto (m, \varphi(x))$ ($m \in \mathbb{Z}$, $x \in \Pi_k$) is a unital ring homomorphism from $\mathcal{A}_{\mathbb{Z},k} = \mathbb{Z} \oplus \Pi_k$ to the set-sized ring S^+ , so $\varphi(\Pi_k) = 0$ by the unital case; this gives parts (i) and (iii). A set-sized unital module M gives a unital map $A \rightarrow \text{End}_{\mathbb{Z}}(M)$, which gives part (iv). $\square$

The implication is logical, not an independent proof: both proofs of [theorem 9.1](#) use the same collision identities as the ring theorem ([section 8](#)). The group theorem is strictly stronger in scope: an abstract group homomorphism on $E_n(A)$ or on P_n need not be induced by a ring or module map, so the theorem also strengthens `osq:cor:faithful` and `osq:prop:matrices`, which concern representations coming from ring maps. The ring theorem is proved in the sibling report for every set field K ; the group theorem is stated here only for $k \in \{\mathbb{R}, \mathbb{C}\}$, as in the sources.

The converse implication is source 11’s: by [corollary 9.3\(i\)](#) and [lemma 9.4](#), part (iii) of the ring theorem implies [theorem 9.1\(iii\)](#) (the third proof above). For $k \in \{\mathbb{R}, \mathbb{C}\}$ the group theorem (iii) and the nonunital ring theorem `osq:thm:universal(iii)` therefore imply each other through finite reductions. Neither implication is an independent proof of its conclusion. Source 12 proves the equivalence for every class ring I without unit with $2I = I$, its Dorroh unitization, and every root system without A_1 component ([theorem 38.5](#)).

Proposition 9.7 (An abstract collision criterion [07]). *Let A be a commutative ring, $I \triangleleft A$, $n \geq 3$, and $E_n(A, I)$ the normal closure of $e_{ij}(I)$ in $E_n(A)$. Fix a set-sized group H . Suppose that for each $0 \neq x \in I$ there are a set $\mathcal{P}$ with $|\mathcal{P}| > |H|$ and elements $u_p \in I$ ($p \in \mathcal{P}$) and $v_{pq} \in I$ ($p \neq q$) with*

$$(u_p - u_q)v_{pq} = x. \quad (9.2)$$

Then every homomorphism $E_n(A, I) \rightarrow H$ is trivial.

Proof. For a normal generator $g e_{ih}(x) g^{-1}$ choose a third index j . The images of $g e_{ij}(u_p) g^{-1}$ cannot all be distinct; a collision at $p \neq q$ kills $g e_{ij}(u_p - u_q) g^{-1}$, and its commutator with $g e_{jh}(v_{pq}) g^{-1}$ is $g e_{ih}(x) g^{-1}$. $\square$

The first proof supplies the criterion with $u_a = \tau a$ and $v_{ab} = x/(\tau(a - b))$, the second with $u_\alpha = \omega^\alpha$ and $v_{\alpha\beta} = x/(\omega^\alpha - \omega^\beta)$. The criterion does not require A to be a field, A/I to split, or H to be linear; the splitting was used only to identify $E_n(A, I)$ with the whole reduction kernel. Source 07 proposes it as the first target for a formalization: it isolates all size and support input in (9.2).

10 A separating proper-class congruence tower

A group with no small images might be mistaken for a simple group; the kernels give an explicit counterexample to that inference (06). Fix $A = \mathbf{Oz}$ or $\mathbf{Oz}[i]$ and $n \geq 3$. For each positive surreal a put

$$P_n^{(a)} = \{g \in P_n : g - 1_n \in M_n(\omega^a A)\}, \quad (10.1)$$

the kernel on P_n of reduction modulo the principal ideal $\omega^a A$, used only as a congruence relation.

Theorem 10.1 (A non-simple invisible group with a large residual tower [06]). *For every $a > 0$, $P_n^{(a)}$ is a nontrivial proper normal subgroup of P_n , and $P_n^{(b)} \subsetneq P_n^{(a)}$ for $0 < a < b$. Moreover*

$$\bigcap_{a \in \mathbf{No}^{>0}} P_n^{(a)} = 1, \quad \bigcap_{a \in S} P_n^{(a)} \neq 1 \quad \text{for every set } S \subset \mathbf{No}^{>0}.$$

Each quotient $P_n/P_n^{(a)}$ is nontrivial and perfect, has no nontrivial homomorphism to a set-sized group, and cannot itself be set-sized.

Proof. Matrices congruent to 1 modulo an ideal form a normal subgroup of $\mathrm{GL}_n(A)$; intersecting with P_n gives normality. $e_{12}(\omega^a) \in P_n^{(a)}$, while $e_{12}(\omega^{a/2}) \notin P_n^{(a)}$ since $\omega^{-a/2} \notin A$. For $b > a$, $\omega^{b-a} \in A$ gives the inclusion, and $e_{12}(\omega^a)$ witnesses strictness. If $g \neq 1$, pick a nonzero entry f of $g - 1$; for $a > \deg_\omega f$, $\omega^{-a}f$ has negative leading exponent, so $g \notin P_n^{(a)}$. For a set S choose b above S (lemma 2.3); then $e_{12}(\omega^b)$ lies in every $P_n^{(a)}$, $a \in S$, including $S = \emptyset$. A homomorphism from $P_n/P_n^{(a)}$ to a set-sized group composes to one from P_n , hence is trivial; a nontrivial set-sized quotient would contradict this through its identity map. Perfectness passes to quotients. $\square$

The tower separates individual matrices only when all surreal scales are allowed. This is a set/class distinction, not a numerical size assigned to P_n (06).

11 The finite side and the rotation groups

The same mechanism applies on the side of infinitesimals. Let $\mathcal{O}_{\mathrm{fin}}(k)$ be the ring of finite elements of F ($k = \mathbb{R}$ for $\mathbf{No}$, $k = \mathbb{C}$ for $\mathbf{No}[i]$), $\mathfrak{m}_k$ its infinitesimals, and $\mathrm{st} : \mathcal{O}_{\mathrm{fin}}(k) \rightarrow k$ the standard part, a surjective ring homomorphism with kernel $\mathfrak{m}_k$ and section $k \subset \mathcal{O}_{\mathrm{fin}}(k)$. No source states the following group analogue of the finite-side ring theorem `osq:thm:finite`; it was observed in the placement dossier and is added here with a complete proof.

Proposition 11.1 (Universal set-sized quotient of SL_n of the finite surreals [merge]). *Let $n \geq 3$ and $Q_n = \ker(\mathrm{st} : \mathrm{SL}_n(\mathcal{O}_{\mathrm{fin}}(k)) \rightarrow \mathrm{SL}_n(k))$. Every class homomorphism from Q_n to a set-sized group is trivial, and every class homomorphism $\mathrm{SL}_n(\mathcal{O}_{\mathrm{fin}}(k)) \rightarrow H$ to a set-sized group factors uniquely through st . Thus $\mathrm{SL}_n(k)$ is the universal set-sized quotient of $\mathrm{SL}_n(\mathcal{O}_{\mathrm{fin}}(k))$.*

Proof. $\mathcal{O}_{\mathrm{fin}}(k)$ is local with maximal ideal $\mathfrak{m}_k$. If x is finite and $\mathrm{st} x \neq 0$, then $|x| \geq |\mathrm{st} x|/2$, so $x^{-1} \in F$ is finite; elements of $\mathfrak{m}_k$ are not units, and $\mathfrak{m}_k$ is an ideal.

$\mathrm{SL}_n(\mathcal{O}_{\mathrm{fin}}(k)) = E_n(\mathcal{O}_{\mathrm{fin}}(k))$. Let $g \in \mathrm{SL}_n(\mathcal{O}_{\mathrm{fin}}(k))$. Since $\det g \notin \mathfrak{m}_k$, some entry of the first column is a unit; elementary row operations move a unit to position $(1, 1)$ (add that row to the first if needed: if $g_{11} \in \mathfrak{m}_k$ and g_{r1} is a unit, the new $(1, 1)$ entry $g_{11} + g_{r1}$ is a unit), then clear the first column and the first row. By induction g becomes $\mathrm{diag}(u_1, \dots, u_n)$ with

$\prod u_i = 1$, and $\text{diag}(u, u^{-1}) = e_{12}(u)e_{21}(-u^{-1})e_{12}(u) (e_{12}(1)e_{21}(-1)e_{12}(1))^{-1}$ (Whitehead's identity, valid over any commutative ring) makes the diagonal matrix elementary.

Q_n is the normal closure of the $e_{ij}(\mathfrak{m}_k)$. This is the argument of [lemma 7.1](#) with the split surjection st in place of ct and $E_n(\mathcal{O}_{\text{fin}}(k)) = \text{SL}_n(\mathcal{O}_{\text{fin}}(k))$, $E_n(k) = \text{SL}_n(k)$.

Killing. Let $\psi : Q_n \rightarrow H$, and consider a normal generator $g e_{ih}(\varepsilon)g^{-1}$ with $g \in \text{SL}_n(\mathcal{O}_{\text{fin}}(k))$ and $0 \neq \varepsilon \in \mathfrak{m}_k$; let $a = -\deg_\omega \varepsilon > 0$. For $0 < \gamma < \delta < a$, $\varepsilon/(\omega^{-\gamma} - \omega^{-\delta}) \in \mathfrak{m}_k$ (`osq:lem:finitediff`: the denominator has leading exponent $-\gamma$, so the quotient has leading exponent $\gamma - a < 0$). Choose a third index j and a set $T \subset (0, a)$ that does not inject into H (for an ordinal Λ that does not inject into H , the exponents $a/\omega^{\xi+1}$, $\xi < \Lambda$, are distinct and lie in $(0, a)$). The elements $g e_{ij}(\omega^{-\gamma})g^{-1}$, $\gamma \in T$, lie in Q_n , so two of them, for $\gamma < \delta$, have equal images, and $\psi(g e_{ij}(d)g^{-1}) = 1$ for $d = \omega^{-\gamma} - \omega^{-\delta} \in \mathfrak{m}_k$. With $b = \varepsilon/d \in \mathfrak{m}_k$, $[g e_{ij}(d)g^{-1}, g e_{jh}(b)g^{-1}] = g e_{ih}(\varepsilon)g^{-1}$ is an identity in Q_n , so the generator is killed.

Factorization. A homomorphism $\varphi : \text{SL}_n(\mathcal{O}_{\text{fin}}(k)) \rightarrow H$ kills Q_n , and $g \text{st}(g)^{-1} \in Q_n$ gives $\varphi = \varphi|_{\text{SL}_n(k)} \circ \text{st}$; uniqueness follows from surjectivity of st . $\square$

Remark 11.2 (Comparison with the surreal rotation groups). The euclidean-three-space report proves that every homomorphism from $\text{SO}(3, \mathbf{No})$ to a set-sized group factors through standard part (`e3:cut:thm:smallquotient`), and the same for $\text{SO}(n, \mathbf{No})$, $n \geq 3$, and $\text{SU}(n, \mathbf{No}[i])$, $n \geq 2$ (`e3:cut:thm:allranks`), while it fails for the circle (`e3:cut:prop:circlecounter`). Those theorems and [theorem 9.1](#) are parallel instances of one phenomenon, a universal set-sized quotient obtained by killing a kernel through a supply of scales, but neither implies the other. The rings differ (all of $\mathbf{No}$, whose compact groups have finite entries, against $\mathbf{Oz}$); the kernels differ (infinitesimal matrices under st against purely infinite matrices under ct); and the mechanisms differ (normal closure at valuation cuts in a compact group followed by Givens elimination, against Steinberg commutators with collisions). $\text{SO}(3, \mathbf{Oz})$ is finite ([section 5](#)), so the rotation group has trivial omnific kernel, and a homomorphism on $E_n(\mathbf{Oz})$ says nothing about $\text{SO}(n, \mathbf{No})$. [Proposition 11.1](#) sits between the two: it concerns the finite side like the rotation theorems but uses the elementary mechanism. It does not imply the rotation theorems either, since a homomorphism defined on the subgroup $\text{SO}(n, \mathbf{No}) \subseteq \text{SL}_n(\mathcal{O}_{\text{fin}}(\mathbb{R}))$ need not extend. The abelian boundaries also correspond: the circle characters of `e3:cut:prop:circlecounter`, the characters of $\text{UT}_n(\Pi_k)$ in [section 6](#), and the rank-two countable detection of [section 15](#).

12 Sharp boundaries of the argument

12.1 A fixed Hahn workspace is not the full class

Let Γ be a nonzero set-sized divisible ordered abelian group, write large monomials X^γ , and put $\Pi_k(\Gamma) = \{f : \text{supp } f \subseteq \Gamma^{>0}\}$ and $\mathcal{A}_{D,k}(\Gamma) = D \oplus \Pi_k(\Gamma)$ inside the Hahn field with set-sized reverse-well-ordered supports; these rings are sets.

The geometric dichotomy of [Part I](#) persists (06): the real Hahn field over a divisible group is real closed and the complex one is algebraically closed [[16](#), Corollary 4]; unit and bounded-coordinate rigidity are unchanged; and a positive exponent gives a nonconstant parameter. So the proofs of [theorems 4.4](#) and [4.5](#) apply to the Hahn-cone rings in place of $\mathcal{B}_{\mathbb{R}}, \mathcal{B}_{\mathbb{C}}$, with integral constant rings as before.

Invisibility cannot persist: the purely infinite elementary group of a nontrivial set-sized workspace is itself a set-sized nontrivial group, and its identity homomorphism is faithful. The full-class proofs need arbitrarily many scales below a support gap (06), and the compression

of [lemma 8.4](#) produces a field that need not lie in any preassigned set-sized subfield or birthday-bounded fragment (07).

Proposition 12.1 (Local cardinal form [06]). *Fix $0 \neq c \in \Pi_k(\Gamma)$ and $0 < \delta < \text{supp } c$. Let $T \subseteq \Gamma^{>0}$ satisfy $m\alpha < \delta$ for all $\alpha \in T$ and ordinary $m \geq 1$. For every homomorphism from the group generated by the purely infinite elementary matrices of rank $n \geq 3$ over this workspace to a group H with $|T| > |H|$, every root element $e_{ij}(c)$ maps to the identity.*

Proof. The second proof of [theorem 9.1](#) applies verbatim: for the two colliding exponents $\alpha > \beta$, $\alpha + m(\alpha - \beta) \leq (m + 1)\alpha < \delta$, so $c/(X^\alpha - X^\beta)$ lies in the workspace with strictly positive support. No normality or perfectness in the workspace is needed. $\square$

This is a conditional lower bound on target size, not an assertion that every exponent group supplies such a T : for an Archimedean group no positive α has all $m\alpha < \delta$ (06). [Proposition 9.7](#) can be applied at a set-sized stage for targets smaller than an available parameter family; an optimal threshold there requires further information about support bounds and the exponent group (07; [question 34.4](#)). For the rings of supports of size $< \kappa$ over $\bigoplus_{\alpha < \kappa} \mathbb{Q}$ and for the sibling report's other set-sized models this information is available, and the optimal threshold is determined in [proposition 21.12](#) and [corollary 21.13](#) [merge].

12.2 Finite supports change the answer

Let Π_k^{fin} consist of the purely infinite normal forms with finite support and $A^{\text{fin}} = D + \Pi_k^{\text{fin}}$; the exponent class is still all of **No**.

Proposition 12.2 (A visible finite-support kernel [06, 11]). *The coefficient sum $\sum_s a_s \omega^s \mapsto \sum_s a_s$ is a ring homomorphism $A^{\text{fin}} \rightarrow k$, and for $n \geq 3$ it maps the group generated by the purely infinite elementary matrices onto the set-sized group $\text{SL}_n(k)$.*

Proof. Sums and products involve finite supports, so expanding a product and summing coefficients gives multiplicativity; it is evaluation $\omega \mapsto 1$ on the group ring $k[\mathbf{No}]$. For $r \in k$ and $a > 0$ the purely infinite $r\omega^a$ maps to r , so every $e_{ij}(r)$ is in the image, and these generate $\text{SL}_n(k)$. $\square$

The finite-support ideal still has the single-product property, so its elementary kernel is perfect and normal; nevertheless it has a large ordinary image. Infinite supports supply the exact divisor (8.1), which finite truncations cannot replace (06). Source 11 proves the nontriviality part (its Proposition 11.3: the same evaluation maps the finite-support kernel nontrivially to $E_n(k)$, since $e_{12}(\omega) \mapsto e_{12}(1)$) and notes that the evaluation does not extend to the full ring, by the ring theorem. This is the group form of the finite-support augmentation in the sibling report's support threshold (`osq:thm:support`).

12.3 Rank two and the full special linear group

All three proofs need a third index to convert a killed additive root parameter into an arbitrary multiplicative one (the third needs it to make the common root kernel an ideal), so none says anything about rank two; and the theorem concerns $E_n(A)$, not $\text{SL}_n(A)$, and uses no stable-range theorem for an omnific ring (06, 07, 11). Source 06 left both issues open, and so does source 11 (its Questions 11.1 and 11.2), whose pin already contained 07's answer in this report; 11 states $n \geq 3$ as a proved range, not as a claim that $n = 2$ is a sharp negative boundary. Rank two is settled by source 07 in the next sections: invisibility fails

there (theorem 15.3). Whether the universal set-sized quotient of $\mathrm{SL}_n(\mathbf{Oz})$, $n \geq 3$, is $\mathrm{SL}_n(\mathbb{Z})$ remains open (question 34.1); for $n = 2$, $\mathrm{SL}_2(\mathbf{Oz}) \neq E_2(\mathbf{Oz})$ (theorem 18.2).

13 Two rows: an amalgam detected by finite and infinite points

From here to the end of section 18 the source is 07, and (D, k) is $(\mathbb{Z}, \mathbb{R})$ or $(\mathbb{Z}[i], \mathbb{C})$, $A = \mathcal{A}_{D,k}$, $K_D = \mathrm{Frac} D$. The proof of theorem 9.1 cannot be repeated with two rows, because it uses a third index. The two-row group is identified explicitly instead. The argument is a specialized elementary form of Cohn's classical two-row normal-form method [4]; it also gives the Gaussian presentation needed later.

13.1 The upper triangular subgroup

Write $G_D = \mathrm{SL}_2(D)$,

$$C_D = \left\{ \begin{pmatrix} u & d \\ 0 & u^{-1} \end{pmatrix} : u \in D^\times, d \in D \right\}, \quad B_D = \left\{ \begin{pmatrix} u & r \\ 0 & u^{-1} \end{pmatrix} : u \in D^\times, r \in A \right\},$$

$U(x) = e_{12}(x)$ and $U(\Pi_k) = \{U(x) : x \in \Pi_k\}$. By lemma 2.1, B_D is the whole upper triangular subgroup of $\mathrm{SL}_2(A)$; each member is uniquely $U(x)c$ with $x \in \Pi_k$ and $c = \mathrm{ct}$ of it in C_D . For c with diagonal u, u^{-1} ,

$$cU(x)c^{-1} = U(u^2x), \quad \text{so} \quad B_D = U(\Pi_k) \rtimes C_D. \quad (13.1)$$

The action is trivial for $D = \mathbb{Z}$ and by the signs ± 1 for $D = \mathbb{Z}[i]$. Both G_D and B_D lie in $E_2(A)$, they generate it (every upper elementary matrix is in B_D , and $SU(x)S^{-1} = e_{21}(-x)$ with $S = \begin{pmatrix} 0 & -1 \\ 1 & 0 \end{pmatrix} \in G_D$), and $G_D \cap B_D = C_D$.

13.2 A projective separation argument

In $\mathbb{P}^1(F) = F \cup \{\infty\}$ let $P_{\mathrm{fin}} = \{z \in F : |z| < m \text{ for some ordinary } m\}$ be the finite points and P_∞ the rest (infinite elements and ∞), with the fractional-linear action of determinant-one matrices.

Lemma 13.1 (Alternation of finite and infinite points [07]).

1. Every $b \in B_D \setminus C_D$ maps P_{fin} into P_∞ and fixes ∞ .
2. Every $g = \begin{pmatrix} a & b \\ c & d \end{pmatrix} \in G_D \setminus C_D$ maps P_∞ into P_{fin} , has $c \neq 0$, and $\mathrm{st}(gz) = a/c \in K_D$ for $z \in P_\infty$ (exactly a/c at $z = \infty$).

Proof. (i) Write $b = U(x)c'$ with $0 \neq x \in \Pi_k$, $c' \in C_D$: c' preserves finite points and fixes ∞ , and adding the nonzero purely infinite x makes a finite point infinite. (ii) $c \neq 0$ since g is not upper triangular; its pole $-d/c$ is constant, hence not in P_∞ . For infinite z ,

$$\frac{az + b}{cz + d} - \frac{a}{c} = \frac{bc - ad}{c(cz + d)} = -\frac{1}{c(cz + d)}, \quad (13.2)$$

which is infinitesimal. □

Theorem 13.2 (The two-row amalgam [07]). *The inclusions induce an isomorphism $E_2(A) \cong G_D *_{C_D} B_D$.*

Proof. Surjectivity is generation. For injectivity use reduced words of the amalgam: a nontrivial reduced word is conjugate to a nonidentity element of one factor or to a cyclically reduced word $g_1 b_1 g_2 b_2 \cdots g_m b_m$ with $g_j \in G_D \setminus C_D$, $b_j \in B_D \setminus C_D$. These finite-word rules are valid for class groups: words are finite, coset representatives are sets, and each shortening step reduces the finite length. Apply the matrix of the cyclic word to ∞ : b_m fixes it, g_m sends it into P_{fin} , b_{m-1} back into P_∞ , and so on (lemma 13.1); the final image is finite, so the matrix is not the identity. Nonidentity elements of the factors are nonidentity matrices, and conjugation preserves this. $\square$

No convergence, continued infinite fraction or action on a set-sized tree is involved, and the Gaussian case needs no order on the coefficients (07). The proof used only that the constant subgroup is $\text{SL}_2(D)$, that nonconstant upper triangular translations are infinite, that all units are constant, and that a non-upper-triangular constant matrix sends infinite points to finite ones. So it applies to the subgroup of $\text{SL}_2(K_D(t))$ generated by G_D and $B'_D = \{U(tq)c : q \in \mathbb{Q}, c \in C_D\}$, using degree at infinity or the injective embedding $K_D(t) \rightarrow F, t \mapsto \omega$:

$$\langle G_D, B'_D \rangle \cong G_D *_{C_D} B'_D. \quad (13.3)$$

14 The two-row kernel as a free product

G_D acts transitively on $\mathbb{P}^1(K_D)$ with stabilizer C_D at ∞ (a point is represented by a coprime pair $(a, c) \in D^2$, which Bézout completes to a first column in $\text{SL}_2(D)$), so $G_D/C_D \cong \mathbb{P}^1(K_D)$, a countable set. Choose a transversal $\mathcal{T} \subset G_D$ of these left cosets containing 1.

Theorem 14.1 (Kernel decomposition [07]). *There is an isomorphism $P_2 \cong *_{t \in \mathcal{T}} \Pi_k^{(t)}$, each $\Pi_k^{(t)}$ a copy of $(\Pi_k, +)$ embedded by $x \mapsto tU(x)t^{-1}$, and*

$$E_2(A) \cong \left(*_{t \in \mathcal{T}} \Pi_k^{(t)} \right) \rtimes G_D,$$

where, if $gt = t'c$ with $t' \in \mathcal{T}$, $c \in C_D$ of first diagonal entry u , the action sends x in factor t to u^2x in factor t' .

Proof. Let $\mathcal{F}$ be the free product with inclusions ι_t , and let G_D act by $g \cdot \iota_t(x) = \iota_{t'}(u^2x)$ when $gt = t'c$. Each $u^2 = \pm 1$ gives an automorphism of the factor; coset factorization is unique, and if $g_2t = t'c_2$, $g_1t' = t''c_1$ then $g_1g_2t = t''c_1c_2$ with multiplier the product, so this is an action. In $\mathcal{F} \rtimes G_D$, $\iota_1(\Pi_k)$ and C_D form $\Pi_k \rtimes C_D$ with the action (13.1), which gives a homomorphism from B_D agreeing with G_D on C_D , hence $G_D *_{C_D} B_D \rightarrow \mathcal{F} \rtimes G_D$. Conversely send G_D to its constant matrices and $\iota_t(x)$ to $tU(x)t^{-1}$; the defining relation of the action is (13.1). The two maps are inverse on generators; under the isomorphism, constant reduction is the projection to G_D , whose kernel is $\mathcal{F}$. Combine with theorem 13.2. $\square$

For $D = \mathbb{Z}$ every $u^2 = 1$ and the action permutes factors indexed by $\mathbb{P}^1(\mathbb{Q})$. In the Gaussian case a change of representative can act on a factor by a sign; omitting it would misdescribe the semidirect product (07).

Corollary 14.2 (Internal abelianization and torsion [07]). $P_2^{\text{ab}} \cong \bigoplus_{q \in \mathbb{P}^1(K_D)} (\Pi_k, +)$. *The group P_2 is torsion-free and not perfect.*

Proof. A free product abelianizes to the direct sum of the abelianized factors, here abelian and nonzero. A cyclically reduced word with one syllable lies in a torsion-free factor; with several syllables its powers concatenate without cancellation. Conjugation preserves finite order. $\square$

Torsion-freeness also follows, by a different route, from source 11's trace argument, which applies to the constant-term kernel of GL_n for every n (theorem 19.1).

This internal abelianization must not be confused with the image of P_2 in $E_2(A)^{\mathrm{ab}}$, where the constant group can identify or kill classes (section 17).

15 Countable representations separating finite sets

The needed additive functionals can be built without a basis of the proper-class vector space Π_k (07).

Lemma 15.1 (A set-based additive functional [07]). *For finitely many nonzero $x_1, \dots, x_m \in \Pi_k$ there is a $\mathbb{Q}$ -linear class map $\ell : \Pi_k \rightarrow \mathbb{Q}$ with $\ell(x_j) \neq 0$ for all j , factoring through finitely many coefficient extractions into the set k^r .*

Proof. Select for each j one exponent where x_j has a nonzero coefficient; list the selected exponents $s_1, \dots, s_r$ and put $\mathbf{p}(x) = ([\omega^{s_1}]x, \dots, [\omega^{s_r}]x) \in k^r$. Each $\mathbf{p}(x_j) \neq 0$. Let $W \subseteq k^r$ be the finite-dimensional $\mathbb{Q}$ -span of these vectors with basis $w_1, \dots, w_d$, and $f_M(w_h) = M^{h-1}$; $f_M(\mathbf{p}(x_j))$ is a nonzero polynomial in M , so some integer M avoids the finitely many roots. Extend f_M $\mathbb{Q}$ -linearly to k^r by extending a basis of the set W , and set $\ell = f_M \circ \mathbf{p}$. $\square$

This is an existence construction with set choice, not a computability claim about arbitrary real or complex coefficients (07).

Proposition 15.2 (Additive specialization [07]). *For every additive $\ell : \Pi_k \rightarrow \mathbb{Q}$ there is a unique homomorphism $\rho_\ell : E_2(A) \rightarrow \mathrm{SL}_2(K_D(t))$ with $\rho_\ell(g) = g$ for $g \in G_D$ and $\rho_\ell(U(x)) = U(t\ell(x))$ for $x \in \Pi_k$.*

Proof. On $B_D = U(\Pi_k) \rtimes C_D$ send $U(x)c \mapsto U(t\ell(x))c$. Root addition is preserved, and compatibility with the C_D -action is $\ell(u^2x) = u^2\ell(x)$, true since $u^2 = \pm 1$. This agrees with the identity on C_D ; theorem 13.2 extends it, and generation gives uniqueness. $\square$

In the Gaussian case ℓ need not be complex-linear or satisfy $\ell(ix) = i\ell(x)$: diagonal conjugation only involves $i^2 = -1$. This is why Gaussian countable representations survive although the Gaussian abelian characters behave differently (07).

Theorem 15.3 (One countable target separates finite patterns [07]). *For every finite $\mathcal{S} \subset E_2(A)$ some ρ_ℓ is injective on $\mathcal{S}$. In particular every nonidentity element of $E_2(A)$ is detected by a homomorphism into the fixed countable group $\mathrm{SL}_2(K_D(t))$ that is the identity on $\mathrm{SL}_2(D)$.*

Proof. Keep nontrivial each $a^{-1}b$, $a \neq b$ in $\mathcal{S}$. Write them in reduced amalgam form; the $B_D \setminus C_D$ syllables are $U(x)c$ with $x \neq 0$, finitely many. Choose ℓ nonzero on all these x (lemma 15.1). Under ρ_ℓ , $G_D \setminus C_D$ syllables are unchanged and $B_D \setminus C_D$ syllables go to $B'_D \setminus C_D$, so reduced words stay reduced in (13.3) and are nontrivial; one-factor words are preserved as well. $K_D(t)$ is countable. $\square$

Example 15.4 (A commutator needing a nonabelian detector [07]). Let ℓ extract the coefficient of ω^1 (followed by a $\mathbb{Q}$ -linear map taking 1 to 1). Then

$$\rho_\ell([U(\omega), e_{21}(\omega)]) = [U(t), e_{21}(t)] = \begin{pmatrix} 1 + t^2 + t^4 & -t^3 \\ t^3 & 1 - t^2 \end{pmatrix} \neq I_2,$$

a countable nonabelian image of a commutator in P_2 , killed of course by every abelian image; valid over both coefficient rings.

Corollary 15.5 (No universal set-sized quotient in rank two [07]). *Neither $E_2(A)$ nor P_2 has a universal set-sized group quotient.*

Proof. If $q : E_2(A) \rightarrow H$ were one, every ρ_ℓ would factor through q , so $\ker q = 1$ by [theorem 15.3](#); but $U(\Pi_k)$ contains more than $|H|$ distinct monomials. Restricting the ρ_ℓ to P_2 gives the same for P_2 . $\square$

There is no contradiction between residual countability and proper-class size: the separating maps are not a set-indexed family whose product is a faithful set-sized representation, and no single faithful countable representation is asserted (07).

16 Finite quotients and the detection trichotomy

Lemma 16.1 (Divisibility kills finite root images [07]). *Every homomorphism from $(\Pi_k, +)$ to a finite group is trivial.*

Proof. With $m = |H|$, $x/m \in \Pi_k$ and $f(x) = f(x/m)^m = 1$. $\square$

Proposition 16.2 (All finite images are arithmetic [07]). *For every $n \geq 2$ every homomorphism from $E_n(A)$ to a finite group factors uniquely through ct , and every homomorphism from P_2 to a finite group is trivial.*

Proof. A finite-target map kills each $e_{ij}(\Pi_k)$ ([lemma 16.1](#)) and hence its normal closure P_n ([lemma 7.1](#)). For P_2 use the free product ([theorem 14.1](#)). $\square$

Lemma 16.3 (Finite detection in the constant group [07]). *$\text{SL}_n(D)$ is residually finite for $n \geq 2$.*

Proof. If $g \neq 1$, a nonzero entry a of $g - 1$ is not in mD for a suitable ordinary $m \geq 2$ (for Gaussian a , take m larger than both integer coordinates), and reduction to $\text{SL}_n(D/mD)$ does not kill g . $\square$

Theorem 16.4 (Finite and set residuals [07]). *For every $n \geq 2$, $\text{Res}_{\text{fin}} E_n(A) = P_n$. The set residual (the class of elements killed by every homomorphism to a set-sized group) satisfies $\text{Res}_{\text{set}} E_2(A) = 1$ and $\text{Res}_{\text{set}} E_n(A) = P_n$ for $n \geq 3$. For the kernels, $\text{Res}_{\text{fin}} P_2 = P_2$, $\text{Res}_{\text{set}} P_2 = 1$ and $\text{Res}_{\text{set}} P_n = P_n$ for $n \geq 3$.*

Proof. Finite maps kill P_n ([proposition 16.2](#)), and elements outside P_n survive a finite map ([lemma 16.3](#)). The set residuals follow from [theorems 9.1](#) and [15.3](#), and those of the kernels also from the last clause of [proposition 16.2](#). $\square$

The finite quotient theory of $E_n(A)$ is exactly that of $\text{SL}_n(D)$; its profinite completion, defined by this set-indexed system, is that of $\text{SL}_n(D)$. No congruence subgroup property is involved: arbitrary finite quotients are identified, and congruence reduction was used only for [lemma 16.3](#) (07). For $n \geq 3$ source 11 proves the finite and set residual statements again (its Theorem 9.1, by the same reduction modulo m), and for **Oz** it then identifies the completion with $\text{SL}_n(\widehat{\mathbb{Z}})$ by importing the congruence subgroup theorem ([corollary 19.7](#)); that corollary is the only place in the report where a congruence subgroup property enters.

Theorem 16.5 (The detection trichotomy [07]). *Let $n \geq 2$ and $1 \neq g \in E_n(A)$. Exactly the applicable alternative holds:*

1. if $\text{ct}(g) \neq 1$, some finite group homomorphism detects g ;
2. if $n = 2$ and $g \in P_2$, no finite group homomorphism detects g , a countable one does, and the least cardinality of a detecting target is $\aleph_0$, also for homomorphisms defined only on P_2 ;
3. if $n \geq 3$ and $g \in P_n$, no homomorphism to any set-sized group detects g , also for homomorphisms defined only on P_n .

Proof. [Lemma 16.3](#), [proposition 16.2](#), and [theorems 9.1](#) and [15.3](#). The last alternative is the nonexistence of any set-sized detecting target, not an “infinite cardinal larger than all cardinals”. $\square$

Element	Smallest detecting target	Reason
$\text{ct}(g) \neq 1$	Some finite group	Reduction modulo m
$1 \neq g \in P_2$	Countably infinite	Additive coefficient maps
$1 \neq g \in P_n, n \geq 3$	No set-sized group	Collision and a third index

17 Abelian characters: a real–Gaussian contrast

Proposition 17.1 (The real abelianization, a classical specialization [07]). $E_2(\mathbf{Oz})^{\text{ab}} \cong \mathbb{Z}/12 \oplus (\Pi, +)$. For every $s \in \mathbf{No}^{>0}$ there is a surjective homomorphism $\chi_s : E_2(\mathbf{Oz}) \rightarrow (\mathbb{R}, +)$ with $\chi_s(\text{SL}_2(\mathbb{Z})) = 0$ and $\chi_s(U(x)) = [\omega^s]x$ for $x \in \Pi$.

Proof. For a split group $\mathcal{F} \rtimes G$, the abelianization is G^{ab} plus the G -coinvariants of $\mathcal{F}^{\text{ab}}$. By [theorem 14.1](#) and [corollary 14.2](#), $\mathcal{F}^{\text{ab}}$ is a direct sum of copies of Π on which $\text{SL}_2(\mathbb{Z})$ acts by permuting the indices transitively without twist; the coinvariants are one copy of Π (sum of the finitely many nonzero coordinates). And $\text{SL}_2(\mathbb{Z})^{\text{ab}} = \mathbb{Z}/12$: from the presentation with generators $S, U(1)$ and relations $S^4 = 1, S^2 = (SU(1))^3$, an abelian quotient has $[S] = -3[U(1)]$ and $12[U(1)] = 0$ and nothing more. Project the second summand to its ω^s -coefficient for χ_s ; $\chi_s(U(r\omega^s)) = r$. $\square$

The formula is not a novelty claim (07). It is the omnific case of Cohn’s abelianization theorem [4] in the formulation of Mirzaii and Torres Pérez [15]: their Corollary 3.3 (credited to Cohn) gives $E_2(R)^{\text{ab}} \cong R/M$ for rings universal for GE_2 , with M the additive subgroup generated by $x(a^2 - 1)$ and $3(b + 1)(c + 1)$ ($x \in R, a, b, c \in R^\times$); their Example 3.4(i) gives $R/M = R/12\mathbb{Z}$ when $R^\times = \{\pm 1\}$; and their Example 1.1(vi) records Cohn’s theorem that discretely ordered rings are universal for GE_2 . These numbers were checked against the arXiv version 3 of [15] for this report. Those statements concern rings that are sets; for the class ring $\mathbf{Oz}$, discretely ordered with units ± 1 , the proof above is self-contained. The quotient is by the additive subgroup $12\mathbb{Z}$, not by the ideal $12\mathbf{Oz} = 12\mathbb{Z} + \Pi$, which would wrongly discard the purely infinite summand (07).

Proposition 17.2 (Gaussian ambient abelianization [07]). Constant reduction induces $E_2(\mathbf{Oz}[i])^{\text{ab}} \cong \text{SL}_2(\mathbb{Z}[i])^{\text{ab}}$. Every homomorphism from $E_2(\mathbf{Oz}[i])$ to an abelian group kills P_2 .

Proof. With $d(i) = \text{diag}(i, -i)$, $d(i)U(y)d(i)^{-1} = U(-y)$, and since $x/2 \in \Pi_{\mathbb{C}}$,

$$[d(i), U(-x/2)] = U(x) \quad (x \in \Pi_{\mathbb{C}}). \quad (17.1)$$

Conjugating by S handles lower roots, and normal closure gives $P_2 \subseteq [E_2, E_2]$. The constant inclusion is an inverse on abelianizations. $\square$

No numerical description of $\mathrm{SL}_2(\mathbb{Z}[i])^{\mathrm{ab}}$ is needed or claimed. The comparison is new in 07: purely infinite Gaussian information is absent from every abelian image but visible in countable nonabelian images ([theorem 15.3](#)). This does not make P_2 perfect: $d(i) \notin P_2$, and P_2^{ab} is still the large sum of [corollary 14.2](#); for $n \geq 3$ the commutators of [proposition 7.2](#) have both entries in P_n .

18 Cusp residues, non-elementary unipotents, and stabilization

Proposition 18.1 (The finite cusp-residue obstruction [07]). *If $h \in E_2(A)$ and $h \cdot \infty$ is finite, then $\mathrm{st}(h \cdot \infty) \in K_D$.*

Proof. By [theorem 13.2](#) write $h = a_0 U(x_1) a_1 \cdots a_{m-1} U(x_m) a_m$ with nonzero $x_j \in \Pi_k$, interior $a_j \in G_D \setminus C_D$ and endpoint $a_0, a_m \in G_D$ (absorb triangular constant parts into neighbours, move interior triangular constants through upper roots by (13.1), merge adjacent roots, and shorten when a parameter vanishes; the process is finite). If $m = 0$ the claim is clear. If $m \geq 1$, a_m sends ∞ to ∞ or to a point of K_D , $U(x_m)$ sends that point into P_∞ , each interior a_j sends P_∞ into P_{fin} with standard part in K_D , and the next root sends it back into P_∞ . If $a_0 \in C_D$ the final point is in P_∞ , contrary to finiteness; otherwise [lemma 13.1\(ii\)](#) applies. $\square$

The obstruction is stronger than looking at constant reduction: a matrix can reduce to the identity and still fail it, because a ratio of its large entries has a nonarithmetic finite standard part. The polynomial pattern $\begin{pmatrix} 1+xy & x^2 \\ -y^2 & 1-xy \end{pmatrix}$ is Cohn's example of a non-elementary matrix over $k[x, y]$ [4, Section 7, p. 26]; neither it nor the block identity below is claimed as new (07).

Theorem 18.2 (A non-elementary omnific unipotent [07]). *Let $\alpha \in k \setminus K_D$, $a > 0$, and*

$$Q_\alpha = \begin{pmatrix} \alpha & 1 \\ -\alpha^2 & -\alpha \end{pmatrix}, \quad M_\alpha = I_2 + \omega^{2a} Q_\alpha.$$

Then $Q_\alpha^2 = 0$, $\det M_\alpha = 1$, $\mathrm{ct}(M_\alpha) = I_2$, M_α has infinite order, and $M_\alpha \in \mathrm{SL}_2(A) \setminus E_2(A)$.

Proof. The square and determinant are polynomial identities; the nonconstant entries have exponent $2a > 0$ and coefficients in k . With $T = \omega^{2a}$, $M_\alpha \cdot \infty = (1 + \alpha T)/(-\alpha^2 T) = -1/\alpha - 1/(\alpha^2 T)$ is finite with standard part $-1/\alpha \notin K_D$, so [proposition 18.1](#) excludes $E_2(A)$. Finally $M_\alpha^m = I_2 + m\omega^{2a} Q_\alpha \neq I_2$. $\square$

For $a = 1/2$ and $\alpha = \sqrt{2}$,

$$M = \begin{pmatrix} 1 + \sqrt{2}\omega & \omega \\ -2\omega & 1 - \sqrt{2}\omega \end{pmatrix} \in \mathrm{SL}_2(\mathbf{Oz}) \setminus E_2(\mathbf{Oz}), \quad (18.1)$$

and the same matrix is non-elementary over $\mathbf{Oz}[i]$ since $\sqrt{2} \notin \mathbb{Q}(i)$. This needs no factorization theory of omnific integers. It is consistent with the omnific Diophantine report's non-Euclidean phenomena (pairs without a gcd, `odg:thm:nogcd`), and with its rationality theorem for primitive constant directions (`odg:thm:realray`), which concerns constant real projective points; the first column of M_α is not one.

Lemma 18.3 (A finite block identity [07]). *Over any commutative ring, for a column v and a row w of length two with $wv = 0$,*

$$\left[\begin{pmatrix} I_2 & v \\ 0 & 1 \end{pmatrix}, \begin{pmatrix} I_2 & 0 \\ w & 1 \end{pmatrix} \right] = \begin{pmatrix} I_2 + vw & 0 \\ 0 & 1 \end{pmatrix}.$$

Proof. The inverses replace v, w by their negatives; multiply the four blocks, and the terms containing wv vanish. $\square$

Theorem 18.4 (One-step stabilization [07]). *With $z = \omega^a$, $v = z(1, -\alpha)^\top$, $w = z(\alpha, 1)$, $X = e_{13}(z)e_{23}(-\alpha z)$ and $Y = e_{31}(\alpha z)e_{32}(z)$, one has $X, Y \in P_3$ and $[X, Y] = \text{diag}(M_\alpha, 1)$. Hence this nonidentity matrix is killed by every homomorphism from P_3 or $E_3(A)$ to a set-sized group.*

Proof. All parameters lie in Π_k . X and Y are the block matrices of lemma 18.3, $wv = z^2(\alpha - \alpha) = 0$ and $vw = z^2 Q_\alpha$. Apply theorem 9.1. $\square$

The commutator is an eight-factor elementary word. The identity itself does not depend on surreals; the surreal content is the coefficient ring, the residue obstruction before stabilization and the invisibility after it (07).

Corollary 18.5 (A strict failure of extension [07]). *Let $j : E_2(A) \rightarrow E_3(A)$, $g \mapsto \text{diag}(g, 1)$. If $\rho : E_2(A) \rightarrow H$, H set-sized, detects a nonidentity element of P_2 , then ρ does not extend along j to a homomorphism into H , nor into any set-sized group containing H .*

Proof. $j(P_2) \subseteq P_3$, which every set-target homomorphism kills (theorem 9.1); an injective enlargement of H preserves $\rho(g) \neq 1$. $\square$

For example $U(\omega)$ is detected in two rows by a map to $U(t)$, but its block inclusion into three rows is killed by every set-sized representation. The embedding is injective; what changes is the class of homomorphisms permitted by the extra relations (07).

The two parts of this report meet at one point. In section 30 bases of $\mathcal{A}^n$ are changed by $\text{GL}_n(\mathcal{A})$; theorem 18.2 shows that already $\text{SL}_2(\mathbf{Oz})$ is not reached by elementary moves, while theorem 30.3 obstructs the *existence* of reduced bases whatever group is used.

19 The kernel for $n \geq 3$: torsion, center and normal subgroups

This section and the next two are source 11's, with two merge results at the end of section 21. Throughout them (D, k) is $(\mathbb{Z}, \mathbb{R})$ or $(\mathbb{Z}[i], \mathbb{C})$, $A = \mathcal{A}_{D,k}$, and P_n is the kernel (7.2); for $n \geq 3$, $P_n = \langle e_{ij}(\Pi_k) \rangle$ and $E_n(A) = P_n \rtimes \text{SL}_n(D)$ (theorem 9.1). Source 11 writes $U_n(\Pi_k)$ for the root-generated subgroup and $N_n(A)$ for the kernel, deliberately never identifies them, and proves its structural statements for both. By proposition 7.3 (06) they are one group, so the statements are printed for P_n ; none of 11's proofs uses the identification.

19.1 Torsion and center

Theorem 19.1 (Torsion-free constant-term kernels [11]). *For every $n \geq 1$ the group $\ker(\text{ct} : \text{GL}_n(A) \rightarrow \text{GL}_n(D))$ is torsion-free. In particular P_n is torsion-free for every $n \geq 2$.*

Proof. Let $g^r = 1_n$ with $r \geq 1$ and $\text{ct}(g) = 1_n$. Work in the algebraically closed field $\mathbf{No}[i]$, which contains both coefficient cases. The polynomial $X^r - 1$ has the r distinct ordinary complex r th roots of unity as its roots, and the minimal polynomial of g divides it; so g is diagonalizable over $\mathbf{No}[i]$ with eigenvalues $\zeta_1, \dots, \zeta_n \in \mathbb{C}$. Hence $\text{tr } g = \sum_j \zeta_j$ is an ordinary complex number, in particular finite; it lies in A , whose finite elements are those of D (lemma 2.1(iii)). So $\text{tr } g = \text{ct}(\text{tr } g) = \text{tr } \text{ct}(g) = n$. Each ζ_j has modulus one, so its real part is at most 1, and the real parts sum to n ; hence every $\zeta_j = 1$, and the diagonalizable g is 1_n . $\square$

For $n = 2$ this is a second route to the torsion-freeness of P_2 in corollary 14.2, where 07 derives it from the free product. The theorem concerns the constant-term kernel only; the constant subgroup $\text{SL}_n(D)$ has torsion.

Theorem 19.2 (Centerlessness [11]). *For $n \geq 3$, P_n has trivial center.*

Proof. Let g be central and fix $0 \neq p \in \Pi_k$. Since $e_{ij}(p) = 1_n + pE_{ij} \in P_n$ and $p \neq 0$ in the field F , $gE_{ij} = E_{ij}g$ for all $i \neq j$. Comparing entries, $g_{ai}\delta_{jb} = \delta_{ai}g_{jb}$: with $b = j$ and $a \neq i$ every off-diagonal entry vanishes, and with $a = i$, $b = j$ one gets $g_{ii} = g_{jj}$. So $g = \lambda 1_n$ with $\lambda, \lambda^{-1} \in A$, hence $\lambda \in D^\times$ (lemma 2.1(ii)), and $\text{ct}(g) = 1_n$ forces $\lambda = 1$. $\square$

P_n is nonabelian: $[e_{12}(\omega), e_{23}(\omega)] = e_{13}(\omega^2) \neq 1_n$.

Corollary 19.3 (Set-sized normal subgroups and conjugacy classes [11]). *For $n \geq 3$, P_n has no nontrivial set-sized normal subgroup, and the conjugacy class in P_n of every nonidentity element is a proper class.*

Proof. If $S \triangleleft P_n$ is a set, conjugation is a homomorphism $P_n \rightarrow \text{Aut}(S)$ into a set-sized group, trivial by theorem 9.1(iii); so S is central and $S = 1$ by theorem 19.2. If the conjugacy class of $x \neq 1$ were a set X , conjugation would give a homomorphism $P_n \rightarrow \text{Sym}(X)$, again trivial; then x would be central. $\square$

This is consistent with the tower of theorem 10.1: its members $P_n^{(a)}$ are nontrivial proper normal subgroups, and so are proper classes; invisibility does not make P_n simple.

Proposition 19.4 (Finite subgroups are detected by constant terms [11]). *Let $n \geq 3$. Every finite subgroup $\Theta \leq E_n(A)$ is mapped injectively into $\text{SL}_n(D)$ by ct . The isomorphism types of finite subgroups of $E_n(A)$ are exactly those of $\text{SL}_n(D)$.*

Proof. The kernel of $\text{ct}|_\Theta$ is a finite subgroup of the torsion-free group P_n (theorem 19.1), hence trivial. Conversely the constant inclusion embeds every finite subgroup of $\text{SL}_n(D)$. $\square$

The statement concerns isomorphism types and injectivity. It is not claimed that every finite subgroup of $E_n(A)$ is conjugate to a constant subgroup; that would need a separate analysis of splittings over finite groups (11).

19.2 A principal congruence chain and common congruence elements

For $0 \neq a \in A$ put $P_n\langle a \rangle = \{g \in P_n : g - 1_n \in M_n(aA)\}$, so that $P_n^{(a)} = P_n\langle \omega^a \rangle$ in (10.1). As there, the ideal aA is used only as a congruence relation; no quotient ring is asserted to be a set.

Theorem 19.5 (A chain at $1 + \omega$ [11]). *Let $n \geq 3$. The subgroups $P_n\langle(1 + \omega)^m\rangle$, $m \geq 0$, are normal in P_n , $P_n\langle 1 \rangle = P_n$, and*

$$P_n\langle(1 + \omega)^m\rangle \supsetneq P_n\langle(1 + \omega)^{m+1}\rangle \quad (m \geq 0),$$

while $e_{12}(\omega^\omega)$ lies in every one of them.

Proof. Matrices congruent to 1_n modulo an ideal form a normal subgroup of $\mathrm{GL}_n(A)$, which gives normality and the inclusions. The parameter $p_m = (1 + \omega)^m \omega^{1/2}$ has support $\{j + 1/2 : 0 \leq j \leq m\}$, so $e_{12}(p_m) \in P_n\langle(1 + \omega)^m\rangle$. If it lay in the next subgroup, $\omega^{1/2}/(1 + \omega)$ would lie in A ; its leading exponent is $-1/2$, which is impossible for a nonzero element of A . For $m \geq 1$,

$$\frac{\omega^\omega}{(1 + \omega)^m} = \sum_{r \geq 0} (-1)^r \binom{m + r - 1}{r} \omega^{\omega - m - r} \in \Pi_k, \quad (19.1)$$

since $(1 + \omega)^{-m} = \omega^{-m}(1 + \omega^{-1})^{-m}$; the exponents $\omega - m - r$ are positive and strictly decreasing, and multiplying by $(1 + \omega)^m$ checks the identity coefficientwise. So $\omega^\omega \in (1 + \omega)^m A$ for every m . $\square$

The comparison with [theorem 10.1](#) is instructive. The tower uses the monomial ideals $\omega^a A$ over the whole class of scales $a > 0$ and separates the elements of P_n ; the chain above is countable, uses powers of a non-monomial element, and does not separate. The second phenomenon is general for sets of principal congruences:

Proposition 19.6 (Common root elements for a set of congruences [11]). *Let $(a_s)_{s \in S}$ be a set-indexed family of nonzero elements of A . There is $h_0 > 0$ such that $\omega^h/a_s \in \Pi_k$ for all $s \in S$ and all $h > h_0$. Consequently, for $n \geq 3$, $i \neq j$ and $h > h_0$, $e_{ij}(\omega^h) \in \bigcap_{s \in S} P_n\langle a_s \rangle$.*

Proof. Each $a_s^{-1} \in F$ has a set support, and the union T of these supports is a set. Choose $h_0 > -t$ for all $t \in T$ ([lemma 2.3](#)); then every exponent $h + t$ of ω^h/a_s is positive for $h > h_0$. Hence $\omega^h = a_s(\omega^h/a_s) \in a_s A$, and $e_{ij}(\omega^h) \in P_n$. $\square$

This generalizes the set clause of [theorem 10.1](#) from the monomials ω^a to arbitrary nonzero elements. It says nothing about the intersection over the proper class of all principal ideals, and makes no claim that the quotient rings $A/a_s A$ are sets (11).

19.3 Finite quotients and the profinite completion

By [theorem 16.4](#) (07; for $n \geq 3$ also 11), $\mathrm{Res}_{\mathrm{fin}} E_n(A) = P_n$, and by [theorem 9.1](#)(iv) every finite quotient of $E_n(A)$ is a finite quotient of $\mathrm{SL}_n(D)$ composed with ct. The source is a class group, so its profinite completion needs a convention; following 11 it is the inverse limit of its finite quotients, and since the whole diagram of finite quotients is that of the set group $\mathrm{SL}_n(D)$, this is an ordinary profinite group.

Corollary 19.7 (The profinite completion [11]). *For $n \geq 3$ there are canonical isomorphisms*

$$\widehat{E_n(\mathbf{Oz})} \cong \widehat{\mathrm{SL}_n(\mathbb{Z})} \cong \mathrm{SL}_n(\widehat{\mathbb{Z}}), \quad \widehat{E_n(\mathbf{Oz}[i])} \cong \widehat{\mathrm{SL}_n(\mathbb{Z}[i])}.$$

Proof. The first isomorphism in each case identifies the finite quotients with those of the constant subgroup. For the second real isomorphism, the congruence subgroup theorem of Bass, Milnor and Serre [2] says that for $n \geq 3$ the principal congruence subgroups of $\mathrm{SL}_n(\mathbb{Z})$ are cofinal among its subgroups of finite index. Reduction $\mathrm{SL}_n(\mathbb{Z}) \rightarrow \mathrm{SL}_n(\mathbb{Z}/m\mathbb{Z})$ is surjective, since elementary elimination works over each local factor of $\mathbb{Z}/m\mathbb{Z}$ and elementary parameters lift to integers. So the completion is the inverse limit of the $\mathrm{SL}_n(\mathbb{Z}/m\mathbb{Z})$, which is $\mathrm{SL}_n(\widehat{\mathbb{Z}})$. $\square$

The last step is a deep imported theorem, not reproved here; the preceding factorization is proved. No isomorphism of $E_n(\widehat{\mathbf{Oz}[i]})$ with $\mathrm{SL}_n(\widehat{\mathbb{Z}[i]})$ is asserted: congruence kernels need care in the totally imaginary case, which the classical source treats separately from fields with a real embedding (11).

20 Steinberg groups

Apart from its extension to all root systems without A_1 component in [theorem 40.1](#) (source 12), no other statement about Steinberg groups or K_2 appears in the collection. For a commutative unital ring B and $n \geq 3$, $\mathrm{St}_n(B)$ is the group with generating symbols $x_{ij}(a)$, $a \in B$, $i \neq j$, and the Steinberg relations

$$x_{ij}(a)x_{ij}(b) = x_{ij}(a+b), \quad (20.1)$$

$$[x_{ij}(a), x_{lm}(b)] = 1 \quad (i \neq m, j \neq l), \quad (20.2)$$

$$[x_{ih}(a), x_{hj}(b)] = x_{ij}(ab) \quad (i, h, j \text{ distinct}). \quad (20.3)$$

$\pi_B : \mathrm{St}_n(B) \rightarrow E_n(B)$, $x_{ij}(a) \mapsto e_{ij}(a)$, is a surjective homomorphism. The symbols x_{ij} are the universal instance of the root maps of [theorem 9.2](#). For a proper-class ring, 11 uses finite words and finite derivations from the relations, and represents an equivalence class of words by Scott's trick, as the set of equivalent words of least rank; the carrier is a class of sets with well-defined operations, and no class of proper classes is formed. Only the presentation and its elementary universal property are used; no theorem on the kernel of π_B or on universal central extensions is needed.

Lemma 20.1 (The split Steinberg kernel [11]). *Let $B = D_0 \oplus I$ be a unital commutative ring with a ring retraction $c : B \rightarrow D_0$ and $I = \ker c$. Then c induces a split surjection $\mathrm{St}_n(B) \rightarrow \mathrm{St}_n(D_0)$ whose kernel is the normal closure of the $x_{ij}(a)$, $a \in I$.*

Proof. The inclusion $D_0 \hookrightarrow B$ and c induce maps of presentations whose composite on $\mathrm{St}_n(D_0)$ is the identity. The normal closure N' lies in the kernel. Modulo N' , (20.1) gives $x_{ij}(d+a) \equiv x_{ij}(d)$, so every word is congruent to the image of its constant-term word; if that word is the identity in $\mathrm{St}_n(D_0)$, so is its image in $\mathrm{St}_n(B)$. $\square$

Theorem 20.2 (Universal set-sized quotient of the Steinberg group [11]). *Let $n \geq 3$ and $P_n^{\mathrm{St}} = \ker(\mathrm{ct} : \mathrm{St}_n(A) \rightarrow \mathrm{St}_n(D))$. The sequence $1 \rightarrow P_n^{\mathrm{St}} \rightarrow \mathrm{St}_n(A) \rightarrow \mathrm{St}_n(D) \rightarrow 1$ is split exact, P_n^{St} is perfect, every homomorphism from P_n^{St} to a set-sized group is trivial, and every homomorphism from $\mathrm{St}_n(A)$ to a set-sized group factors uniquely through $\mathrm{St}_n(D)$. Moreover $\pi_A(P_n^{\mathrm{St}}) = P_n$.*

Proof. Let $V \leq \mathrm{St}_n(A)$ be generated by the $x_{ij}(a)$, $a \in \Pi_k$. Relations (20.1) and (20.3) are the hypotheses of [theorem 9.2](#); with the ring theorem `osq:thm:universal`(iii), [corollary 9.3](#) kills every set-sized image of V and, with [proposition 2.4](#), makes V perfect. By [lemma 20.1](#) P_n^{St} is the normal closure of V , and [lemma 9.4](#) transfers both properties. Factorization is as in [theorem 9.1](#)(iv). Finally π_A maps the generators of V onto the root elements $e_{ij}(\Pi_k)$ and is onto $E_n(A)$, so it maps the normal closure of V onto the normal closure of those root elements, which is P_n ([lemma 7.1](#)). $\square$

Remark 20.3 (No K -theoretic conclusion [11]). The quotient $\mathrm{St}_n(D)$ is a set (countable here: its alphabet is countable and its words are finite), and it must not be replaced by $\mathrm{SL}_n(D)$ without accounting for the kernel of π_D . P_n^{St} is the kernel of the constant term, not of $\pi_A : \mathrm{St}_n(A) \rightarrow E_n(A)$, which is not computed. The theorem does not calculate K_2 , assert

a universal central extension in any unstable rank, establish an excision theorem, or claim anything about algebraic K -theory in general; nor does it assert that P_n^{St} is centerless or torsion-free: the proofs of [theorems 19.1](#) and [19.2](#) are about matrices and are not transferred.

21 Set-sized fragments and exact thresholds

In a fixed set-sized ring the identity map is a faithful set-sized representation, so only cardinal bounds can survive ([section 12](#)). Source 11 replaces the class of scales by a regular cardinal. Fix an uncountable regular cardinal κ and let $\Gamma_\kappa = \bigoplus_{\alpha < \kappa} \mathbb{Q}e_\alpha$, ordered by the sign of the coefficient at the least index where it is nonzero; so $e_\alpha > 0$ and $me_\beta < e_\alpha$ for $\alpha < \beta$ and ordinary $m \geq 1$. Let $\Pi_{<\kappa}$ be the set of Hahn series in X over Γ_κ with coefficients in k whose support is a reverse well-ordered subset of $\Gamma_\kappa^{>0}$ of cardinality $< \kappa$, and $R_{<\kappa} = D \oplus \Pi_{<\kappa}$. The map $X^g \mapsto \omega^{\iota(g)}$, with $\iota(e_\alpha) = \omega^{-\alpha}$ for the ordinal α , embeds $R_{<\kappa}$ in A ; there are two levels of exponents, and $X^{e_\alpha} = \omega^{\omega^{-\alpha}}$, for instance $X^{e_0} = \omega$ and $X^{e_1} = \omega^{1/\omega}$. This is the ring R_κ^{11} of the sibling report, with its ideal Π_κ^{11} , the same group (`osq:prop:lex`) and the same embedding (`osq:rel:lem:embedding`); source 11 describes it only as related to that report's constructions and proves the facts it needs directly. Put $P_{n,<\kappa} = \ker(\text{ct} : E_n(R_{<\kappa}) \rightarrow E_n(D))$ (11's $N_{n,\kappa,D}$).

21.1 Cardinal rigidity below κ

Lemma 21.1 (Closure and small scales [11]).

1. $R_{<\kappa}$ is a set-sized subring of A , ct retracts it onto D , and its units are $D^\times$.
2. For every $S \subseteq \Gamma_\kappa^{>0}$ with $|S| < \kappa$ there is $\beta < \kappa$ with $me_\beta < s$ for all $s \in S$ and ordinary $m \geq 1$.
3. Every element of $\Pi_{<\kappa}$ is a product of two elements of $\Pi_{<\kappa}$.

Proof. (i) Unions and sumsets of fewer than κ exponents have fewer than κ elements, so Hahn multiplication preserves the support bound; positivity and the multiplicativity of ct are unchanged, the coefficient field and exponent group are sets, and the leading-exponent argument of [lemma 2.1\(ii\)](#) gives the units. (ii) The least nonzero coordinate indices of the members of S are fewer than κ ordinals below κ , hence bounded below κ by regularity; for β above all of them, the leading coordinate of each $s \in S$ is positive and occurs before β , so s dominates every me_β . (iii) For $p \neq 0$ take β from (ii) for $S = \text{supp } p$ and write $p = X^{e_\beta}(X^{-e_\beta}p)$. $\square$

Parts (i) and (ii) are also `osq:rel:lem:embedding(iii)` and (ii).

Lemma 21.2 (No small ring image of the fragment ideal [11]). *Every additive multiplicative map from $\Pi_{<\kappa}$ to a ring of cardinality $< \kappa$ is zero.*

Proof. Let ρ be such a map and $0 \neq p \in \Pi_{<\kappa}$; take β from [lemma 21.1\(ii\)](#) for $\text{supp } p$ and put $\delta = e_\beta$. The κ exponents e_γ , $\beta < \gamma < \kappa$, lie in $(0, \delta)$, so two of the monomials X^{e_γ} have the same image; call their exponents $\eta < \varepsilon$. By the division lemma `osq:lem:division` (11's Lemma 3.1, whose proof uses only the ordered-group inequalities $\varepsilon + r(\varepsilon - \eta) < (r+1)\delta < s$, reverse well ordering and Hahn multiplication), $p = (X^\varepsilon - X^\eta)q$ with q of positive support of cardinality at most $\max(\aleph_0, |\text{supp } p|) < \kappa$, since κ is uncountable. So $q \in \Pi_{<\kappa}$ and $\rho(p) = \rho(X^\varepsilon - X^\eta)\rho(q) = 0$. $\square$

Regularity bounds the leading indices in [lemma 21.1\(ii\)](#); uncountability keeps the countable geometric tail within the support bound. 11 does not extend the lemma to singular cardinals or to finite supports. It is the nonunital form, for this ring, of the sibling report's cointiality theorem `osq:thm:cointial` (through `osq:prop:lex`), and it is weaker than that report's exact ring threshold `osq:prop:card11`; the merge uses the latter in [proposition 21.12](#).

Theorem 21.3 (Cardinal elementary rigidity [11]). *Let $n \geq 3$. Every homomorphism $P_{n,<\kappa} \rightarrow H$ with $|H| < \kappa$ is trivial, and every homomorphism $E_n(R_{<\kappa}) \rightarrow H$ with $|H| < \kappa$ factors uniquely through $\text{ct} : E_n(R_{<\kappa}) \rightarrow \text{SL}_n(D)$. The group $P_{n,<\kappa}$ is nontrivial, nonabelian, torsion-free, centerless and perfect.*

Proof. Let $U_{<\kappa} = \langle e_{ij}(\Pi_{<\kappa}) \rangle$. By [corollary 9.3\(ii\)](#) and [lemma 21.2](#) it has no nontrivial image of cardinality $< \kappa$, and by [corollary 9.3\(iii\)](#) and [lemma 21.1\(iii\)](#) it is perfect. $P_{n,<\kappa}$ is its normal closure, by the split-kernel argument of [lemma 7.1](#) (11's Lemma 5.1, for any split ring), and [lemma 9.4](#) transfers both properties; factorization is as in [theorem 9.1\(iv\)](#). (By [proposition 7.2](#) and [lemma 21.1\(iii\)](#), in fact $U_{<\kappa} = P_{n,<\kappa}$.) The monomial X^{e_0} gives $e_{12}(X^{e_0}) \neq 1_n$ and $[e_{12}(X^{e_0}), e_{23}(X^{e_0})] = e_{13}(X^{2e_0}) \neq 1_n$. Torsion-freeness holds because $P_{n,<\kappa}$ lies in the kernel of [theorem 19.1](#). The proof of [theorem 19.2](#) applies inside $R_{<\kappa}$, whose root matrices are present and whose units are $D^\times$. $\square$

Corollary 21.4 (Cardinal Steinberg rigidity [11]). *Let $n \geq 3$. Every homomorphism from $\text{St}_n(R_{<\kappa})$ to a group of cardinality $< \kappa$ factors through $\text{St}_n(D)$. Its constant-term kernel $P_{n,<\kappa}^{\text{St}}$ is perfect and has no nontrivial image in a group of cardinality $< \kappa$.*

Proof. Apply [corollary 9.3\(ii\),\(iii\)](#) and [lemma 21.2](#) to the subgroup generated by the $x_{ij}(\Pi_{<\kappa})$, and take its normal closure by [lemmas 9.4](#) and [20.1](#). $\square$

Proposition 21.5 (Non-simple fragments [11]). *Let $n \geq 3$, $t_0 = X^{e_0}$ and $t_1 = X^{e_1}$. The subgroups $\{g \in P_{n,<\kappa} : g - 1_n \in M_n((1+t_1)^m R_{<\kappa})\}$, $m \geq 0$, are normal and strictly decreasing, and $e_{12}(t_0)$ lies in all of them. The group $P_{n,<\kappa}$ is linear over the set field $\text{Frac}(R_{<\kappa})$.*

Proof. Normality follows from reduction modulo an ideal. The parameter $(1+t_1)^m X^{e_1/2}$ witnesses strictness, since $X^{e_1/2}/(1+t_1)$ has leading exponent $-e_1/2 < 0$. For $m \geq 1$, $t_0/(1+t_1)^m = \sum_{r \geq 0} (-1)^r \binom{m+r-1}{r} X^{e_0-(m+r)e_1} \in \Pi_{<\kappa}$: the exponents are positive in the lexicographic order and the support is countable. Finally $R_{<\kappa}$ is a set-sized integral domain, and the matrix inclusion into its fraction field is faithful. $\square$

21.2 Concrete cardinals, actions and representations

Proposition 21.6 (The countable-support case [11]). *For $\kappa = \aleph_1$ and $n \geq 3$, $|R_{<\aleph_1}| = |P_{n,<\aleph_1}| = \mathfrak{c}$, and $P_{n,<\aleph_1}$ has no nontrivial homomorphism to a countable group.*

Proof. $|\Gamma_{\aleph_1}| = \aleph_1 \leq \mathfrak{c}$, and a countable support with a countable list of coefficients is coded in a set of size $\mathfrak{c}^{\aleph_0} = \mathfrak{c}$. The series rX^{e_0} , $r \in \mathbb{R}$, give $\mathfrak{c}$ distinct root matrices in $P_{n,<\aleph_1}$, and finite matrices over a ring of size $\mathfrak{c}$ are at most $\mathfrak{c}$ in number. The last assertion is [theorem 21.3](#). $\square$

Proposition 21.7 (Action thresholds [11]). *Let $n \geq 3$. Every action of $P_{n,<\kappa}$ on a set X with $|\text{Sym}(X)| < \kappa$ is trivial, in particular for finite X and for infinite X with $2^{|X|} < \kappa$; every action of $E_n(R_{<\kappa})$ on such a set factors through $\text{SL}_n(D)$.*

Proof. Apply [theorem 21.3](#) to the target $\text{Sym}(X)$. For infinite $|X| = \lambda$, $|\text{Sym}(X)| = 2^\lambda$: $\lambda^\lambda \leq (2^\lambda)^\lambda = 2^\lambda$, and swapping the members of any subfamily of λ disjoint pairs gives 2^λ permutations. $\square$

A target of cardinality $< \kappa$ and an action on a set of cardinality $< \kappa$ are different conditions; the action threshold uses the cardinality of the symmetric group (11).

Corollary 21.8 (Ordinary groups invisible to bounded actions [11]). *For every infinite cardinal λ there is an explicit set-sized nontrivial, torsion-free, centerless, perfect group with no nontrivial action on any set of cardinality at most λ : $P_{3,<\kappa}$ with $D = \mathbb{Z}$ and $\kappa = (2^\lambda)^+$.*

Proof. A successor cardinal μ^+ is regular: a cofinal family of at most μ ordinals below μ^+ would have union of size at most $\mu \cdot \mu = \mu$. For $|X| \leq \lambda$, $2^{|X|} \leq 2^\lambda < \kappa$. Apply [theorem 21.3](#) and [proposition 21.7](#). $\square$

Corollary 21.9 (Finite-dimensional complex representations [11]). *For $\kappa = \mathfrak{c}^+$ and $n \geq 3$, every finite-dimensional complex representation of $P_{n,<\kappa}$ is trivial, and every such representation of $E_n(R_{<\kappa})$ factors through $\mathrm{SL}_n(D)$.*

Proof. $|\mathrm{GL}_m(\mathbb{C})| = \mathfrak{c} < \kappa$ for $m \geq 1$; the case $m = 0$ is trivial. $\square$

Remark 21.10 (No overstatement at $\aleph_1$ [11]). The absence of a countable image in [proposition 21.6](#) does not by itself exclude an action on a countably infinite set, since $|\mathrm{Sym}(\mathbb{N})| = \mathfrak{c}$, nor every finite-dimensional complex representation. The larger cardinals in [corollaries 21.8](#) and [21.9](#) are intentional. The exact threshold $\mathfrak{c}$ at $\aleph_1$ obtained in [proposition 21.12](#) does not change this: $\mathrm{Sym}(\mathbb{N})$ and $\mathrm{GL}_m(\mathbb{C})$ have cardinality exactly $\mathfrak{c}$.

21.3 Exact thresholds

Proposition 21.11 (An exact-threshold criterion [11]). *Let I be a set-sized commutative ring, possibly without unit, of infinite cardinality θ , such that every nonzero additive multiplicative map from I to a ring has image of cardinality at least θ . Let $A_0 = D_0 \oplus I$ be a split unital commutative ring with $I \neq 0$ an ideal and $|D_0| \leq \theta$, and let $n \geq 3$ and $N = \ker(E_n(A_0) \rightarrow E_n(D_0))$. Then $|N| = \theta$, every homomorphism from N that does not kill a given nonidentity element has a target of cardinality at least θ , and the identity of N attains this bound for every nonidentity element. In particular the least cardinality of a group receiving a nontrivial homomorphic image of N is θ .*

Proof. N is the normal closure of $U = \langle e_{ij}(I) \rangle$ by the split-kernel argument ([lemma 7.1](#); 11's Lemma 5.1). If $f : N \rightarrow H$ is nontrivial, it is nontrivial on some conjugate gUg^{-1} , and $u \mapsto f(gug^{-1})$ is a nontrivial homomorphism $U \rightarrow H$. By [theorem 9.2](#) there is a nonzero ring image R_φ of I with $\theta \leq |R_\varphi| \leq |H|^{n(n-1)}$; as θ is infinite, H is infinite and $|H| \geq \theta$. The root matrices inject I into N , and $N \subseteq M_n(A_0)$ has at most θ elements, so $|N| = \theta$. $\square$

Source 11 notes that the hypothesis is stronger than [lemma 21.2](#) and does not claim that its fragments have exact threshold κ or $\kappa^{<\kappa}$: any application must verify the ring hypothesis. The sibling report's field bound verifies it for $R_{<\kappa}$. The next two results are the merge's.

Proposition 21.12 (Exact thresholds for the fragments [merge]). *Let κ be uncountable and regular and $n \geq 3$.*

1. *Every nonzero additive multiplicative map from $\Pi_{<\kappa}$ to a ring (not necessarily unital or commutative) has image of cardinality at least $\kappa^{<\kappa}$, and $|\Pi_{<\kappa}| = |R_{<\kappa}| = \kappa^{<\kappa}$.*
2. *For every $1 \neq g \in P_{n,<\kappa}$ the least cardinality of a group H admitting a homomorphism $P_{n,<\kappa} \rightarrow H$ that does not kill g is exactly $\kappa^{<\kappa}$; the same holds for homomorphisms*

defined on $E_n(R_{<\kappa})$. Every homomorphism from $E_n(R_{<\kappa})$ to a group of cardinality $< \kappa^{<\kappa}$ factors uniquely through ct , and every action of $P_{n,<\kappa}$ on a finite set, or on an infinite set X with $2^{|X|} < \kappa^{<\kappa}$, is trivial.

3. The least cardinality of a group detecting a nonidentity element of $P_{n,<\kappa}^{\text{St}}$, by a homomorphism defined on $P_{n,<\kappa}^{\text{St}}$, is also exactly $\kappa^{<\kappa}$.
4. For $\kappa = \aleph_1$ the value is $\mathfrak{c}$: $P_{n,<\aleph_1}$ has no nontrivial homomorphic image of cardinality below $\mathfrak{c}$.

Proof. (i) Let $\rho : \Pi_{<\kappa} \rightarrow S$ be additive and multiplicative with $\rho(x) \neq 0$. By lemma 21.1(ii) there is $\gamma = e_\beta > 0$ strictly below $\text{supp } x$; put $c = \gamma/2 \in \Gamma_\kappa$. Then $X^\gamma \in \Pi_{<\kappa}$, and $y = X^{-\gamma}x$ has positive support of the same cardinality, so $y \in \Pi_{<\kappa}$; since $\rho(X^\gamma)\rho(y) = \rho(x) \neq 0$, $\rho(X^{2c}) \neq 0$. Let $\Gamma_\kappa^{\ll c} = \{h \in \Gamma_\kappa : m|h| < c \text{ for all ordinary } m \geq 1\}$ (the sibling report's $H_c(\Gamma)$) and let $\mathbb{L}_c$ be the set of Hahn series over Γ_κ with coefficients in k and fewer than κ exponents, all in $\Gamma_\kappa^{\ll c}$. By `osq:prop:card11` (with $K = k$) and its proof, $\mathbb{L}_c$ is a field with at least $\kappa^{<\kappa}$ elements. For $f \in \mathbb{L}_c$ every exponent $c+h$ of $X^c f$ is positive, since $|h| < c$, and there are fewer than κ of them; so $X^c \mathbb{L}_c \subseteq \Pi_{<\kappa}$. For distinct $f, f' \in \mathbb{L}_c$ the element $X^c/(f - f')$ lies in $X^c \mathbb{L}_c \subseteq \Pi_{<\kappa}$, and in $\Pi_{<\kappa}$

$$(X^c f - X^c f') \cdot \frac{X^c}{f - f'} = X^{2c}.$$

If $\rho(X^c f) = \rho(X^c f')$, additivity and multiplicativity give $\rho(X^{2c}) = 0$, a contradiction. So $f \mapsto \rho(X^c f)$ is injective on $\mathbb{L}_c$ and $|\rho(\Pi_{<\kappa})| \geq \kappa^{<\kappa}$. This is the scaled-field collision lemma `osq:lem:collision(i)`, whose proof, as that report notes, uses only that the source is closed under addition and multiplication and contains $X^c \mathbb{L}_c$; it therefore applies to maps defined on $\Pi_{<\kappa}$ alone. Since $|k| = \mathfrak{c} \leq \max(\kappa, \mathfrak{c})$, `osq:prop:card11` gives $|R_{<\kappa}| = \kappa^{<\kappa}$; and $|\Pi_{<\kappa}| \geq \kappa^{<\kappa}$ by the case $\rho = \text{id}$.

(ii) Apply proposition 21.11 with $I = \Pi_{<\kappa}$, $\theta = \kappa^{<\kappa}$ and $D_0 = D$, which is countable. A homomorphism on $E_n(R_{<\kappa})$ not killing g restricts to one on $P_{n,<\kappa}$, and the inclusion $E_n(R_{<\kappa}) \subseteq \text{GL}_n(R_{<\kappa})$, a group of cardinality $\kappa^{<\kappa}$, detects g . A homomorphism from $E_n(R_{<\kappa})$ to a group of cardinality $< \kappa^{<\kappa}$ therefore kills $P_{n,<\kappa}$, and factors uniquely through ct by the split decomposition $E_n(R_{<\kappa}) = P_{n,<\kappa} \rtimes \text{SL}_n(D)$ as in theorem 9.1(iv). For actions, $|\text{Sym}(X)|$ is finite or $2^{|X|}$ (proof of proposition 21.7).

(iii) Let $f : P_{n,<\kappa}^{\text{St}} \rightarrow H$ not kill g . By lemma 20.1, $P_{n,<\kappa}^{\text{St}}$ is the normal closure of the subgroup V generated by the $x_{ij}(\Pi_{<\kappa})$, so f is nontrivial on a conjugate of V , and composing with conjugation gives a nontrivial homomorphism $V \rightarrow H$. By (20.1), (20.3) and theorem 9.2, $\Pi_{<\kappa}$ has a nonzero ring image of cardinality at most $|H|^{n(n-1)}$, so $|H| \geq \kappa^{<\kappa}$ by (i). Conversely $\text{St}_n(R_{<\kappa})$ is a quotient of the free group on $n(n-1)|R_{<\kappa}| = \kappa^{<\kappa}$ symbols, so its kernel has at most $\kappa^{<\kappa}$ elements, and its identity map detects g .

(iv) $\aleph_1^{<\aleph_1} = \aleph_1^{\aleph_0}$, and $2^{\aleph_0} \leq \aleph_1^{\aleph_0} \leq (2^{\aleph_0})^{\aleph_0} = 2^{\aleph_0}$. □

When $\kappa^{<\kappa} > \kappa$, for instance for $\kappa = \aleph_1$ when the continuum hypothesis fails, the exact value is strictly larger than the bound κ of theorem 21.3. The value agrees with the size $\mathfrak{c}$ of proposition 21.6. 11's non-claim about exact thresholds remains a true description of 11; the value is obtained here by combining 11's criterion with `osq:prop:card11` and the scaled-field collision, which 11 did not use.

The same argument applies to the other set-sized rings of the sibling report.

Corollary 21.13 (Exact thresholds in the sibling report's set-sized models [merge]). *Let A_0 be one of the five set-sized rings of `osq:thm:thresholds`, or one of the Gaussian*

versions named there, with constant ring D_0 , and let $n \geq 3$. For every nonidentity g in $\ker(\text{ct} : E_n(A_0) \rightarrow E_n(D_0))$ the least cardinality of a group detecting g , by a homomorphism defined on this kernel or on $E_n(A_0)$, is exactly $|A_0|$.

Proof. In the notation of the sibling report's `osq:sec:fixed`, $A_0 = D_0 + \Pi(\mathbb{F})$ for a subfield $\mathbb{F}$ of a Hahn field over an ordered set group Γ containing the coefficients and every monomial, with $\Pi(\mathbb{F})$ the elements of positive support (the report writes F ; here F is the ambient field `No` or `No[i]`). The proof of `osq:thm:thresholds` establishes, for each of these rings, that Γ is divisible, that every nonzero $x \in \Pi(\mathbb{F})$ has a positive strict lower bound $\gamma \in \Gamma$ of its support, and that for every $c > 0$ the field $L_c(\mathbb{F})$ of members of $\mathbb{F}$ supported in $\Gamma^{\ll c} = \{h \in \Gamma : m|h| < c \text{ for all ordinary } m \geq 1\}$ has exactly $|A_0|$ elements, while $X^c L_c(\mathbb{F}) \subseteq \Pi(\mathbb{F})$ (`osq:prop:fieldbound(i)`). With $c = \gamma/2$, $X^\gamma \in \Pi(\mathbb{F})$ and $X^{-\gamma}x \in \Pi(\mathbb{F})$, the argument of [proposition 21.12\(i\)](#) shows that every nonzero additive multiplicative map from $\Pi(\mathbb{F})$ to a ring has image of cardinality at least $|A_0|$. Moreover $|\Pi(\mathbb{F})| = |A_0|$, which is infinite, since $X^c L_c(\mathbb{F}) \subseteq \Pi(\mathbb{F})$, and $|D_0| \leq |A_0|$. Apply [proposition 21.11](#) with $\theta = |A_0|$, and use the inclusion into $\text{GL}_n(A_0)$ for homomorphisms defined on $E_n(A_0)$. $\square$

Together, [proposition 21.12](#) and [corollary 21.13](#) answer [question 34.4](#) for $n \geq 3$ in these rings: the least detecting target cardinal of every nonidentity relative elementary matrix is the cardinality of the ring, as for ring and module targets in `osq:thm:thresholds`. Other exponent groups and support restrictions remain open.

Part III

Shortest vectors and missing infima

This part is source 10's. An ordinary positive-definite quadratic form on $\mathbb{Z}^n$ has a shortest nonzero vector: a positive lower bound by the Euclidean norm combines with the finiteness of bounded subsets of $\mathbb{Z}^n$. Over `Oz` neither positive definiteness nor uniform separation settles the question. The mechanism is visible in one example: for infinitesimal $\varepsilon > 0$ let

$$\mathcal{E}_\varepsilon(x, y) = (x - \sqrt{2}y)^2 + \varepsilon(x^2 + y^2), \quad (x, y) \in \mathbf{Oz}^2. \quad (21.1)$$

No nonzero ordinary integer vector lies on the kernel $x = \sqrt{2}y$ of the dominant form, but for every $\gamma > 0$ the vector $\omega^\gamma(\sqrt{2}, 1)/\sqrt{3}$ has omnific coordinates, lies on that line and has energy $\varepsilon\omega^{2\gamma}$; making γ smaller never produces a finite vector. This creates a missing infimum. Unimodular vectors behave differently: `ct` sends a unimodular omnific vector to a nonzero unimodular ordinary vector, an admissible competitor, whereas the constant term of a nonzero vector may be zero.

Throughout, $(k, D, F, \mathcal{A})$ is $(\mathbb{R}, \mathbb{Z}, \mathbf{No}, \mathbf{Oz})$ or $(\mathbb{C}, \mathbb{Z}[i], \mathbf{No}[i], \mathbf{Oz}[i])$; conjugation is trivial in the real row and fixes `No` in the complex row, and $\|x\|^2 = \sum_\ell \overline{x_\ell} x_\ell$. *Terminology.* A vector is *unimodular* if its coordinates generate the unit ideal. Source 10 calls such vectors “primitive”; the omnific Diophantine report uses *primitive* for the weaker “no common nonunit divisor” (`odg:thm:realray`), so the word is not used for 10's notion here. The energy is written $\mathcal{E}$, since E_n is the elementary group.

22 Positive forms contract under constant-term descent

For an ordinary symmetric or Hermitian positive-semidefinite matrix Q write $q(x) = x^*Qx$, on k^n and on F^n . The ordinary spectral theorem gives an ordinary B with $Q = B^*B$, so $q(x) = \|Bx\|^2$; nonnegativity and the kernel persist after scalar extension to F .

Theorem 22.1 (Constant-term contraction [10]). *For $x \in \mathcal{A}^n$ and ordinary $a \in k^n$, $q(x - a) \geq q(\text{ct}(x) - a)$, with equality exactly when $x - \text{ct}(x)$ lies in the scalar-extended kernel of Q ; otherwise $q(x - a)$ is positive infinite while $q(\text{ct}(x) - a)$ is an ordinary real.*

Proof. Write $x = z + j$ with $z \in D^n$, $j \in \Pi_k^n$ (lemma 2.1). Then $B(x - a) = B(z - a) + Bj$ with $B(z - a)$ ordinary and Bj purely infinite. If $Bj \neq 0$ one coordinate is infinite (lemma 2.1(iii)), its squared modulus is positive infinite and the others are nonnegative. Finally $Bj = 0$ iff $Qj = 0$. $\square$

Corollary 22.2 (Strict descent for a positive family [10]). *Let $Q_0, \dots, Q_m$ be ordinary positive-semidefinite with common kernel 0 and $\lambda_0, \dots, \lambda_m$ positive surreals. For $\mathcal{E}(x) = \sum_i \lambda_i x^* Q_i x$ and $a \in k^n$, $\mathcal{E}(x - a) \geq \mathcal{E}(\text{ct}(x) - a)$ with equality exactly when $x \in D^n$.*

Proof. Each difference in theorem 22.1 is nonnegative, and for $j = x - \text{ct}(x) \neq 0$ at least one is positive. $\square$

Remark 22.3 (What the argument does not assert [10]). ct is not order preserving on $\mathcal{B}_k$, the contraction is not a Pythagorean decomposition (mixed terms may be nonzero), and the weights λ_i need not lie in $\mathcal{B}_k$, so applying ct to the energy itself is not legitimate. The argument works because an ordinary sum-of-squares map either annihilates the whole positive-exponent part or makes the norm infinite. For closest points the contraction always preserves admissibility; for shortest *nonzero* vectors it need not, since $x \neq 0$ does not imply $\text{ct}(x) \neq 0$; for unimodular vectors it does (proposition 26.1).

For a definite Q the contraction gives a Tarski-free proof that omnific points on a finite level of an ordinary definite form are ordinary, the conclusion of `odg:cor:definite` in the omnific Diophantine report.

23 Separated positive pencils

Definition 23.1 (Separated positive pencil [10]). Let $n \geq 1$, $m \geq 0$. A separated positive pencil consists of ordinary positive-semidefinite matrices $Q_0, \dots, Q_m$ and positive surreals $\varepsilon_0, \dots, \varepsilon_m$ with $\varepsilon_{j+1} \ll \varepsilon_j$. Put $q_j(x) = x^* Q_j x$,

$$K_j = \bigcap_{i < j} \ker_k Q_i, \quad L_j = K_j \cap D^n, \quad K_0 = k^n,$$

and require $K_m \neq 0$ and $K_{m+1} = 0$. The energy is $\mathcal{E}(x) = \sum_{j=0}^m \varepsilon_j q_j(x)$ for $x \in F^n$. For an ordinary subspace K , K_F is its scalar extension. Repeated kernels and zero forms are allowed.

The condition $K_m \neq 0 = K_{m+1}$ says that the last index is the first at which the cumulative kernel vanishes; no finer forms are appended after it, and this convention is part of the scope of the exact minimum formula below. For $m = 0$ it says Q_0 is positive definite (10).

Lemma 23.2 (Coercivity at the last scale [10]). *There are ordinary $c, C > 0$ with $c\varepsilon_m \|x\|^2 \leq \mathcal{E}(x) \leq C\varepsilon_0 \|x\|^2$ on F^n , and q_m is positive definite on K_m .*

Proof. $S = \sum_j Q_j$ is positive definite; choose $cI \leq S \leq CI$ over k , which persists over F by sum-of-squares factorization, and use $\varepsilon_m \leq \varepsilon_j \leq \varepsilon_0$. A vector of K_m killed by q_m lies in $K_{m+1} = 0$. $\square$

Lemma 23.3 (Exact lexicographic comparison [10]). *For ordinary $a, z, z' \in k^n$, the sign of $\mathcal{E}(z - a) - \mathcal{E}(z' - a)$ is the sign of the first nonzero entry of $(q_j(z - a) - q_j(z' - a))_{j=0}^m$, and the energies are equal if all entries vanish.*

Proof. If the first nonzero entry is d_j , divide by ε_j : the result is d_j plus the infinitesimal $\sum_{i>j} (\varepsilon_i/\varepsilon_j) d_i$. $\square$

This is exact for fixed ordinary vectors, not uniform in all ordinary integer coordinates; an infinitesimal weight times any one ordinary real is infinitesimal, which is central to the Pell example (10).

24 The unrestricted shortest-vector dichotomy

For a nonempty class $S \subseteq \mathbf{No}^{>0}$ let $\text{LB}_{\geq 0}(S) = \{b \in \mathbf{No}_{\geq 0} : b \leq s \text{ for all } s \in S\}$. A subclass $T \subseteq S$ is coinital if every $s \in S$ has some $t \in T$ with $t \leq s$. A surreal infimum is a greatest lower bound in $\mathbf{No}$, not an adjoined gap. Let $\mathcal{V} = \{\mathcal{E}(x) : x \in \mathcal{A}^n \setminus \{0\}\}$.

Theorem 24.1 (Minimum or exact missing-infimum cut [10]). *For a separated positive pencil exactly one of the following holds.*

- (a) *If $L_m \neq 0$, then $\mu = \min\{q_m(z) : z \in L_m \setminus \{0\}\}$ is an attained positive ordinary real, $\min \mathcal{V} = \varepsilon_m \mu$, and the minimizers are the finitely many $z \in L_m \setminus \{0\}$ with $q_m(z) = \mu$.*
- (b) *If $L_m = 0$, then $\mathcal{E}(x)/\varepsilon_m$ is positive infinite for every nonzero $x \in \mathcal{A}^n$, and $\text{LB}_{\geq 0}(\mathcal{V}) = \varepsilon_m(\mathcal{O}_{\text{fin}})_{\geq 0}$. In particular $\mathcal{V}$ has neither a minimum nor a surreal infimum.*

Proof. If $x \notin D^n$, $\mathcal{E}(x)/\varepsilon_m \geq c\|x\|^2$ is positive infinite (lemmas 2.1 and 23.2). If $x \in D^n \setminus K_m$, some $q_i(x) > 0$ with $i < m$ is an ordinary positive real and $\mathcal{E}(x)/\varepsilon_m \geq (\varepsilon_i/\varepsilon_m)q_i(x)$ is infinite. If $x \in L_m$, $\mathcal{E}(x) = \varepsilon_m q_m(x)$.

(a) q_m is positive definite on K_m , so the ordinary lattice vectors with $q_m(z) \leq q_m(z_0)$, for one $z_0 \in L_m \setminus \{0\}$, have bounded norm and are finitely many; their least positive value is μ . The classification gives the rest.

(b) The classification makes every nonnegative finite multiple of ε_m a lower bound. Conversely let $b \geq 0$ with $h = b/\varepsilon_m$ infinite; choose ordinary $w \in K_m$ with $q_m(w) = 1$ and $\gamma > 0$ with $2\gamma < \deg_\omega h$. Then $x = \omega^\gamma w \in \Pi_k^n \setminus \{0\}$, the earlier forms vanish on it, and $\mathcal{E}(x) = \varepsilon_m \omega^{2\gamma} \ll \varepsilon_m h = b$, so b is not a lower bound. The class $\varepsilon_m(\mathcal{O}_{\text{fin}})_{\geq 0}$ has no greatest member ($b + \varepsilon_m$ is larger), and a greatest lower bound would be nonnegative, so no infimum exists. $\square$

Theorem 24.2 (Improvement of every set-sized family [10]). *Assume $L_m = 0$. For every set-indexed family $(x_s)_{s \in S} \subseteq \mathcal{A}^n \setminus \{0\}$ there are an ordinary $w \in K_m$ with $q_m(w) = 1$ and $\gamma > 0$ such that $x = \omega^\gamma w$ satisfies $\mathcal{E}(x) \ll \mathcal{E}(x_s)$ for every s . Consequently $\mathcal{V}$ has no set-sized coinital subclass.*

Proof. Each $h_s = \mathcal{E}(x_s)/\varepsilon_m$ is positive infinite; lemma 2.3(iii) gives $\gamma > 0$ with $\omega^{2\gamma} \ll h_s$ for all s . The construction applies directly to a set of energy values, through their leading exponents, without choosing vector representatives. $\square$

Corollary 24.3 (No ordinary-factor approximation [10]). *If $L_m = 0$, for every nonzero $y \in \mathcal{A}^n$ there is a nonzero $x \in \mathcal{A}^n$ with $\mathcal{E}(x) < c\mathcal{E}(y)$ for every ordinary $c > 0$.*

The witness is a single monomial vector; no sum or limit is taken, and replacing the missing infimum by the limit of a chosen ordinary sequence would be incorrect. In the full surreal field no countable list of improving vectors is coinital in $\mathcal{V}$ (10). The monomial witness $\omega^\gamma w$ is the mechanism of the homogeneous cone criterion and the zero Pell fiber of the omnific Diophantine report (`odg:thm:cone`, `odg:prop:zeropell`).

25 Recovering the kernel flag from low energies

For $0 \leq j \leq m$ let $\text{Low}_j = \{x \in \mathcal{A}^n : \mathcal{E}(x) \ll \varepsilon_j\}$, with $0 \in \text{Low}_j$. (Source 10 writes V_j ; the letter is left to the radical flag of the Hahn–Tate report.)

Theorem 25.1 (Field span and integral trace of small energies [10]). *For every j : $\text{Low}_j \subseteq (K_{j+1})_F$, $\text{span}_F \text{Low}_j = (K_{j+1})_F$, and $\text{ct}(\text{Low}_j) = L_{j+1}$.*

Proof. For $x \in \text{Low}_j$ and $i \leq j$, $0 \leq q_i(x) \leq \mathcal{E}(x)/\varepsilon_j \in \mathfrak{m}$; by the sum-of-squares argument of [theorem 22.1](#), $q_i(x)$ is an ordinary nonnegative real or positive infinite, so it is 0. This gives the inclusion, and applying ordinary matrices and ct gives $\text{ct}(x) \in L_{j+1}$. Every $z \in L_{j+1}$ has $\mathcal{E}(z)/\varepsilon_j = \sum_{i>j} (\varepsilon_i/\varepsilon_j) q_i(z) \in \mathfrak{m}$. For the span, take an ordinary basis w_ℓ of K_{j+1} ; each $h_\ell = \mathcal{E}(w_\ell)/\varepsilon_j$ is positive infinitesimal, and for $0 < 2\gamma < -\deg_\omega h_\ell$ (all ℓ) the vectors $\omega^\gamma w_\ell$ lie in Low_j and span $(K_{j+1})_F$. $\square$

An irrational kernel can have a large field span of small-energy vectors and a small integral trace; in the binary example the trace is 0 (10).

26 Unimodular vectors

Let $\text{Um}(\mathcal{A}^n)$ be the unimodular vectors: those x with $\sum_\ell a_\ell x_\ell = 1$ for some $a_\ell \in \mathcal{A}$.

Proposition 26.1 (Unimodular descent [10]). *If $x \in \text{Um}(\mathcal{A}^n)$ then $\text{ct}(x) \in \text{Um}(D^n)$, and $\text{Um}(D^n) \subseteq \text{Um}(\mathcal{A}^n)$. Hence $\mathcal{E}(x) \geq \mathcal{E}(\text{ct}(x))$ on $\text{Um}(\mathcal{A}^n)$, strictly unless x is ordinary; the energies of ordinary unimodular vectors form a countable coinital subset of the unimodular energies; and every unimodular minimizer, if any, is ordinary.*

Proof. Apply the ring homomorphism ct to a relation $\sum a_\ell x_\ell = 1$; a relation over D is one over $\mathcal{A}$. Then use [corollary 22.2](#) at target 0; D^n is countable. $\square$

The converse “ $\text{ct}(x)$ unimodular implies x unimodular” for arbitrary $x \in \mathcal{A}^n$ is neither needed nor asserted (10). Descent of unimodularity under ct is the mechanism of the omnific Diophantine report’s unimodular Fermat theorem and rational-direction theorem (`odg:thm:unimodularfermat`, `odg:thm:realray`).

Corollary 26.2 (Countable coinitality [10]). *If the unimodular energy has no minimum, its coinitality is exactly $\aleph_0$. In particular [theorem 24.2](#) fails if every vector in it is required to be unimodular.*

Proof. [Proposition 26.1](#) gives a countable coinital set; a finite one would contain a minimum. $\square$

Lemma 26.3 (Discreteness of a projected lattice). *Let V be a finite-dimensional real vector space, Λ a full lattice in V and $T : V \rightarrow W$ real linear. Then $T(\Lambda)$ is discrete iff $\ker T = \text{span}_{\mathbb{R}}(\ker T \cap \Lambda)$. If $T(\Lambda)$ is not discrete, it has nonzero points arbitrarily close to 0, it is not closed, and there is $h \in \overline{T(\Lambda)} \setminus T(\Lambda)$. The same holds for complex-linear maps and Gaussian lattices, through the underlying real spaces.*

Proof. The equivalence is `tate:theta:lem:latticeprojection` of the Hahn–Tate report (for $\Lambda = \mathbb{Z}^d$, which is the general case after choosing a basis); source 10 proves it independently by the same saturation and rank count. The remaining assertions are 10’s. A nondiscrete subgroup contains nonzero $g_j \rightarrow 0$; passing to a subsequence with $g_j / \|g_j\| \rightarrow u$, for real t let a_j be an integer nearest to $t / \|g_j\|$; then $a_j g_j \rightarrow tu$, so a closed such subgroup would contain $\mathbb{R}u$, contradicting countability. For the complex case, a complex subspace is spanned over $\mathbb{C}$ by its Gaussian integral vectors iff it is spanned over $\mathbb{R}$ by them in real coordinates, because the Gaussian lattice and its intersection with a complex subspace are closed under multiplication by i . $\square$

Let r be the largest index in $\{0, \dots, m\}$ with $L_r \neq 0$ (it exists since $L_0 = D^n$), $\Lambda = L_r$, $W = \text{span}_k \Lambda$ and $W' = W \cap K_{r+1}$ (10’s H). By maximality $K_{r+1} \cap D^n = 0$, and Λ is a full lattice in W (in real coordinates in the complex case).

Theorem 26.4 (Exact unimodular attainment criterion [10]). *For a separated positive pencil the following are equivalent:*

1. $\mathcal{E}$ has a minimum on $\text{Um}(\mathcal{A}^n)$;
2. $\mathcal{E}$ has a minimum on $D^n \setminus \{0\}$;
3. q_r is positive definite on W ;
4. $W' = 0$.

When they hold, the minimizers in (i) and (ii) are the same finite set of ordinary unimodular vectors, obtained by minimizing q_r on $\Lambda \setminus \{0\}$ and breaking ties lexicographically by $q_{r+1}, \dots, q_m$. When they fail, the unimodular energies have coinitality $\aleph_0$.

Proof. The kernel of q_r on $W \subseteq K_r$ is W' , so (iii) $\Leftrightarrow$ (iv). A nonzero ordinary vector outside K_r has a positive earlier coefficient and is worse than every fixed nonzero vector of Λ (lemma 23.3), so ordinary minimization may be restricted to $\Lambda \setminus \{0\}$.

If $W' = 0$, q_r has a positive minimum on $\Lambda \setminus \{0\}$ attained at finitely many vectors, and lexicographic tie-breaking leaves a finite nonempty set of minimizers on $D^n \setminus \{0\}$. Each is unimodular: D is Euclidean, and dividing a non-unimodular ordinary vector by a nonunit gcd g gives a nonzero ordinary vector of energy $\mathcal{E}(z)/|g|^2 < \mathcal{E}(z)$. So (ii) holds, and proposition 26.1 gives (i) with no extra nonordinary minimizers.

If $W' \neq 0$, factor $q_r|_W = \|T \cdot\|^2$ with an ordinary T of kernel W' . Since $W' \cap \Lambda = 0$, lemma 26.3 makes $T(\Lambda)$ nondiscrete, while $q_r(z) > 0$ on $\Lambda \setminus \{0\}$ because $L_{r+1} = 0$; so q_r has infimum 0 and no minimum there. Dividing by the coordinate gcd keeps vectors in Λ , makes them unimodular and does not increase q_r , so arbitrarily small positive q_r values are attained by unimodular vectors, and any ordinary candidate in Λ is beaten (lemma 23.3); candidates outside K_r were already worse. So there is no ordinary nonzero or unimodular minimum, and proposition 26.1 excludes a nonordinary one. The coinitality is corollary 26.2. $\square$

The criterion uses the span of the actual integral intersection, not the whole deepest kernel, and can hold when theorem 24.1 gives unrestricted nonattainment (example 29.4).

27 Closest points

27.1 What is already in the Hahn–Tate report

Part II of the Hahn–Tate uniformization report (`docs/surcomplex/hahn-tate-uniformization, tate:part:theta`), present in the collection at 10’s pin, studies lexicographically valued quadratic lattice energies $\mathcal{E}_j(n) = \frac{1}{2}\mathcal{B}_j(n, n) + b_j(n)$ with Amini–Nicolussi higher-rank forms [1]. It already contains the ordinary backbone of this section:

- the lattice-projection lemma `tate:theta:lem:latticeprojection` (lemma 26.3) and the one-level quadratic criterion `tate:theta:lem:onelevel`;
- the exact flag criterion `tate:theta:thm:flag` and its corollary `tate:theta:cor:domain`: for an AN-admissible form, the compatible linear terms are exactly those giving an energy with a global lattice minimum;
- the parity bound: at most 2^g minimizers, attained (`tate:theta:thm:minimizers`, with the example $\sum_j n_j(n_j - 1)$);
- the Pell example `tate:theta:ex:irrational`: the form $((m - \sqrt{2}n)^2, m^2 + n^2)$ and the Pell vectors $p_k + q_k\sqrt{2} = (1 + \sqrt{2})^k$, along which the lexicographic energies strictly decrease.

Source 10 cites the Lean module that formalizes part of that report (`Surreal/Algebra/LatticeEnergyCertificate.lean`) but not the report itself; its repository search for “shortest vector” found no hit, which is literally true and misleading (Appendix A.3). These results are cited here, not reprinted; 10’s own proofs are kept only where they prove more (the non-closedness in lemma 26.3, the converse in theorem 27.2, the Gaussian bounds, arbitrary surreal targets) or where the omnific statement needs them.

The dictionary is as follows. For ordinary PSD forms q_j on all of k^n and an ordinary center a , the profile $z \mapsto (q_j(z - a))_{j=0}^m$ on $D^n = \mathbb{Z}^n$ (real case) is, up to additive constants, a lexicographic energy with $r = m + 1$ coordinates, $\mathcal{B}_{j+1}(z, z') = 2z^\top Q_j z'$ and $b_{j+1}(z) = -2a^\top Q_j z$ (the Hahn–Tate forms are indexed from 1). Its radical flag `tate:theta:eq:flag` is $V_j = K_j$, ending with $V_{m+1} = K_{m+1} = 0$; the global radical condition `tate:theta:eq:globalradical` holds automatically because $K_{j+1} \subseteq \ker Q_j$; and the compatibility $b_{j+1}(V_{j+1}) = -2a^\top Q_j K_{j+1} = 0$ holds for every center for the same reason. AN admissibility then amounts to rationality of every K_j . The Gaussian case is the real case for the rank- $2n$ lattice $\mathbb{Z}[i]^n \cong \mathbb{Z}^{2n}$ (a Hermitian PSD form is a real PSD form on the underlying space, with the same kernel).

27.2 Reduction to an ordinary lexicographic problem

For $a \in F^n$ a closest point is a minimizer of $\mathcal{E}(x - a)$ on $\mathcal{A}^n$, with no nonzero or unimodular restriction.

Proposition 27.1 (Exact reduction for an ordinary target [10]). *Every closest point to $a \in k^n$, if any, is ordinary, and it is exactly a lexicographic minimizer over $z \in D^n$ of $(q_0(z - a), \dots, q_m(z - a))$. Equivalently, with $S_{-1} = D^n$ and $S_j = \arg \min_{z \in S_{j-1}} q_j(z - a)$, a closest point exists iff every stage is nonempty, and then S_m is the finite set of closest points.*

Proof. Strict descent (corollary 22.2) gives ordinarity and lemma 23.3 the lexicographic description. On S_m all $q_j(z - a)$ are fixed, their sum is an ordinary positive-definite form, so $\|z - a\|$ is bounded and S_m finite. $\square$

27.3 Rational flags

Call an ordinary subspace $K \subseteq k^n$ *D-rational* if $K = \text{span}_k(K \cap D^n)$: ordinary rationality for $k = \mathbb{R}$, Gaussian rationality for $k = \mathbb{C}$; no rationality of a displayed basis or matrix is required.

Theorem 27.2 (Closest points for every ordinary center [10]; sufficiency also in the Hahn–Tate report). *For a separated positive pencil, every $a \in k^n$ has a closest point in $\mathcal{A}^n$ iff every prefix kernel K_j is D-rational. If so, after stage j the staged sets of proposition 27.1 are finite unions of cosets of L_{j+1} . If not, an ordinary center can be chosen for which one stage has an unattained ordinary infimum 0.*

Proof. Sufficiency. By proposition 27.1 it is the existence of a lexicographic minimum of the profile on D^n , which is `tate:theta:cor:domain` with `tate:theta:thm:flag` under the dictionary of subsection 27.1. The coset description is 10’s induction, which is also the inductive step of the proof of `tate:theta:thm:flag`: on a coset $v + L_j$ of S_{j-1} , the stage- j function $\ell \mapsto \|B_j(v - a) + B_j\ell\|^2$ has kernel K_{j+1} on K_j , which is rational, so B_jL_j is a lattice (lemma 26.3) and the minimum is attained at finitely many image points, each with preimage one coset of L_{j+1} (`tate:theta:lem:onelevel`); retaining the smallest value over the finitely many cosets proves the induction, and $L_{m+1} = 0$ makes the final set finite.

Necessity [10]. Let K_{j+1} be the first irrational kernel, so K_j is rational and L_j is a full lattice in K_j . With $q_j = \|B_j \cdot\|^2$, $G = B_jL_j$ is not discrete (lemma 26.3); choose $h \in \overline{G} \setminus G$. Since B_jK_j is closed and contains $\overline{G}$, there is ordinary $a \in K_j$ with $B_ja = h$. For $z \in L_j$ the earlier $q_i(z - a)$ vanish and $q_j(z - a) = \|B_jz - h\|^2 > 0$ has infimum 0 and no minimum, while ordinary $z \notin K_j$ are worse than every fixed candidate in L_j . So no ordinary lexicographic minimizer exists, and strict descent excludes nonordinary closest points. $\square$

The converse, an explicit bad center for an irrational flag, is not in the Hahn–Tate report, which proves nonsummability for non-admissible forms rather than the absence of a minimum for some center. Both 10 and that report credit the rational-filtration mechanism to higher-rank Voronoi theory [1].

Corollary 27.3 (Nearest-point multiplicity [10]). *For an ordinary target with closest points, their number is at most 2^n in the real case and 4^n in the Gaussian case, and both bounds are attained.*

Proof. The real bound is the midpoint argument of `tate:theta:thm:minimizers`: distinct minimizers z, z' congruent modulo $2D^n$ would have midpoint in D^n and $\mathcal{E}(\frac{z+z'}{2} - a) = \frac{\mathcal{E}(z-a) + \mathcal{E}(z'-a)}{2} - \frac{\mathcal{E}(z-z')}{4}$ smaller. The Gaussian bound is the same bound for the rank- $2n$ lattice $\mathbb{Z}[i]^n$. For the Euclidean form with one weight, the target $(1/2, \dots, 1/2)$ has 2^n closest integer points and the target with every coordinate $(1+i)/2$ has 4^n closest Gaussian points (10). $\square$

The parity argument is a standard finite-lattice argument; source 10 includes it as a consequence, not as a priority claim.

27.4 Arbitrary surreal targets

Taking the standard part of a target and using its ordinary closest points is not enough: infinitesimal displacements can decide between branches before the next energy scale (10). For $a \in F^n$ split supports coordinatewise:

$$a = a^+ + a^0 + a^-, \quad a^+ \in \Pi_k^n, \quad a^0 \in k^n, \quad a^- \in \mathfrak{m}_k^n, \quad (27.1)$$

so $b = a^0 + a^-$ is finite with standard part a^0 . This is a support decomposition, not a claim that ct is a homomorphism on F .

Lemma 27.4 (Matching the target's infinite part [10]). *For $a \in F^n$, $x \in \mathcal{A}^n$, $y = x - a^+$ and $z = \text{ct}(y) \in D^n$, $\mathcal{E}(x - a) \geq \mathcal{E}(a^+ + z - a) = \mathcal{E}(z - b)$, strictly unless $x = a^+ + z$. Every closest point to a has purely infinite part exactly a^+ .*

Proof. $y \in \mathcal{A}^n$; write $y = z + j$ and $q_i = \|B_i \cdot\|^2$. $B_i(z - b)$ is finite and $B_i j$ purely infinite; if $B_i j \neq 0$, $\|B_i(y - b)\|^2$ is infinite and exceeds $\|B_i(z - b)\|^2$. All weighted differences are nonnegative, and one is positive when $j \neq 0$. $\square$

Theorem 27.5 (Finite candidates for every target [10]). *Assume every prefix kernel is D -rational. For every $a^0 \in k^n$ there is a finite nonempty $C(a^0) \subseteq D^n$, depending only on a^0 and $q_0, \dots, q_m$, such that for every $a \in F^n$ with decomposition (27.1) the closest points of a are exactly*

$$a^+ + \arg \min_{z \in C(a^0)} \mathcal{E}(z - a^0 - a^-).$$

$|C(a^0)| \leq 2^n$ (real) or 4^n (Gaussian), and $C(a^0)$ does not depend on a^- or on the separated weights. Consequently closest points exist for every $a \in F^n$ iff they exist for every $a \in k^n$ iff every prefix kernel is D -rational.

Proof. Build a finite tree of cosets: the root is D^n ; at level j , for each retained coset $v + L_j$, minimize $q_j(z - a^0)$ on it and retain *all* minimizing cosets of L_{j+1} as children (finitely many and at least one, by the proof of theorem 27.2). Unlike the staged sets, no parent is discarded because another has a smaller value. At level $m + 1$ the cosets are singletons; $C(a^0)$ is the set of leaves.

Every ordinary z is dominated by a leaf for the target $b = a^0 + a^-$: if z is in a retained parent $v + L_j$ but not in a minimizing child, pick z' in one; for $i < j$, $q_i(z - b) = q_i(z' - b)$ exactly because $z - z' \in L_j$; the difference $q_j(z - b) - q_j(z' - b)$ has positive ordinary standard part $q_j(z - a^0) - q_j(z' - a^0)$; later differences are finite, so after division by ε_j their weighted sum is infinitesimal and $\mathcal{E}(z - b) > \mathcal{E}(z' - b)$. Repeating reaches a leaf, strictly better unless z was a leaf. Lemma 27.4 handles nonordinary candidates.

For the bound let $d_j = \dim_k K_j - \dim_k K_{j+1}$. The children of a parent are closest points in the projected lattice $B_j L_j$ of real rank d_j or $2d_j$; two congruent modulo twice that lattice would have a better midpoint, as in corollary 27.3; so each parent has at most 2^{d_j} or 4^{d_j} children, and $\sum_j d_j = n$. The equivalences follow from theorem 27.2. $\square$

Example 27.6 (The ordinary minimizer set is too small [10]). Let $0 < \varepsilon \in \mathfrak{m}$, $\eta = \sqrt{\varepsilon}$, and $\mathcal{E}(x, y) = x^2 + \varepsilon(y - x/2)^2$; the flag is rational. The ordinary target $a^0 = (1/2, 1/4)$ has the unique closest point $(0, 0)$, but the tree retains $C(a^0) = \{(0, 0), (1, 0), (1, 1)\}$. For $a = (1/2 + \eta, 1/4)$, $\mathcal{E}((1, 0) - a) - \mathcal{E}((0, 0) - a) = -2\eta + \varepsilon(1/4 - \eta/2) < 0$ and $\mathcal{E}((1, 1) - a) - \mathcal{E}((1, 0) - a) = \varepsilon\eta > 0$, so the unique closest point is $(1, 0)$: a branch losing the ordinary comparison must be kept.

This is a finite mathematical reduction, not an algorithm for arbitrary surreal inputs; constructing or comparing unrestricted surreals is a separate representational problem (10).

28 A binary model with two different missing cuts

Consider, in both rows,

$$\mathcal{E}_\varepsilon(x, y) = |x - \sqrt{2}y|^2 + \varepsilon(|x|^2 + |y|^2), \quad 0 < \varepsilon \in \mathfrak{m}, \quad (28.1)$$

with matrix $Q_\varepsilon = \begin{pmatrix} 1+\varepsilon & -\sqrt{2} \\ -\sqrt{2} & 2+\varepsilon \end{pmatrix}$, $\det Q_\varepsilon = \varepsilon(3 + \varepsilon) > 0$. Here $K_1 = k(\sqrt{2}, 1)$, $K_2 = 0$, and $K_1 \cap D^2 = 0$ since $\sqrt{2} \notin \mathbb{Q}(i)$. The form and the Pell vectors below are those of `tate:theta:ex:irrational`, where the lexicographic energy $((m - \sqrt{2}n)^2, m^2 + n^2)$ is shown to decrease strictly along the Pell vectors, so that its theta series is not Hahn summable. Source 10 uses the same example for a different purpose: to compare two omnific lower cuts.

Theorem 28.1 (Unrestricted binary cut [10]). *The nonzero-vector energies of (28.1) have nonnegative lower bounds exactly $\varepsilon(\mathcal{O}_{\text{fin}})_{\geq 0}$, no surreal infimum and no set-sized coinital family; every set-sized collection of nonzero candidates is improved by some $x_\gamma = \omega^\gamma(\sqrt{2}, 1)/\sqrt{3}$, with $\mathcal{E}_\varepsilon(x_\gamma) = \varepsilon\omega^{2\gamma}$.*

Proof. Theorems 24.1 and 24.2 with $q_1 = |x|^2 + |y|^2$. $\square$

Define ordinary integers by $p_0 = 1$, $q_0 = 0$, $p_{n+1} = p_n + 2q_n$, $q_{n+1} = p_n + q_n$. By induction

$$p_n \pm q_n\sqrt{2} = (1 \pm \sqrt{2})^n, \quad p_n^2 - 2q_n^2 = (-1)^n, \quad p_n q_{n+1} - q_n p_{n+1} = (-1)^n, \quad (28.2)$$

so consecutive $u_n = (p_n, q_n)$ form an ordinary unimodular basis and every u_n is unimodular over $\mathbf{Oz}$ and $\mathbf{Oz}[i]$, with

$$\mathcal{E}_\varepsilon(u_n) = (\sqrt{2} - 1)^{2n} + \varepsilon(p_n^2 + q_n^2). \quad (28.3)$$

These identities come from the recurrence, not from a classification of Pell solutions over the omnific integers (10); Pell rigidity in the omnific Diophantine report (only ordinary solutions at nonzero levels) is consistent with them.

Theorem 28.2 (Unimodular cut and countable coinitality [10]). *For (28.1) the nonnegative lower bounds of the unimodular energies are exactly $\mathbf{m}_{\geq 0}$; there is no infimum and no minimum, and $(\mathcal{E}_\varepsilon(u_n))_{n \in \mathbb{N}}$ is strictly decreasing and coinital in the unimodular energies, in both rows.*

Proof. For $x \in \text{Um}(\mathcal{A}^2)$ and $z = \text{ct}(x)$, z is a nonzero ordinary unimodular vector and $r_z = |z_1 - \sqrt{2}z_2|^2 > 0$ is an ordinary real, so $\mathcal{E}_\varepsilon(x) \geq \mathcal{E}_\varepsilon(z) > r_z$ (theorem 22.1); every nonnegative infinitesimal is a lower bound. A positive noninfinitesimal b exceeds an ordinary $r > 0$; choose n with $(\sqrt{2} - 1)^{2n} < r/2$, and since $p_n^2 + q_n^2$ is ordinary, $\mathcal{E}_\varepsilon(u_n) < r < b$. The class $\mathbf{m}_{\geq 0}$ has no greatest member. For coinitality apply this with $r < r_z$. Consecutive energies differ in the dominant coefficient by a positive ordinary real and in the rest by an ordinary multiple of ε , so the sequence decreases (lemma 23.3). $\square$

For a fixed positive ordinary weight the second term of (28.3) eventually stops the improvement; for infinitesimal ε no ordinary index reaches that balance, and substituting an infinite omnific index into the recurrence is not part of its definition and would not justify a limit (10).

With $\varepsilon = \omega^{-1}$ the two cuts differ visibly: $\text{LB}_{\geq 0}(\text{all nonzero energies}) = \omega^{-1}(\mathcal{O}_{\text{fin}})_{\geq 0}$ and $\text{LB}_{\geq 0}(\text{unimodular energies}) = \mathbf{m}_{\geq 0}$. For example $\omega^{-1/2}$ bounds every unimodular energy but not $\mathcal{E}(x_{1/8}) = \omega^{-3/4}$. Every unrestricted witness with small γ beats the whole Pell sequence, yet it is not unimodular, since its constant term is 0. “A coinital Pell family” is correct for unimodular energies and false for unrestricted ones (10).

29 Three attainment properties with strict implications

Corollary 29.1 (A strict hierarchy [10]). *For separated positive pencils: closest points for every target in $F^n \Rightarrow$ a shortest nonzero vector $\Rightarrow$ a shortest unimodular vector; neither implication reverses.*

Proof. If every prefix kernel is rational, $K_m \neq 0$ contains a nonzero integral vector and [theorem 24.1](#) applies; its ordinary minimizers are unimodular, since division by a nonunit gcd would give a shorter nonzero vector. Strictness: the examples below. $\square$

Example 29.2 (All three properties [10]). $\mathcal{E}(x, y) = x^2 + \varepsilon y^2$ on $\mathbf{Oz}^2$: all kernels rational, shortest vectors $\pm(0, 1)$ of energy ε ; the Gaussian version has the four unit multiples of $(0, 1)$.

Example 29.3 (A shortest vector without universal closest points [10]). On $\mathbf{Oz}^3$ let $\mathcal{E} = (z - \sqrt{2}y)^2 + \varepsilon(x^2 + y^2 + z^2)$. The last kernel is $K_1 = \text{span}_{\mathbb{R}}\{(1, 0, 0), (0, 1, \sqrt{2})\}$ with $K_1 \cap \mathbb{Z}^3 = \mathbb{Z}(1, 0, 0)$, so the shortest vectors are $\pm(1, 0, 0)$ with minimum ε ; but K_1 is not rational. For the center $(0, 0, 1/2)$ the dominant term on integer candidates is $(z - \sqrt{2}y - 1/2)^2$, with infimum 0 never attained: $\mathbb{Z} + \sqrt{2}\mathbb{Z}$ contains the arbitrarily small nonzero $(1 - \sqrt{2})^n$, so it is dense, and $1/2 \notin \mathbb{Z} + \sqrt{2}\mathbb{Z}$.

Example 29.4 (Unimodular attainment without unrestricted attainment [10]). On $\mathbf{Oz}^3$ let $\mathcal{E} = (z - \sqrt{2}y)^2 + \varepsilon x^2 + \varepsilon^2(y^2 + z^2)$. Then $K_1 = \text{span}_{\mathbb{R}}\{(1, 0, 0), (0, 1, \sqrt{2})\}$, $K_2 = \mathbb{R}(0, 1, \sqrt{2})$, $L_1 = \mathbb{Z}(1, 0, 0)$ and $L_2 = 0$; the unrestricted energies have no infimum, with lower cut $\varepsilon^2(\mathcal{O}_{\text{fin}})_{\geq 0}$. For the unimodular criterion $r = 1$, $W = \mathbb{R}(1, 0, 0)$ and $W \cap K_2 = 0$: the shortest unimodular vectors are exactly $\pm(1, 0, 0)$, of energy ε . Directly: an ordinary vector off the x -axis has positive dominant coefficient, ordinary unimodular vectors on the axis are $\pm(1, 0, 0)$, and nonordinary unimodular vectors are worse than their constant terms. The Gaussian analogue has four unimodular minimizers.

The binary form [\(28.1\)](#) has neither minimum ([section 28](#)).

Real example	All centers have closest points	Shortest nonzero	Unimodular shortest
$x^2 + \varepsilon y^2$	yes	yes	yes
Example 29.3	no	yes	yes
Example 29.4	no	no	yes
Form (28.1)	no	no	no

The table is not a classification of arbitrary surreal forms; it records all patterns allowed by the implication chain within the pencil class (10).

30 Reduction obstructions

Nonattainment can rule out the existence of the reduced basis itself, not only the termination of an algorithm. A basis of $\mathcal{A}^n$ means the columns of a matrix in $\text{GL}_n(\mathcal{A})$; an invertible matrix over F is not a change of basis of the same $\mathcal{A}$ -module (10).

Lemma 30.1 (What Gauss inequalities would imply [10]). *Over $\mathbf{Oz}$, with $\beta(u, v) = u^T Q v$ the bilinear form of the energy $\mathcal{E}$, a basis with $\mathcal{E}(b_1) \leq \mathcal{E}(b_2)$ and $2|\beta(b_1, b_2)| \leq \mathcal{E}(b_1)$ has b_1 as a shortest nonzero vector.*

Proof. For $s, t \in \mathbf{Oz}$ not both 0, $\mathcal{E}(sb_1 + tb_2) \geq (s^2 - |st| + t^2)\mathcal{E}(b_1)$. If one coefficient vanishes the other has modulus at least 1 ([lemma 2.1](#)); otherwise $s^2 - |st| + t^2 = (|s| - |t|)^2 + |st| \geq 1$. $\square$

Corollary 30.2 (No Gauss-reduced basis [10]). *A binary separated positive pencil with $L_m = 0$, for instance [\(28.1\)](#), has no Gauss-reduced $\mathbf{Oz}$ -basis.*

For a real basis $b_1, \dots, b_n$ perform finite Gram–Schmidt orthogonalization, $b_j = b_j^* + \sum_{i < j} \mu_{ji} b_i^*$ with $\Delta_i = \mathcal{E}(b_i^*) > 0$ (10 writes D_i ; all divisions are in $\mathbf{No}$). The δ -LLL conditions [9] are

$$|\mu_{ji}| \leq \frac{1}{2} \quad (i < j), \quad \Delta_j + \mu_{j,j-1}^2 \Delta_{j-1} \geq \delta \Delta_{j-1} \quad (j \geq 2). \quad (30.1)$$

Theorem 30.3 (No ordinary-parameter LLL basis [10]). *Let $\mathcal{E}$ be a separated positive pencil on $\mathbf{Oz}^n$ with $L_m = 0$. For no ordinary $\delta \in (1/4, 1]$ does an $\mathbf{Oz}$ -basis satisfy (30.1).*

Proof. With $c = \delta - 1/4 \in (0, 3/4]$ ordinary, the size bound and the Lovász inequality give $\Delta_j \geq c \Delta_{j-1} \geq c^{j-1} \Delta_1$. For nonzero $v = \sum a_i b_i$, $a_i \in \mathbf{Oz}$, with a_k the last nonzero coefficient, the b_k^* -coefficient of v is a_k , so $\mathcal{E}(v) \geq a_k^2 \Delta_k \geq \Delta_k \geq c^{n-1} \mathcal{E}(b_1)$, using $|a_k| \geq 1$ and $c \leq 1$. The factor c^{n-1} is ordinary, contradicting corollary 24.3 for b_1 . $\square$

This is an existence obstruction. No assertion is made about a modified reduction notion with different constants or an infinitesimal gap $\delta - 1/4$ (10). It applies also to example 29.4, which has a shortest unimodular vector: LLL conditions control every nonzero module vector, including the non-unimodular monomial vectors.

For $F = \mathbf{No}[i]$ use $\langle u, v \rangle = u^* Q v$, conjugate-linear in the first argument, $\mu_{ji} = \langle b_i^*, b_j \rangle / \Delta_i$, and the explicitly stated conditions

$$|\Re \mu_{ji}| \leq \frac{1}{2}, \quad |\Im \mu_{ji}| \leq \frac{1}{2} \quad (i < j), \quad \Delta_j + |\mu_{j,j-1}|^2 \Delta_{j-1} \geq \delta \Delta_{j-1} \quad (j \geq 2), \quad (30.2)$$

stated to avoid relying on a convention for complex LLL parameters (10).

Theorem 30.4 (Gaussian reduction obstruction [10]). *For a separated positive pencil on $\mathbf{Oz}[i]^n$ with $L_m = 0$, no $\mathbf{Oz}[i]$ -basis satisfies (30.2) for an ordinary $\delta \in (1/2, 1]$.*

Proof. The size conditions give $|\mu_{j,j-1}|^2 \leq 1/2$; with $c = \delta - 1/2 \in (0, 1/2]$, $\Delta_j \geq c^{j-1} \Delta_1$, and $|a_k| \geq 1$ gives $\mathcal{E}(v) \geq c^{n-1} \mathcal{E}(b_1)$, contradicting corollary 24.3. $\square$

31 Closed, uniformly separated Euclidean lattices

For ordinary real α and positive ε let

$$T_{\alpha, \varepsilon} = \begin{pmatrix} \sqrt{1+\varepsilon} & -\alpha/\sqrt{1+\varepsilon} \\ 0 & \sqrt{\varepsilon(1+\alpha^2+\varepsilon)/(1+\varepsilon)} \end{pmatrix}, \quad T_{\alpha, \varepsilon}^* T_{\alpha, \varepsilon} = \begin{pmatrix} 1+\varepsilon & -\alpha \\ -\alpha & \alpha^2+\varepsilon \end{pmatrix},$$

with positive surreal square roots and determinant $\sqrt{\varepsilon(1+\alpha^2+\varepsilon)} \neq 0$. The module $\mathcal{L}_\varepsilon = T_{\sqrt{2}, \varepsilon} \mathcal{A}^2 \subseteq F^2$ satisfies $\|T_{\sqrt{2}, \varepsilon} x\|^2 = \mathcal{E}_\varepsilon(x)$, so the binary example is a literal Euclidean lattice problem in surreal coordinates (10).

Proposition 31.1 (Uniform separation without a shortest vector [10]). *For positive infinitesimal ε , $\mathcal{L}_\varepsilon$ is uniformly separated and closed in the $\mathbf{No}$ -valued Euclidean topology, yet its nonzero distances to 0 have neither a minimum nor a surreal infimum.*

Proof. $\mathcal{E}_\varepsilon(x) \geq \varepsilon \|x\|^2$ and a nonzero $x \in \mathcal{A}^2$ has a coordinate of modulus at least 1, so distinct points are at distance at least $\rho = \sqrt{\varepsilon}$. For $p \notin \mathcal{L}_\varepsilon$ the ball $B(p, \rho/3)$ contains at most one lattice point ℓ , and the ball of radius $\min(\rho/3, \|p - \ell\|/2)$ contains none; so the complement is open, without any set-indexed limit. Squaring is an order isomorphism of the nonnegative surreals, so theorem 28.1 transfers. $\square$

Lemma 23.2 gives uniform separation for every pencil; the missing ingredient for an attained minimum is ordinary compactness and finiteness, not positive definiteness. No contradiction to a Minkowski theorem with a surreal volume is claimed, since no such volume theory is constructed; $\mathcal{L}_\varepsilon$ is a proper-class module, and its two generators are a basis over $\mathcal{A}$, not over $\mathbb{Z}$ (10).

32 Scope and sharp boundaries

The full surreal class matters. The failure of an infimum and the failure of set-sized coinitality are distinct; the latter uses a positive exponent below an arbitrary set and must not be transferred to a fixed Hahn field (10). In the Hahn field $k((t^\mathbb{R}))$ (t positive infinitesimal, well-ordered supports) let $A^{\text{Hahn}} = D + k((\mathbb{R}_{<0}))$, the series with nonpositive exponents and constant coefficient in D (here $k((\mathbb{R}_{<0}))$ denotes series supported in $\mathbb{R}_{<0}$, not a field). For the binary form with $\varepsilon = t^\eta$, $\eta > 0$ real, the vectors $v_n = t^{-1/n}(\sqrt{2}, 1)/\sqrt{3}$ have energies $t^{\eta-2/n}$; every nonzero energy divided by ε is still positive infinite, and if its leading term is ct^{-d} , $d > 0$, then n with $2/n < d$ gives $\mathcal{E}(v_n)/\mathcal{E}(x)$ infinitesimal. So (v_n) is a countable coinital family there; the energies still have no infimum. In **No** the exponent $\gamma = 1/\omega$ lies below every $1/n$, and $\omega^{1/\omega}(\sqrt{2}, 1)/\sqrt{3}$ defeats the whole sequence under $t = \omega^{-1}$.

Ordinary coefficient forms are essential. For the positive-definite surreal form $q(x, y) = (x - \omega y)^2 + y^2$ and $v = (\omega, 1) \in \mathbf{Oz}^2$, $q(v) = 1$ but $q(\text{ct } v) = q(0, 1) = \omega^2 + 1$: contraction fails as strongly as possible. This form is even reduced by the unimodular $\mathbf{Oz}$ -basis $(1, 0), (\omega, 1)$ (both energies 1, orthogonal), so it is not asserted that all positive-definite surreal forms lack reduced bases. A general asymptotic expansion of a positive-definite matrix need not have positive-semidefinite coefficients and cannot be inserted into the hypotheses (10).

What is and is not uniform. **Theorem 27.5** handles every target, but it does not say that infinitesimal perturbations leave the nearest point unchanged (**example 27.6**). The unrestricted theorem improves every set-sized family, not a family indexed by the whole class, and in the unimodular problem even a countable coinital family cannot be improved simultaneously by another unimodular vector. The LLL obstructions need an ordinary parameter strictly above the critical value (10).

Relations, questions, and records

33 Relations to neighbouring reports

- `docs/surreal/omnific-diophantine-geometry (odg:)`. Part I answers its question `odg:q:affine` for closed subgroup schemes of GL_N over $\mathbb{Z}$ (and over $\mathbb{Z}[i]$ for $\mathbf{Oz}[i]$) and contains `odg:cor:unipotent` and `odg:cor:orthogonal` as special cases (**remark 4.6**); **theorem 3.3** and `odg:thm:bounded` use the same transfer and neither contains the other. The anisotropic example $x^2 + y^2 - 3z^2$ is covered by `odg:cor:unipotent` and is printed here only for its rational part (**section 5**). Part III's constant-term contraction gives `odg:cor:definite` without Tarski transfer, its monomial witnesses are the mechanism of `odg:thm:cone`, and its unimodular descent is that of `odg:thm:unimodularfermat` and `odg:thm:realray`. The word *primitive* keeps that report's meaning; 10's primitive vectors are called unimodular here.

- `docs/surreal/set-sized-quotients-of-omnific-integers` (`osq:`). [Theorem 9.1](#) strengthens `osq:thm:universal`, `osq:cor:faithful` and `osq:prop:matrices` from ring and module maps to abstract group maps and implies the ring theorem for $K \in \{\mathbb{R}, \mathbb{C}\}$ ([remark 9.6](#)); its first two proofs use that report's collision lemmas `osq:lem:division` and `osq:lem:collision`. The finite-support boundary ([proposition 12.2](#)) is the group form of its support threshold `osq:thm:support`, and [proposition 11.1](#) is the linear-group form of its finite-side theorem `osq:thm:finite`. Source 11's third proof derives [theorem 9.1\(iii\)](#) from that report's `osq:thm:universal(iii)` ([remark 9.6](#)); 11's set-sized fragments are that report's ring R_κ^{11} (`osq:prop:lex`, `osq:rel:lem:embedding`), and its exact ring threshold `osq:prop:card11`, with `osq:lem:collision`, gives the exact group thresholds of [proposition 21.12](#); the five set-sized models of `osq:thm:thresholds` give [corollary 21.13](#). Two manuscripts of the batch of `f4c9504` have since been merged into that report: manuscript 05, on countable-domain representations detecting the extended ideal, is a ring-level relative of the countable detection of [section 15](#); manuscript 01, on relation modules $\ker(\omega^\gamma C)$, is a relative of the rational kernel flags of [Part III](#). Neither is used here. [Part IV](#) (source 12) uses `osq:thm:universal(iii)` as the ring input for every root system without A_1 component and for current Lie rings, shows that ring and group invisibility are equivalent for every Dorroh unitization ([theorem 38.5](#)), and with `osq:prop:card11` and `osq:thm:thresholds` gives the exact thresholds of [corollary 43.7](#).
- `docs/surcomplex/first-kappa-coefficients` (batch 32). Its omnific integers O_κ with fewer than κ terms have the same set-sized quotients as $\mathbf{Oz}$; they are the rings $\mathcal{A}_{\mathbb{Z}, \mathbb{R}}^{<\kappa}$ of the sibling report's support threshold, and for uncountable κ [corollary 44.1](#) [merge] makes $E_\Phi(\mathbb{Z})$ the universal set-sized quotient of $E_\Phi(O_\kappa)$ for every Φ without A_1 component. That report is cited by name only; nothing is imported from it.
- `docs/surreal/euclidean-three-space` (`e3:`), [Part III](#) (`e3:cut:part`). The standard-part theorems for $\mathrm{SO}(n, \mathbf{No})$ and $\mathrm{SU}(n, \mathbf{No}[i])$ are parallel to [theorem 9.1](#) but neither implies the other ([remark 11.2](#)). Its open question on a uniform reductive classification (`e3:cut:sub:open`) concerns the surreal field and is not answered here.
- `docs/surcomplex/hahn-tate-uniformization` (`tate:`), [Part II](#) (`tate:part:thet a`). The ordinary backbone of [section 27](#) and the Pell example are there and are cited, not reprinted ([subsection 27.1](#)); `Surreal/Algebra/LatticeEnergyCertificate.lean` formalizes part of it. The omnific statements of [Part III](#) are not in that report.
- `docs/surcomplex/surcomplex-field-automorphisms` (`saut:`). The exponent rescaling Φ_h of [lemma 8.3](#) is its dilation (`saut:eq:dilation`).
- `docs/surreal/omnific-preserving-automorphisms`, written concurrently from the same batch. Its manuscript on algebraic-parameter rigidity treats algebraic group actions on $\mathbf{Oz}$, not points of groups over $\mathbf{Oz}$; it is cited by directory only.

34 Questions

The sources call these questions suggested by their proofs, not previously published named open problems (06, 07, 10, 11).

Question 34.1 (The full special linear groups [06, 07, 11]). For $n \geq 3$, is $\mathrm{SL}_n(\mathbb{Z})$ the universal set-sized quotient of $\mathrm{SL}_n(\mathbf{Oz})$, and $\mathrm{SL}_n(\mathbb{Z}[i])$ that of $\mathrm{SL}_n(\mathbf{Oz}[i])$? More generally, what can set-sized groups detect on $\mathrm{SL}_n(\mathcal{A}_{D,k})$ outside $E_n(\mathcal{A}_{D,k})$, and can the cusp-residue obstruction be organized into a classification of elementary orbits or of extensions of two-row representations?

Status: open. Non-elementary matrices and relative algebraic K -theory are the additional obstruction (06). [Theorem 18.2](#) shows that the issue cannot be bypassed by declaring all unimodular matrices elementary, and [theorem 18.4](#) shows only that one family becomes invisible after one stabilization; it does not classify the remaining obstructions (07). Source 11 asks the first part again (its Question 11.1) and notes that elementary generation, or an independent elimination of the unstable obstruction, would settle it; neither step is hidden in any proof here. Source 12 asks it for every split semisimple group and for full rank-one points ([questions 45.1](#) and [45.2](#)).

Question 34.2 (Rank two [06, 11]). Does an analogue of kernel invisibility hold for E_2 ?

Status: answered negatively by 07 ([theorems 15.3](#) and [16.5](#)): every nonidentity element of P_2 is detected by a countable group, although no finite group detects it. Source 11 asks for the set-sized images of $E_2(\mathbf{Oz})$, $E_2(\mathbf{Oz}[i])$ and their kernels (its Question 11.2), at a pin where this answer was already in the report: there is no universal set-sized quotient ([corollary 15.5](#)), finite images are arithmetic ([proposition 16.2](#)), and one countable group separates finite sets ([theorem 15.3](#)). Source 12 reproves the negative answer, with a second proof of the amalgam theorem, and extends it to every root system with an A_1 component ([theorem 41.1](#)).

Question 34.3 (Other root systems [07, 11]). For which elementary Chevalley-type groups over $\mathcal{A}_{D,k}$ does a collision force the whole relative kernel to have no set-sized images, and which low-rank systems keep additive specializations like [proposition 15.2](#)?

Status when first recorded: open. Other root systems have extra root factors and integer structure constants in their commutator formulas, which a formal analogy does not control (07). Source 11 suggests the same for elementary symplectic and other split Chevalley groups through root-kernel reconstruction, and notes that root lengths and commutator constants must be addressed; it also leaves the kernel of $\mathrm{St}_n(\mathcal{A}_{D,k}) \rightarrow E_n(\mathcal{A}_{D,k})$ uncomputed ([remark 20.3](#)). *Status* (batch 32): answered for split elementary groups by source 12 ([Part IV](#)). For every unital $D \subseteq k$ and every finite reduced root system Φ without A_1 component, a collision forces the whole relative kernel P_Φ of the simply connected elementary group to have no set-sized images ([theorem 38.2](#)), also for the Steinberg group and for the relative kernels of the other split isogeny forms, which are isomorphic to P_Φ ([theorem 40.1](#) and [proposition 40.3](#)); the extra root factors and the constants 2 and 3 are controlled by treating long roots first and halving only the free parameter ([theorem 37.2](#)). For $D = \mathbb{Z}, \mathbb{Z}[i]$ this is exact: the low-rank systems that keep additive specializations like [proposition 15.2](#) are exactly those with an A_1 component, whose factor P_2 is detected by the countable maps of [theorem 15.3](#), and then no universal set-sized quotient exists ([theorem 41.1](#)). Open: twisted, nonsplit and relative-rank-one groups ([question 45.4](#)), non-elementary points ([questions 34.1](#) and [45.1](#)), the rank-one converse for non-simply-connected forms of the whole group, and the kernel of $\mathrm{St}_\Phi \rightarrow E_\Phi$ ([question 45.3](#)).

Question 34.4 (Thresholds at set-sized stages [07; 06]). For a given exponent group and support restriction, what is the least target cardinal detecting a prescribed nonzero relative elementary matrix?

Status: partly answered [merge]. [Propositions 9.7](#) and [12.1](#) give lower bounds; a matching upper bound needs an actual detecting representation at that stage (06, 07). The ring analogue is studied in the sibling report's fixed-group thresholds. Source 11 proves the lower bound κ for the rings of supports of size $< \kappa$ over $\bigoplus_{\alpha < \kappa} \mathbb{Q}$ ([theorem 21.3](#)) and an abstract criterion for exactness ([proposition 21.11](#)); combined with the sibling report's field bound, the least detecting cardinal of every nonidentity relative elementary matrix, $n \geq 3$, is exactly $\kappa^{<\kappa}$ there, and the size of the ring in each of the sibling report's five set-sized models ([proposition 21.12](#)

and [corollary 21.13](#)). Other exponent groups and support restrictions remain open, and nothing is said here about $n = 2$ at set-sized stages. *Status* (batch 32): source 12's cardinal transfer ([theorem 43.3](#) and [corollary 43.4](#)) holds for every infinite cardinal and every root system without A_1 component, and with [proposition 21.12](#) the same exact values hold in these rings for every such root system, for Steinberg kernels and for perfect current Lie rings [[merge](#)] ([corollary 43.7](#)); the question is otherwise unchanged ([question 45.5](#)).

Question 34.5 (Arbitrary positive-definite surreal matrices [10]). Classify shortest-vector existence for arbitrary positive-definite surreal matrices on $\mathbf{Oz}^n$, for instance by classifying which changes of coordinates in $\mathrm{GL}_n(\mathbf{Oz})$ bring a form into a separated positive pencil.

Status: open. The form $(x - \omega y)^2 + y^2$ shows that the kernel-flag criterion does not apply to arbitrary decompositions ([section 32](#)).

Question 34.6 (Reduction toward cuts [10]). Is there a theory of reduced bases whose targets are cuts rather than attained successive minima, distinguishing changes of basis over $\mathbf{Oz}$ from scalar-field diagonalization, and not assuming that a shortest unimodular vector controls every nonzero module vector?

Status: open.

Question 34.7 (Fixed exponent groups [10]). In a fixed Hahn field, what is the least size of a coinital family of energies, in terms of the coinitality of the positive exponent group?

Status: open. The real-exponent comparison of [section 32](#) suggests the question; an exact theorem must state divisibility and coinitality hypotheses (10).

The question `odg:q:affine` of the omnific Diophantine report is answered here for closed subgroup schemes of GL_N . *Status* (batch 31): that report now answers it for geometrically integral curves, except over $\mathbf{Oz}$ for real curves with normalization $\mathbb{A}_{\mathbb{R}}^1$ whose integer points all lack a real normalization preimage (`odg:sg:q:real`); it stays open for general affine varieties of dimension at least two ([remark 4.6](#)).

Part IV

Chevalley groups and current Lie rings

This part is source 12's ([subsection 1.2](#)). [Part II](#) proves, for $n \geq 3$, that the purely infinite kernel of $E_n(\mathcal{A}_{D,k})$ has no set-sized images, and that the answer reverses for $n = 2$. Source 12 asks which feature of the matrix size is responsible, and answers: the absence of a root component of type A_1 . For every finite reduced root system without such a component, including C_2 , G_2 and the exceptional types, the constant-term kernel of the elementary Chevalley group has no nontrivial homomorphism to a set-sized group; for $D = \mathbb{Z}, \mathbb{Z}[i]$ an A_1 component, even inside $A_1 \times A_1$, destroys every universal set-sized quotient. For current Lie rings the dividing line is different: rational perfection. The part stands after the questions so that the numbering of the earlier parts is unchanged; its answer to [question 34.3](#) and its bearing on [question 34.4](#) are recorded in status notes there.

35 Conventions, results, and what is printed once

35.1 Standing objects

Throughout this part $k \in \{\mathbb{R}, \mathbb{C}\}$, $D \subseteq k$ is any unital subring (automatically a set), and $A = \mathcal{A}_{D,k} = D \oplus \Pi_k$ as in (2.3); when a statement needs $D \in \{\mathbb{Z}, \mathbb{Z}[i]\}$ it says so. Let Φ be a nonempty finite reduced crystallographic root system, and let G_Φ be the split simply connected semisimple group scheme over $\mathbb{Z}$ with root system Φ and a fixed pinning, with root subgroups $x_\alpha : \mathbb{G}_a \rightarrow G_\Phi$ ($\alpha \in \Phi$). For a commutative unital ring B ,

$$E_\Phi(B) = \langle x_\alpha(b) : \alpha \in \Phi, b \in B \rangle \leq G_\Phi(B)$$

(finite words), $\text{St}_\Phi(B)$ is the Steinberg group with symbols $\tilde{x}_\alpha(b)$, the additive relations and the Chevalley commutator relations for nonopposite roots, and

$$P_\Phi(A) = \ker(\text{ct} : E_\Phi(A) \rightarrow E_\Phi(D)), \quad P_\Phi^{\text{St}}(A) = \ker(\text{ct} : \text{St}_\Phi(A) \rightarrow \text{St}_\Phi(D)),$$

where ct is induced by the ring retraction $\text{ct} : A \rightarrow D$ of lemma 2.1. Fix a faithful representation $G_\Phi \hookrightarrow \text{GL}_{N,\mathbb{Z}}$; the entries of $x_\alpha(t)$ are polynomials in t with integral coefficients, so ct is applied entrywise and commutes with the representation. Φ has no A_1 component means that every irreducible component of Φ has rank at least two. For $\Phi = A_{n-1}$, $n \geq 3$, one has $G_\Phi = \text{SL}_n$, $x_{\varepsilon_i - \varepsilon_j}(a) = e_{ij}(a)$, $E_\Phi = E_n$, $P_\Phi(A) = P_n$, $\text{St}_\Phi = \text{St}_n$ and $P_\Phi^{\text{St}}(A) = P_n^{\text{St}}$: the notation of this part extends that of Part II without changing it. For $\Phi = A_1$, $E_\Phi = E_2$ and $P_\Phi(A) = P_2$.

35.2 Reserved letters

Source 12's notation collides with the report's in many letters. The following choices hold throughout this part; the renamings are also listed in table 3.

- Φ is a root system only. The exponent rescaling Φ_h of lemma 8.3 (source 12's T_δ , its “exponent compression”) is not used in this part; Φ_h with a surreal subscript is never a root system.
- α, β, γ are roots, not exponents as in Part II; exponents are written η . Root elements $x_\alpha(a)$ carry a root subscript; the abstract root maps and Steinberg symbols x_{ij} of Part II carry two indices, and Steinberg symbols here are $\tilde{x}_\alpha$. Ring elements are a, b, c, r, s, t ; no bare x is a ring element in this part.
- I is an abstract commutative ring without unit (as in proposition 7.2 and theorem 9.2); source 12 also writes I for Π_k , which is written Π_k here.
- A homomorphism to a target group H is φ (source 12's h). The group generated by root maps is Σ (its U), the common root kernel J_φ (its J), the tuple map and the reconstructed ring $\vartheta_\varphi : I \rightarrow R_\varphi$ (its T , and its $\tau : I \rightarrow R_h$), exactly as in theorem 9.2. Source 12's τ is not the monomial $\tau = \omega^{2\delta}$ of lemma 8.5, and its h is neither the third index nor the rescaling factor of Part II.
- In rank one, $G_D = \text{SL}_2(D)$, C_D and B_D are those of section 13 (source 12's G_0 , B_0 and $B(A)$); C_D is the constant upper triangular group and must not be confused with the ring $\mathcal{B}_k = k \oplus \Pi_k$. Source 12's detection field $F \in \{\mathbb{Q}, \mathbb{Q}(i)\}$ is $K_D = \text{Frac } D$; F remains the ambient field **No** or **No** $[i]$, and source 12's set field F of its collision data is the field L of lemma 8.4.

- A Lie lattice is $\mathfrak{l}$, a finite-rank free $\mathbb{Z}$ -module with an integral Lie bracket (source 12's Λ); $\mathfrak{l}_{\mathbb{Q}} = \mathfrak{l} \otimes_{\mathbb{Z}} \mathbb{Q}$ (its $\mathfrak{g}$), and $\mathfrak{l}(B) = \mathfrak{l} \otimes_{\mathbb{Z}} B$ for a commutative ring B , possibly without unit (its $L(B)$). The letter Λ keeps its meanings of an ordinal ([lemma 8.1](#)) and of a Euclidean lattice ([section 26](#)); L is a compressed field and L_j are integral points. A basis of $\mathfrak{l}$ is $\mathbf{b}_1, \dots, \mathbf{b}_d$ (its b_i), the structure integers are z_{lij} (its n_{lij}), the standard basis of $\mathfrak{sl}_2(\mathbb{Z})$ is $\mathbf{e}, \mathbf{f}, \mathbf{h}$ (its e, f, h ; e_{ij} are elementary matrices and e_{α} basis vectors of Γ_{κ}), a target Lie ring is $\mathfrak{H}$ (its M), and the integral functional of [theorem 42.6](#) is ϖ (its λ , a cardinal in [Part II](#)).
- The matrices of the explicit rank-two identities ([subsection 44.5](#)) are $\mathbf{N}_1, \dots, \mathbf{N}_4$ and $\mathbf{G}_1, \dots, \mathbf{G}_6, \mathbf{G}_1^-, \mathbf{G}_2^-, \mathbf{H}_1, \mathbf{H}_2$ (source 12's N, M, L, K ; $A, \dots, F, A_-, B_-, H_A, H_B$), because $A, B, C, D, E, F, H, K, L, M$ and N all name rings, groups, fields or subgroups here.
- The congruence kernel of [proposition 39.5](#) is $P_{\Phi}\langle a \rangle$, extending $P_n\langle a \rangle$ of [section 19](#) (source 12's N). A cardinal below κ is ν (its μ).

Three readings must be kept apart. $2\Pi_k = \Pi_k$ does *not* give $2 \in A^{\times}$: $\omega/2 \in \mathbf{Oz}$, but $1/2 \notin \mathbf{Oz}$. “No A_1 component” is *not* “rank at least two”: $A_1 \times A_1$ has rank two and fails ([theorem 41.1](#)). And ct is the constant coefficient, not the standard part ([subsection 1.4](#)).

35.3 Results of source 12

Source 12 states four principal theorems. In this report they are:

- (A) *Higher-rank arithmetic quotient* (for every unital $D \subseteq k$): if Φ has no A_1 component, every homomorphism from $P_{\Phi}(A)$ to a set-sized group is trivial, and $\text{ct} : E_{\Phi}(A) \rightarrow E_{\Phi}(D)$ is a split universal set-sized quotient; the same holds for $\text{St}_{\Phi}(A) \rightarrow \text{St}_{\Phi}(D)$ ([theorems 38.2](#) and [40.1](#) and [corollary 38.3](#)). The root reconstruction needs only $2\Pi_k = \Pi_k$.
- (B) *Exact root-component boundary* (for $D \in \{\mathbb{Z}, \mathbb{Z}[i]\}$): $E_{\Phi}(A)$ has a universal set-sized quotient if and only if Φ has no A_1 component, and then it is $E_{\Phi}(D)$ ([theorem 41.1](#)).
- (C) *Lie-ring classification* (for every unital $D \subseteq k$): the current Lie ring $\mathfrak{l}(A)$ has a universal set-sized Lie-ring quotient if and only if $\mathfrak{l}_{\mathbb{Q}}$ is perfect, and then it is $\mathfrak{l}(D)$ ([theorems 42.3](#) and [42.6](#)). In particular $\mathfrak{sl}_2(\mathbf{Oz}) \rightarrow \mathfrak{sl}_2(\mathbb{Z})$ is universal although $E_2(\mathbf{Oz})$ has no universal set-sized quotient.
- (D) *Transfer of detection thresholds*: for a split ring $D \oplus I$ with $2I = I$ and every infinite cardinal κ , if no nonzero ring homomorphism from I has a target of cardinality $< \kappa$, no nonidentity element of P_{Φ} has a detecting group of cardinality $< \kappa$; if $|A| = \kappa$ the least detecting group of each has cardinality exactly κ ([theorems 43.3](#) and [43.5](#) and [corollary 43.4](#)).

It adds a ring-group equivalence for Dorroh unitizations ([theorem 38.5](#)), the structure of the relative kernel ([section 39](#)), other isogeny forms ([proposition 40.3](#)), explicit rank-two matrices ([subsection 44.5](#)) and twelve questions ([section 45](#)). The merge adds exact thresholds for every root system without A_1 component in the set-sized rings of [section 21](#) ([corollary 43.7](#)), the same quotient theorems for countably or boundedly supported coefficients ([corollary 44.1](#)), and a comparison with [Part I](#) ([remark 42.8](#)).

35.4 Printed once, credited, and second routes

Source 12 credits the report’s antecedents and reproves what it uses. The following are not reprinted.

- Its normal-form facts (its Propositions 3.1, 3.2 and the example $\omega/2 \in \mathbf{Oz}$, $1/2 \notin \mathbf{Oz}$) are [lemma 2.1](#) and [proposition 2.4](#).
- Its scalar theorem (its Theorem 4.3: every ring homomorphism from Π_k to a set-sized ring is zero, and every one from A factors through ct) is `osq:thm:universal(ii)`, (iii) of the sibling report for $K \in \{\mathbb{R}, \mathbb{C}\}$. Its proof is the scaled-field collision of that report: its exponent compression (its Lemma 4.1) is Φ_δ of [lemma 8.3](#); its collision data (its Lemma 4.2: a set field L with $|L| > \nu$, compressed as in [lemma 8.4](#), a positive monomial $w = \omega^{\delta M}$ with $wL \subseteq \Pi_{\mathbb{R}}$, and $a = w^2 v$ with $v \in \Pi_k$) feed `osq:lem:collision(i)` with $m = w$, which kills w^2 and hence a . Source 07’s variant ([lemma 8.5](#)) reaches the prescribed parameter in one step instead of through w^2 . Its remark that a fixed set-sized Hahn ring is a counterexample (its Remark 4.4, the identity map) is in [section 12](#).
- For $\Phi = A_{n-1}$, $n \geq 3$, its main theorem is [theorem 9.1\(iii\)](#), and its proof there coincides with the third route of [section 9](#) (source 11): the root-kernel reconstruction [theorem 9.2](#) with the ring theorem. It is therefore not a fourth independent proof of [theorem 9.1](#). Its general statements extend, in order, [theorem 9.2](#) (to [theorems 37.2](#) and [37.4](#)), [lemma 7.1](#) (to [lemma 38.1](#)), [remark 9.6](#) and source 11’s converse (to [theorem 38.5](#)), [theorems 19.1](#) and [19.2](#), [corollary 19.3](#), and [proposition 19.4](#) (to [proposition 39.3](#) and [corollary 39.4](#)), [theorem 20.2](#) (to [theorem 40.1](#)), [corollary 9.3\(ii\)](#) and [proposition 21.11](#) (to [theorem 43.3](#) and [corollary 43.4](#), now for every infinite cardinal).
- In rank one, its amalgam theorem (its Theorem 10.1) is [theorem 13.2](#) (07); its proof, by the leading term of a reduced word, is genuinely different from 07’s projective separation and is kept as a second route ([section 41](#)). Its free-product description of the kernel (its Corollary 10.2) is [theorem 14.1](#); its equivariant additive functional (its Lemma 10.3) is [lemma 15.1](#) with values in K_D instead of $\mathbb{Q}$; its residual countability and the nonexistence of a universal quotient in rank one (its Theorem 10.4) are [theorem 15.3](#) and [corollary 15.5](#). Its countable target $G_D *_{C_D} (K_D \rtimes C_D)$ is, for $K_D = \mathbb{Q}$, isomorphic to 07’s target ([13.3](#)); the two routes use the same mechanism.
- Its ring collision criterion (its Proposition 12.2) is printed as [proposition 43.2](#); it is `osq:lem:collision(i)` followed by the factorization $a = w^2 v$, and [proposition 9.7](#) (07) is its group analogue.
- Its finite-support boundary is [proposition 12.2](#) (06, 11), its visible subgroups are source 11’s non-claim 11.16 and the characters of [section 6](#), and its distinction of ct from the standard part is [subsection 1.4](#); they are recalled in [section 44](#) for all root systems.

Source 12’s search of the repository for “Chevalley” is stale: the word occurs twice at its pin, in [question 34.3](#) and its status note, where sources 07 and 11 had asked for exactly this extension ([Appendix A.3](#)).

36 Universal quotients and two obstructions

The foundations are those of [subsection 1.3](#). Source 12 works in Gödel–Bernays class theory whose set part satisfies ZFC; global choice is harmless but unused, no inaccessible cardinal or

universe is assumed, and a class homomorphism may use arbitrary class parameters, so no definability is imposed.

Definition 36.1 ([12]). A *universal set-sized quotient* of a class group G is a surjective class homomorphism $q : G \rightarrow Q$ onto a set-sized group such that every class homomorphism $f : G \rightarrow H$ into a set-sized group factors uniquely as $f = \bar{f} \circ q$. The same definition is used for rings and Lie rings. A group is *set-invisible* if every homomorphism from it to a set-sized group is trivial; a ring without unit or a Lie ring is set-invisible if every homomorphism from it to a set-sized ring, respectively Lie ring, is zero.

A universal quotient, when it exists, is unique up to a unique compatible isomorphism; [theorem 9.1\(iv\)](#) says that it is one for $E_n(A)$, $n \geq 3$. Displays $\text{Hom}(G, H)$ with a proper-class source abbreviate the factorization statement; no set or class whose members are proper classes is formed (12).

Lemma 36.2 (Descent to quotients [12]). *If a class group G has a universal set-sized quotient and B is a quotient of G by a normal subclass N , then B has a universal set-sized quotient. The same holds for Lie rings, with ideals.*

Proof. Let $q : G \rightarrow Q$ be universal. As q is onto, $q(N)$ is normal in Q , and $B \rightarrow Q/q(N)$ is well defined and onto. A homomorphism from B to a set-sized group pulls back to one from G that kills N ; its factor through Q kills $q(N)$ and so factors uniquely through $Q/q(N)$. $\square$

Lemma 36.3 (Residual detection obstruction [12]). *A proper-class group in which every nonidentity element survives some homomorphism to a set-sized group has no universal set-sized quotient. The same holds for Lie rings.*

Proof. The kernel of a universal quotient lies in every kernel of a map to a set-sized target, hence is trivial; the proper class would inject into a set, which replacement applied to the inverse on the image excludes. $\square$

The proof of [corollary 15.5](#) (07) is this argument for $E_2(A)$. In the reconstructions below no family of class-sized cosets is collected into a set: the parameters are first mapped into a finite power of an actual set, the image is the carrier, and the ring operations are shown to descend to it. For Steinberg groups the words and Scott's trick of [section 20](#) are used (11, 12).

37 Root identities and the root-kernel ideal

Let I be a commutative ring without unit, possibly a class, with $2I = I$, and let Σ be a group generated by additive maps $x_\alpha : (I, +) \rightarrow \Sigma$, $\alpha \in \Phi$, satisfying the Chevalley commutator relations used below; the subgroup of $E_\Phi(A)$ generated by the $x_\alpha(\Pi_k)$ is the example. The commutator is $[x, y] = xyx^{-1}y^{-1}$ as in [table 1](#); the signs $\pm$ below depend on the pinning, are units, and play no role. The integral identities are classical [6, 8]; matrix realizations of the nonsimply laced cases are in [subsection 44.5](#).

Type A_2 . For roots α, β spanning an A_2 subsystem with $\alpha + \beta$ a root,

$$[x_\alpha(r), x_\beta(s)] = x_{\alpha+\beta}(\pm rs), \quad (37.1)$$

and no other positive combination of α, β is a root; the same holds in every closed A_2 subsystem.

Type $B_2 = C_2$. For short roots α, β with long sum,

$$[x_\alpha(r), x_\beta(s)] = x_{\alpha+\beta}(\pm 2rs); \quad (37.2)$$

for a long root α and a short root β with $\alpha + \beta$ short and $\alpha + 2\beta$ long,

$$[x_\alpha(r), x_\beta(s)] = x_{\alpha+\beta}(\pm rs) x_{\alpha+2\beta}(\pm rs^2). \quad (37.3)$$

Type G_2 . The long roots form a closed A_2 subsystem. For short roots α, β at angle 120° , with $\alpha + \beta$ short, the other positive combinations that are roots are the long roots $2\alpha + \beta$ and $\alpha + 2\beta$, and

$$[x_\alpha(r), x_\beta(s)] = x_{\alpha+\beta}(\pm 2rs) x_{2\alpha+\beta}(\pm 3r^2s) x_{\alpha+2\beta}(\pm 3rs^2). \quad (37.4)$$

This is the short–short formula, not the longer one for a simple short root paired with a simple long root; all its error terms lie in long-root subgroups, which the A_2 relations handle first, and no division by 3 is needed (12).

Lemma 37.1 (Coverage of all roots [12]). *Every root of an irreducible reduced root system of rank at least two is the target $\alpha + \beta$ of one of the configurations (37.1)–(37.4), as follows.*

Type	Roots covered by A_2	Remaining configuration
$A_\ell, D_\ell, E_6, E_7, E_8$	all roots	none
$B_\ell, \ell \geq 3$	long roots	(37.3) for short roots
$C_\ell, \ell \geq 3$	short roots	(37.2) for long roots
$B_2 = C_2$	none needed	long roots first, then short
F_4	both lengths	none
G_2	long roots	(37.4) for short roots

Proof. In a simply laced irreducible system of rank at least two, adjacent simple roots span an A_2 subsystem and the Weyl group is transitive on roots. In orthonormal coordinates the long roots $\pm \varepsilon_i \pm \varepsilon_j$ of B_ℓ lie in A_2 subsystems for $\ell \geq 3$, and a short root such as ε_1 is the sum of the long root $\varepsilon_1 - \varepsilon_2$ and the short root ε_2 , with error root $\varepsilon_1 + \varepsilon_2$. In C_ℓ the short roots $\pm \varepsilon_i \pm \varepsilon_j$ lie in A_2 subsystems for $\ell \geq 3$, and $2\varepsilon_1 = (\varepsilon_1 - \varepsilon_2) + (\varepsilon_1 + \varepsilon_2)$ is the long configuration. In rank two these are the two B_2 cases. The long roots of F_4 contain D_4 ; for the short roots, $\alpha = \frac{1}{2}(\varepsilon_1 + \varepsilon_2 + \varepsilon_3 + \varepsilon_4)$ and $\beta = \frac{1}{2}(-\varepsilon_1 - \varepsilon_2 - \varepsilon_3 + \varepsilon_4)$ have sum ε_4 , and $\pm\alpha, \pm\beta, \pm(\alpha + \beta)$ form a closed A_2 subsystem. The Weyl group is transitive on each root length. In G_2 the long roots form A_2 , and transitivity on short roots moves the short–short configuration to any short root. Transporting a configuration is a statement about the finite root system and its identities; it does not assume that the kernel of a homomorphism is Weyl invariant. $\square$

Theorem 37.2 (Root-kernel ideal [12]). *Let I and Σ be as above and suppose that Φ has no A_1 component. For a homomorphism $\varphi : \Sigma \rightarrow H$ put*

$$J_\varphi = \{a \in I : \varphi(x_\alpha(a)) = 1 \text{ for every } \alpha \in \Phi\}.$$

Then J_φ is an ideal of I .

Proof. Additivity of the root maps makes J_φ an additive subgroup. Let $r \in J_\varphi$ and $s \in I$; every root image of rs must be shown trivial. *A_2 target:* by (37.1) the root image of $\pm rs$ is the image of a commutator with a trivial entry; additive inverses remove the sign. *Long target in B_2 :* choose $t \in I$ with $2t = s$ and apply (37.2) to r, t ; this holds for every long root and every

second parameter in I . *Short target in B_2* : in (37.3) with parameters r, s the commutator has trivial image, and so does its long factor, whose parameter is $\pm r \cdot s^2$ with $s^2 \in I$, by the long case; hence so does the short factor. *Long target in G_2* : the long-root A_2 subsystem kills the long root images of rt for every $t \in I$. *Short target in G_2* : choose t with $2t = s$ and apply (37.4) to r, t . The long parameters are integer multiples of $r(rt)$ and $r(t^2)$, killed by the long case and additivity; so the short factor, with parameter $\pm 2rt = \pm rs$, is killed. By lemma 37.1 these cases reach every root of every component, so $rs \in J_\varphi$, and commutativity gives a two-sided ideal. $\square$

Remark 37.3 (Where divisibility is used [12]). The proof never infers $r/2 \in J_\varphi$ from $r \in J_\varphi$, which could fail when H has elements of order two; it halves only the unrestricted second parameter, which lies in I . So $2I = I$ suffices and no torsion condition on H is hidden. The factor 3 in G_2 multiplies parameters already killed. In the simply laced case no divisibility is used at all, as in theorem 9.2 for type A .

Theorem 37.4 (Set-image reconstruction [12]). *In the setting of theorem 37.2, let H be a set-sized group. There are a set-sized commutative ring R_φ , possibly without unit, and a surjective ring homomorphism $\vartheta_\varphi : I \rightarrow R_\varphi$ with kernel J_φ and $|R_\varphi| \leq |H^\Phi|$. If every ring homomorphism from I to a set-sized ring is zero, then φ is trivial.*

Proof. Put $\vartheta_\varphi(a) = (\varphi(x_\alpha(a)))_{\alpha \in \Phi} \in H^\Phi$ and $R_\varphi = \vartheta_\varphi(I)$, an actual subset of the set H^Φ . By additivity $\vartheta_\varphi(a) = \vartheta_\varphi(b)$ iff $a - b \in J_\varphi$. Define $\vartheta_\varphi(a) + \vartheta_\varphi(b) = \vartheta_\varphi(a + b)$, $\vartheta_\varphi(a)\vartheta_\varphi(b) = \vartheta_\varphi(ab)$ and $-\vartheta_\varphi(a) = \vartheta_\varphi(-a)$; multiplication is well defined because $ab - a'b' = (a - a')b + a'(b - b') \in J_\varphi$ by theorem 37.2, and the axioms are inherited from I . This multiplication is transported from I ; it is not a product in the group H^Φ . If I is set-invisible, $\vartheta_\varphi = 0$, so every root generator is killed and φ is trivial. $\square$

For $\Phi = A_{n-1}$ this is theorem 9.2 (11). The theorem turns an arbitrary group homomorphism into a ring obstruction without asserting that the homomorphism comes from a ring map (12).

38 Elementary Chevalley groups over omnific rings

Lemma 38.1 (The relative kernel is a normal closure [12]). *Let $B = D_0 \oplus I$ be a commutative unital ring with a ring retraction onto D_0 and kernel I . For every Φ ,*

$$\ker(E_\Phi(B) \rightarrow E_\Phi(D_0)) = \langle\langle x_\alpha(a) : \alpha \in \Phi, a \in I \rangle\rangle_{E_\Phi(B)}.$$

Proof. The normal closure N' lies in the kernel. Since $x_\alpha(d + a) = x_\alpha(d)x_\alpha(a)$, every finite word is congruent modulo N' to a word in constant root elements; for a word in the kernel, that constant word is the identity in $E_\Phi(D_0)$, hence in $E_\Phi(B)$ under the split inclusion. $\square$

This is lemma 7.1 (07, 11) for every root system, by the same finite-word replacement.

Theorem 38.2 (Intrinsic higher-rank invisibility [12]). *Let $D \subseteq k$ be any unital subring, $A = \mathcal{A}_{D,k}$, and let Φ have no A_1 component. Every class homomorphism $\varphi : P_\Phi(A) \rightarrow H$ into a set-sized group is trivial.*

Proof. Fix $g \in E_\Phi(A)$. The elements $gx_\alpha(a)g^{-1}$, $a \in \Pi_k$, lie in $P_\Phi(A)$ and satisfy the same root relations. Restrict φ to the subgroup they generate and apply theorem 37.4 with $I = \Pi_k$: $2\Pi_k = \Pi_k$ by proposition 2.4 (division by 2 keeps supports), and Π_k is set-invisible by osq:thm:universal(iii) (subsection 35.4). So the restriction is trivial for every g , and these

conjugated subgroups generate $P_\Phi(A)$ by [lemma 38.1](#). Neither normality of the unconjugated root-generated subgroup nor an extension of φ to $E_\Phi(A)$ is used. $\square$

Corollary 38.3 (Arithmetic quotient and split extension [12]). *Under the hypotheses of [theorem 38.2](#), $\text{ct} : E_\Phi(A) \rightarrow E_\Phi(D)$ is a universal set-sized quotient, $E_\Phi(A) \cong P_\Phi(A) \rtimes E_\Phi(D)$, and for every set-sized group H composition with ct is a bijection $\text{Hom}(E_\Phi(D), H) \rightarrow \text{Hom}(E_\Phi(A), H)$.*

Proof. A homomorphism from $E_\Phi(A)$ to H kills $P_\Phi(A)$ by restriction and [theorem 38.2](#), so it factors through ct , which is split by the constant root elements; surjectivity gives uniqueness. $\square$

Example 38.4 (Exceptional omnific groups [12]). For $\Phi \in \{G_2, F_4, E_6, E_7, E_8\}$ every abstract homomorphism $E_\Phi(\mathbf{Oz}) \rightarrow H$ into a set-sized group factors through $E_\Phi(\mathbb{Z})$, and every one from $E_\Phi(\mathbf{Oz}[i])$ through $E_\Phi(\mathbb{Z}[i])$. Neither algebraicity of the map, linearity of H nor compatibility with a valuation is assumed. These arithmetic groups are countable, so every set-sized image of the omnific group is countable, however large the target.

Theorem 38.5 (Ring-group transfer equivalence [12]). *Let I be a commutative ring without unit, possibly a proper class, with $2I = I$, and let $A_I = \mathbb{Z} \rtimes I$ be its Dorroh unitization. For a fixed Φ without A_1 component the following are equivalent:*

1. *every ring homomorphism from I to a set-sized ring is zero;*
2. *every homomorphism from $P_\Phi(A_I) = \ker(E_\Phi(A_I) \rightarrow E_\Phi(\mathbb{Z}))$ to a set-sized group is trivial.*

If they hold for one such Φ , they hold for all.

Proof. (i) $\Rightarrow$ (ii) is the proof of [theorem 38.2](#), which uses only (i), the split retraction and $2I = I$. Conversely let $f : I \rightarrow S$ be a nonzero ring homomorphism into a set-sized ring; replace S by the commutative ring $f(I)$. Then $(n, a) \mapsto (n, f(a))$ is a unital ring homomorphism $\mathbb{Z} \rtimes I \rightarrow \mathbb{Z} \rtimes S$, and functoriality gives a homomorphism $E_\Phi(\mathbb{Z} \rtimes I) \rightarrow E_\Phi(\mathbb{Z} \rtimes S)$ into a set-sized group. If $f(a) \neq 0$, the relative root element $x_\alpha((0, a))$ maps to $x_\alpha((0, f(a))) \neq 1$, since root parametrizations are injective over every ring. Condition (i) does not mention Φ . $\square$

The converse is stated for the integral unitization on purpose: a nonunital map on I need not be D -linear for a larger constant ring D , so extending it by the identity on D would not be justified (12). For $\Phi = A_{n-1}$, $n \geq 3$, and $I = \Pi_k$, the two implications are [remark 9.6](#) (the merge's) and source 11's converse recorded there; the theorem proves them for every class ring I with $2I = I$ and every root system without A_1 component. Since $\mathcal{A}_{\mathbb{Z},k} = \mathbb{Z} \oplus \Pi_k$ is the Dorroh unitization of Π_k , it applies to $\mathcal{A}_{\mathbb{Z},k}$, for instance to $\mathbf{Oz}$; like [remark 9.6](#) it is a logical equivalence, not an independent proof of either side.

39 The relative kernel: perfectness, torsion, center and non-simplicity

In this section Φ is nonempty without A_1 component and $A = \mathcal{A}_{D,k}$.

Proposition 39.1 (Perfectness and rootwise commutator bounds [12]). *The subgroup $\Sigma = \langle x_\alpha(\Pi_k) : \alpha \in \Phi \rangle$ and its normal closure $P_\Phi(A)$ are perfect. Each root element $x_\alpha(a)$, $a \in \Pi_k$, is a product of at most three commutators of elements of Σ : one for an A_2 target or a long B_2 target, two for a short B_2 target, three for a short G_2 target.*

Proof. Write $a = bc$ with $b, c \in \Pi_k$ ([proposition 2.4](#)). Then [\(37.1\)](#) gives one commutator for an A_2 target, and [\(37.2\)](#) with second parameter $c/2$ one for a long B_2 target. By [\(37.3\)](#) a short target is one commutator times the inverse of a long root element, itself one commutator. In G_2 , [\(37.4\)](#) with second parameter $c/2$ gives a short root element as one commutator times two long root elements, each one commutator. Signs and inverses do not change these counts. So the generators of Σ lie in $[\Sigma, \Sigma]$; conjugates of a perfect subgroup are perfect, and they generate $P_\Phi(A)$ ([lemma 38.1](#)), as in [lemma 9.4](#). $\square$

Remark 39.2 (Rootwise, not global [12]). The bound three concerns a root element, not an arbitrary element of $P_\Phi(A)$, which may need many conjugated root factors. No bounded commutator width and no bounded generation follow ([question 45.7](#)).

Proposition 39.3 (Torsion and center [12]). *$P_\Phi(A)$ is torsion-free and centerless, and it is a nontrivial proper-class group.*

Proof. Use the faithful representation $G_\Phi \hookrightarrow \mathrm{GL}_{N, \mathbb{Z}}$ of [subsection 35.1](#). If $g \in P_\Phi(A)$ has finite order, its matrix is diagonalizable over $\mathbf{No}[i]$, its minimal polynomial dividing some $X^m - 1$, and its eigenvalues are ordinary roots of unity. So its characteristic polynomial has ordinary coefficients; these lie in A , hence equal their constant terms ([lemma 2.1](#)), and $\mathrm{ct}(g) = 1$ makes the polynomial $(X - 1)^N$. A diagonalizable matrix with this characteristic polynomial is 1. (Since ct commutes with the representation, $P_\Phi(A)$ also lies in the constant-term kernel of $\mathrm{GL}_N(A)$, which is torsion-free by [theorem 19.1](#), source 11's trace argument.)

Let g be central. It commutes with $x_\alpha(a)$ for all $a \in \Pi_k$ and all α . The entries of $g x_\alpha(T) - x_\alpha(T)g$ are polynomials in T with coefficients in the field F vanishing at the infinitely many points of Π_k , hence zero; so g commutes with every root subgroup over F . The root subgroups generate G_Φ , so g lies in the center of G_Φ , a finite group scheme whose points in characteristic zero have ordinary algebraic coordinates. Thus g is an ordinary constant point, and $\mathrm{ct}(g) = 1$ gives $g = 1$. Finally the $x_\alpha(\omega^\eta)$, $\eta > 0$, are a proper class of distinct elements of $P_\Phi(A)$. $\square$

For $\Phi = A_{n-1}$ this is [theorems 19.1 and 19.2 \(11\)](#), whose center argument compares matrix entries with the $e_{ij}(p)$; the argument above replaces that by the center of the group scheme.

Corollary 39.4 (Normal subgroups, conjugacy classes, finite subgroups [12]). *$P_\Phi(A)$ has no nontrivial set-sized normal subgroup, the conjugacy class of each of its nonidentity elements is a proper class, and every finite subgroup of $E_\Phi(A)$ is mapped injectively by ct .*

Proof. Conjugation on a set-sized normal subgroup S is a homomorphism $P_\Phi(A) \rightarrow \mathrm{Aut}(S)$, trivial by [theorem 38.2](#), so S is central and trivial. A set-sized conjugacy class would give a trivial action on it, so its element would be central. A finite subgroup meets the torsion-free kernel trivially. $\square$

This is [corollary 19.3](#) and [proposition 19.4 \(11\)](#) for every Φ without A_1 component.

Proposition 39.5 (Nonsimplicity [12]). *The subgroup*

$$P_\Phi \langle 1 + \omega \rangle = P_\Phi(A) \cap \ker(G_\Phi(A) \rightarrow G_\Phi(A/(1 + \omega)A))$$

is a nontrivial proper normal subgroup of $P_\Phi(A)$; in particular the nontrivial perfect group $P_\Phi(A)$ is not simple.

Proof. Normality is clear. The subgroup contains $x_\alpha((1+\omega)\omega) \neq 1$, since $(1+\omega)\omega \in \Pi_k \cap (1+\omega)A$. It does not contain $x_\alpha(\omega)$: if $\omega \in (1+\omega)A$, the quotient $\omega/(1+\omega) = 1 - 1/(1+\omega)$ in the ambient field would lie in A , but it is finite and not an ordinary constant (lemma 2.1(iii)); and root parametrizations are injective over the ring $A/(1+\omega)A$, which is used only through this congruence relation, as in section 19. Only properness of this one congruence subgroup is proved; no ideal or normal subgroup is classified. $\square$

For $\Phi = A_{n-1}$ the subgroup is $P_n\langle 1+\omega \rangle$ of theorem 19.5 (11), and theorem 10.1 (06) gives a separating proper-class tower there.

Remark 39.6 (Actions, representations and finite quotients [12]). Every action of $E_\Phi(A)$ on a set factors through $E_\Phi(D)$, since $\text{Sym}(X)$ is a set, and every representation on a set-sized vector space over a set-sized field factors through constants. There is no conflict with the faithful representation over F , whose field, space and matrix group are proper classes. The finite quotients and finite-index normal subgroups of $E_\Phi(A)$ are exactly those of $E_\Phi(D)$ composed with ct , so the set-sized profinite completions agree; identifying this completion with a congruence completion is a separate arithmetic theorem about D and Φ , and no congruence subgroup claim, Gaussian or other, is made. For $\Phi = A_{n-1}$ these are corollary 9.5 and theorem 16.4, and for $D = \mathbb{Z}$ the completion is identified in corollary 19.7 with the Bass–Milnor–Serre theorem.

40 Steinberg groups and other isogeny forms

Theorem 40.1 (Steinberg arithmetic quotient [12]). *Let Φ have no A_1 component and $A = \mathcal{A}_{D,k}$. Every homomorphism from $P_\Phi^{\text{St}}(A)$ to a set-sized group is trivial, $\text{ct} : \text{St}_\Phi(A) \rightarrow \text{St}_\Phi(D)$ is a split universal set-sized quotient, and $P_\Phi^{\text{St}}(A)$ is perfect.*

Proof. The constant inclusion and ct respect the Steinberg presentation and split each other, and the finite-word replacement of lemma 38.1 shows that $P_\Phi^{\text{St}}(A)$ is the normal closure of the symbols $\tilde{x}_\alpha(a)$, $a \in \Pi_k$, as in lemma 20.1. Conjugated relative symbols satisfy the relations used in theorem 37.2, so theorem 37.4 and the ring theorem kill every set-sized image of each conjugated relative subgroup; these generate the kernel. Factorization follows as in corollary 38.3, and the proof of proposition 39.1 applies verbatim to the symbols. $\square$

For $\Phi = A_{n-1}$ this is theorem 20.2 (11).

Remark 40.2 (Not a computation of K_2 [12]). $P_\Phi^{\text{St}}(A)$ is the kernel of the constant term, not of $\text{St}_\Phi(A) \rightarrow E_\Phi(A)$, and must not be called $K_2(A)$. No torsion-freeness or centerlessness is claimed for it. A subgroup of a set-invisible group need not be set-invisible (subsection 44.2), so no statement about K_2 follows by restriction (compare remark 20.3).

Proposition 40.3 (Other split semisimple forms [12]). *Let $G_{\text{sc}} \rightarrow G$ be a central isogeny from $G_\Phi = G_{\text{sc}}$ to another split semisimple group scheme with root system Φ , and let $E_G(B)$ be generated by the images of the root subgroups. The relative kernel of $\text{ct} : E_G(A) \rightarrow E_G(D)$ is isomorphic to $P_\Phi(A)$. If Φ has no A_1 component, $\text{ct} : E_G(A) \rightarrow E_G(D)$ is a universal set-sized quotient.*

Proof. $E_{\text{sc}}(A) \rightarrow E_G(A)$ is onto by definition. Its kernel consists of points of the finite central kernel of the isogeny, whose coordinates in the characteristic-zero field F are ordinary algebraic numbers, hence lie in D (lemma 2.1); such a point of $E_{\text{sc}}(A)$ equals its constant term, which lies in $E_{\text{sc}}(D)$. So this kernel lies in the constant subgroup and meets $P_\Phi(A)$ trivially. If $y \in E_G(A)$ has $\text{ct}(y) = 1$, lift it to $x \in E_{\text{sc}}(A)$; then $\text{ct}(x)$ maps to 1, and $x \text{ct}(x)^{-1} \in P_\Phi(A)$

still maps to y . So $P_\Phi(A)$ maps isomorphically onto the relative kernel, and [theorem 38.2](#) gives the quotient as in [corollary 38.3](#). $\square$

The proposition concerns root-generated groups in these forms, not all points of the group scheme, and it asserts no rank-one converse for central quotients other than the simply connected one: [theorem 41.1](#) is stated in the simply connected elementary category (12).

41 Rank one and the componentwise classification

Here $(D, k) \in \{(\mathbb{Z}, \mathbb{R}), (\mathbb{Z}[i], \mathbb{C})\}$, $A = \mathcal{A}_{D,k}$, and $G_D = \mathrm{SL}_2(D) = E_2(D)$, C_D , B_D , $U(a) = e_{12}(a)$ are as in [section 13](#); by [\(13.1\)](#), $B_D = U(\Pi_k) \rtimes C_D$, a constant upper triangular matrix with diagonal u, u^{-1} acting by u^2 . The rank-one theory is source 07's ([sections 13 to 15](#)); source 12 reproves it to see where the higher-rank argument breaks, with a different proof of the amalgam theorem.

Second proof of [theorem 13.2](#): the leading term [12]. Surjectivity of $G_D *_{C_D} B_D \rightarrow E_2(A)$ is generation. For injectivity, the reduced-word normal form of the amalgam and $B_D = U(\Pi_k) \rtimes C_D$ bring a word not in G_D to the form

$$g_0 U(a_1) g_1 U(a_2) \cdots g_{m-1} U(a_m) g_m, \quad (41.1)$$

with $m \geq 1$, all $a_j \in \Pi_k \setminus \{0\}$, and interior $g_j \in G_D \setminus C_D$ ($1 \leq j < m$), so that $(g_j)_{21} \neq 0$. Expand each $U(a_j) = 1 + a_j E_{12}$. Since $E_{12} g_j E_{12} = (g_j)_{21} E_{12}$, the term using $a_j E_{12}$ from every factor is

$$\left(\prod_{j=1}^m a_j \right) \left(\prod_{j=1}^{m-1} (g_j)_{21} \right) g_0 E_{12} g_m, \quad (41.2)$$

a nonzero matrix whose entries have leading exponent $\sum_j \deg_\omega a_j > 0$. Every other term omits some a_j , so its entries have strictly smaller leading exponent; the leading coefficient cannot cancel, and the word is not the identity. Nonidentity words in G_D stay nonidentity under the constant inclusion. $\square$

Only finitely many terms of the matrix product are expanded; the parameters may have arbitrary set-sized supports, and no finite-support restriction enters (12). 07's proof (projective separation of finite and infinite points) and this one share only the reduced-word normal form.

Theorem 41.1 (Componentwise classification [12], (ii) [merge]). *Let $(D, k) \in \{(\mathbb{Z}, \mathbb{R}), (\mathbb{Z}[i], \mathbb{C})\}$ and $A = \mathcal{A}_{D,k}$.*

1. $E_\Phi(A)$ has a universal set-sized group quotient if and only if Φ has no A_1 component, and then $\mathrm{ct} : E_\Phi(A) \rightarrow E_\Phi(D)$ is that quotient.
2. $P_\Phi(A)$ is set-invisible if and only if Φ has no A_1 component. If Φ has an A_1 component, every nonidentity element of that factor $P_2 \leq P_\Phi(A)$ is detected by a homomorphism from $P_\Phi(A)$ into the fixed countable group $\mathrm{SL}_2(K_D(t))$.

Proof. The simply connected group is the product of those of the irreducible components, and $E_\Phi(A) = \prod_i E_{\Phi_i}(A)$, since root subgroups of different components commute and generate; likewise $P_\Phi(A) = \prod_i P_{\Phi_i}(A)$. If no component is A_1 , [corollary 38.3](#) and [theorem 38.2](#) apply. If one is, projection onto that factor is a surjection $E_\Phi(A) \rightarrow E_2(A)$; a universal quotient of $E_\Phi(A)$ would descend to $E_2(A)$ ([lemma 36.2](#)), contradicting [corollary 15.5](#). For (ii), compose the projection $P_\Phi(A) \rightarrow P_2$ with the restrictions to P_2 of the maps ρ_ℓ of [theorem 15.3](#). $\square$

Part (i) is source 12's Theorem B; its rank-one input is [corollary 15.5](#), which source 12 reproves through [lemma 36.3](#) with the countable target $G_D *_{C_D} (K_D \rtimes C_D)$ ([subsection 35.4](#)). Part (ii), the kernel form, is immediate and is recorded by the merge because it is the form asked in [question 34.3](#). Total rank is not the criterion: $E_{A_1 \times A_1}(A) = E_2(A) \times E_2(A)$ has no universal set-sized quotient. The root-kernel reconstruction cannot be imitated in A_1 : no third root direction carries the product rs , and the additive coefficient functionals survive into countable amalgam targets (12).

42 Current Lie rings

Let $\mathfrak{l}$ be a Lie lattice with basis $\mathbf{b}_1, \dots, \mathbf{b}_d$. For a class-sized ring I , $\mathfrak{l}(I)$ means the class I^d with coordinatewise addition and the bracket $[\mathbf{b}_i a, \mathbf{b}_j b] = [\mathbf{b}_i, \mathbf{b}_j]ab$ determined by the finitely many integral structure constants; no class-sized tensor construction is used. Targets are arbitrary Lie rings, not necessarily rational Lie algebras, so torsion targets are allowed.

Lemma 42.1 (Clearing the bracket lattice [12]). *If $\mathfrak{l}_{\mathbb{Q}}$ is perfect, there are a positive ordinary integer m and integers z_{lij} with*

$$m \mathbf{b}_l = \sum_{i,j} z_{lij} [\mathbf{b}_i, \mathbf{b}_j] \quad (1 \leq l \leq d). \quad (42.1)$$

Proof. Perfection writes each $\mathbf{b}_l$ as a rational combination of the finitely many brackets $[\mathbf{b}_i, \mathbf{b}_j]$; clear all denominators. Equivalently the subgroup generated by the brackets has full rank, and m annihilates its finite cokernel. $\square$

Theorem 42.2 (Lie-kernel reconstruction [12]). *Let I be a commutative ring without unit with $mI = I$ for an m as in (42.1), and let $\varphi : \mathfrak{l}(I) \rightarrow \mathfrak{H}$ be a Lie-ring homomorphism. Then $J = \{r \in I : \varphi(\mathbf{b}_i r) = 0 \text{ for all } i\}$ is an ideal of I , and if $\mathfrak{H}$ is a set, the tuple image of $r \mapsto (\varphi(\mathbf{b}_1 r), \dots, \varphi(\mathbf{b}_d r)) \in \mathfrak{H}^d$ carries a set-sized commutative ring quotient of I with kernel J .*

Proof. J is an additive subgroup. For $r \in J$, $s \in I$ choose $t \in I$ with $mt = s$; multiplying (42.1) by rt gives

$$\mathbf{b}_l r s = \sum_{i,j} z_{lij} [\mathbf{b}_i r, \mathbf{b}_j t], \quad (42.2)$$

and each term on the right is killed by φ , because $\varphi(\mathbf{b}_i r) = 0$. So $rs \in J$. The carrier is built as in [theorem 37.4](#). $\square$

The identity (42.2) divides by m in the source ring, never in the target, which is why torsion targets are covered; treating φ as $\mathbb{Q}$ -linear would be an unjustified shortcut (12).

Theorem 42.3 (Perfect current Lie rings [12]). *If $\mathfrak{l}_{\mathbb{Q}}$ is perfect, every homomorphism from $\mathfrak{l}(\Pi_k)$ to a set-sized Lie ring is zero, and for every unital $D \subseteq k$ the constant term $\mathfrak{l}(\mathcal{A}_{D,k}) \rightarrow \mathfrak{l}(D)$ is a split universal set-sized Lie-ring quotient.*

Proof. Take m from [lemma 42.1](#); $m\Pi_k = \Pi_k$. The ring quotient of [theorem 42.2](#) is zero by `osq:thm:universal`(iii), so every $\varphi(\mathbf{b}_i r)$ vanishes, and these elements generate $\mathfrak{l}(\Pi_k)$ additively. A homomorphism from $\mathfrak{l}(\mathcal{A}_{D,k})$ to a set-sized Lie ring therefore kills the ideal $\mathfrak{l}(\Pi_k)$ and factors uniquely through the split quotient. $\square$

Corollary 42.4 ([12]). *If $\mathfrak{l}_{\mathbb{Q}}$ is perfect, $\mathfrak{l}(\Pi_k)$ is perfect as a Lie ring.*

Proof. For $a \in \Pi_k$ write $a = bc$ with $b, c \in \Pi_k$ and put $t = c/m$; (42.2) with $r = b$, $s = c$ writes $b_t a$ as an integral sum of brackets. $\square$

Lemma 42.5 (Additive detection [12]). *For every unital $D \subseteq k$, the additive group of $\mathcal{A}_{D,k}$, with zero bracket, is a proper-class abelian Lie ring in which every nonzero element is detected by an additive map into $\mathbb{Q}$.*

Proof. Choose one exponent (possibly 0) with nonzero coefficient; coefficient extraction into the set k , followed by a $\mathbb{Q}$ -linear functional $k \rightarrow \mathbb{Q}$ nonzero there (extend from the rational span of that coefficient along a rational basis of k), detects the element. The positive monomials form a proper class. $\square$

Theorem 42.6 (Nonperfect obstruction [12]). *If $\mathfrak{l}_{\mathbb{Q}}$ is not perfect, $\mathfrak{l}(\mathcal{A}_{D,k})$ has no universal set-sized Lie-ring quotient, for every unital $D \subseteq k$.*

Proof. A nonzero rational functional on $\mathfrak{l}_{\mathbb{Q}}$ vanishing on all brackets exists; clearing denominators and dividing by the gcd gives a surjective homomorphism $\varpi : \mathfrak{l} \rightarrow \mathbb{Z}$ with $\varpi([l, l]) = 0$. Its coefficientwise extension $\mathfrak{l}(\mathcal{A}_{D,k}) \rightarrow (\mathcal{A}_{D,k}, +)$ is a surjective Lie-ring homomorphism onto the abelian Lie ring of lemma 42.5, which has no universal set-sized quotient by lemma 36.3. Lemma 36.2 rules one out for the source. $\square$

Together, theorems 42.3 and 42.6 give source 12's Theorem C for every unital $D \subseteq k$, not only $\mathbb{Z}$ and $\mathbb{Z}[i]$. The criterion is perfection, not semisimplicity: a nonsemisimple perfect rational Lie algebra is covered. The converse uses a genuine surjection onto the abelian additive class and the descent of universal quotients, not merely a nonzero character (12).

Example 42.7 (The rank-one contrast [12]). For $\mathfrak{l} = \mathfrak{sl}_2(\mathbb{Z})$ with $e = E_{12}$, $f = E_{21}$, $h = \text{diag}(1, -1)$,

$$[h, e] = 2e, \quad [h, f] = -2f, \quad [e, f] = h, \quad (42.3)$$

so $m = 2$ works. If r is killed in all three directions and $s \in \Pi_k$, then $ers = [hr, e(s/2)]$, $f rs = -[hr, f(s/2)]$ and $h rs = [er, fs]$ are killed: the three brackets reconstruct multiplication in Lie rank one. Hence $\mathfrak{sl}_2(\mathbf{Oz}) \rightarrow \mathfrak{sl}_2(\mathbb{Z})$ is a universal set-sized Lie-ring quotient, while $E_2(\mathbf{Oz})$ has none (corollary 15.5). There is no contradiction with differentiation or exponentiation: abstract group maps need not differentiate, the additive directions of the amalgam keep many nonalgebraic maps, and no global exponential equivalence between these class groups and Lie rings is assumed.

Remark 42.8 (Unipotent lattices and Part I [merge]). A nilpotent Lie lattice $\mathfrak{l} \neq 0$, for instance the strictly upper triangular matrices $\mathfrak{n}_n(\mathbb{Z})$, $n \geq 2$, is not perfect over $\mathbb{Q}$, so $\mathfrak{n}_n(\mathcal{A}_{D,k})$ has no universal set-sized Lie-ring quotient (theorem 42.6). This is the Lie counterpart of theorem 6.2: the group $\text{UT}_n(\Pi_k)$, the constant-term kernel of the unitriangular group (proposition 6.1), has abelianization $(\Pi_k, +)^{n-1}$ and nonzero coefficient characters. The comparison is by statement only; neither result is derived from the other.

43 Cardinal transfer

The proper-class theorems have set-sized analogues. Source 12 states them as a transfer: the size argument belongs to the coefficient ideal, and the root and bracket reconstructions cost only a finite power of the target.

Definition 43.1 ([12]). Let κ be an infinite cardinal. A commutative ring I without unit is *ring-invisible below κ* if every ring homomorphism from I into a ring of cardinality $< \kappa$ is zero.

Proposition 43.2 (A ring collision criterion [12]). *Let I be a subring, possibly without unit, of a field $\mathbb{E}$. Suppose that for every $0 \neq a \in I$ and every cardinal $\nu < \kappa$ there are a set subfield $L \subseteq \mathbb{E}$ with $|L| > \nu$, an element $w \neq 0$ with $wL \subseteq I$, and $v \in I$ with $a = w^2v$. Then I is ring-invisible below κ .*

Proof. For $\psi : I \rightarrow S$ with $|S| < \kappa$ take the data for $\nu = |S|$. Two of the $wf, f \in L$, have equal images; for $f \neq f'$ with $\psi(wf) = \psi(wf')$, $w/(f - f') \in wL \subseteq I$ and $\psi(w^2) = \psi(w(f - f'))\psi(w/(f - f')) = 0$, so $\psi(a) = \psi(w^2)\psi(v) = 0$. Every element used lies in I . $\square$

This is the scaled-field collision `osq:lem:collision(i)` with $m = w$, followed by the factorization $a = w^2v$; [proposition 9.7](#) (07) is its group-level analogue for $n \geq 3$.

Theorem 43.3 (Group cardinal transfer [12]). *Let $B = D_0 \oplus I$ be a split commutative unital ring with $2I = I$, and suppose that I is ring-invisible below the infinite cardinal κ . If Φ has no A_1 component, every homomorphism from $\ker(E_\Phi(B) \rightarrow E_\Phi(D_0))$, or from $\ker(\text{St}_\Phi(B) \rightarrow \text{St}_\Phi(D_0))$, to a group of cardinality $< \kappa$ is trivial.*

Proof. For each conjugated relative root-generated subgroup the reconstructed ring of [theorem 37.4](#) is a subset of H^Φ , and $|H^\Phi| < \kappa$: it is finite for finite H and has cardinality $|H|$ for infinite H . So it is zero. Conclude as in [theorems 38.2](#) and [40.1](#), using [lemma 38.1](#) and its Steinberg form. $\square$

Corollary 43.4 (Exact detecting size [12]). *If moreover $|B| = \kappa$, then for every nonidentity g in either kernel the least cardinality of a group detecting g is exactly κ .*

Proof. The lower bound is [theorem 43.3](#). $E_\Phi(B)$ is a set of finite words in finitely many root families parameterized by B , of cardinality at most κ , and its identity detects g ; $\text{St}_\Phi(B)$ is a quotient of the free group on at most κ symbols. $\square$

Theorem 43.5 (Lie cardinal transfer [12]). *Let $\mathfrak{l}_\mathbb{Q}$ be perfect and let m satisfy (42.1). If $mI = I$ and I is ring-invisible below κ , every homomorphism from $\mathfrak{l}(I)$ to a Lie ring of cardinality $< \kappa$ is zero. If $|I| = \kappa$, every nonzero element of $\mathfrak{l}(I)$ has least detecting Lie-ring cardinality exactly κ .*

Proof. The reconstructed ring is a subset of $\mathfrak{H}^d$, of cardinality $< \kappa$; the upper bound is the identity of $\mathfrak{l}(I)$, of cardinality κ . $\square$

Remark 43.6 (What the transfer does not assume [12]). No regularity of κ is used in these transfer proofs. Regularity, cofinality and support-length restrictions may be needed to establish the set collision property in a particular Hahn fragment, and must be checked there. An action on a set X is a homomorphism to $\text{Sym}(X)$, of cardinality $2^{|X|}$ for infinite X : the transfer gives triviality when $2^{|X|} < \kappa$, and this cannot be weakened to $|X| < \kappa$ without a further argument (as in [proposition 21.7](#)). For $\Phi = A_{n-1}$, $n \geq 3$, [theorem 43.3](#) is [corollary 9.3\(ii\)](#), stated there for uncountable κ , and [corollary 43.4](#) is [proposition 21.11](#).

Source 12 states the transfer for any verified instance of its hypotheses. The sibling report's field bound supplies one, which the merge records.

Corollary 43.7 (Exact thresholds for every root system [merge]). *Let Φ be a root system without A_1 component, let (D, k) be $(\mathbb{Z}, \mathbb{R})$ or $(\mathbb{Z}[i], \mathbb{C})$, and let $\mathfrak{l}$ be a Lie lattice with $\mathfrak{l}_\mathbb{Q}$ perfect.*

1. Let κ be uncountable and regular, and $R_{<\kappa}$, $\Pi_{<\kappa}$ as in [section 21](#). Every nonidentity element of $P_\Phi(R_{<\kappa}) = \ker(E_\Phi(R_{<\kappa}) \rightarrow E_\Phi(D))$, and every nonidentity element of the Steinberg kernel, has least detecting group of cardinality exactly $\kappa^{<\kappa}$, also for homomorphisms defined on $E_\Phi(R_{<\kappa})$ or $\text{St}_\Phi(R_{<\kappa})$; every homomorphism from $E_\Phi(R_{<\kappa})$ to a group of cardinality $< \kappa^{<\kappa}$ factors uniquely through ct ; and every nonzero element of $\mathfrak{l}(\Pi_{<\kappa})$ has least detecting Lie-ring cardinality exactly $\kappa^{<\kappa}$. For $\kappa = \aleph_1$ the value is $\mathfrak{c}$.
2. The same holds, with $|A_0|$ in place of $\kappa^{<\kappa}$, for each of the sibling report's set-sized rings $A_0 = D_0 \oplus \Pi(\mathbb{F})$ of `osq:thm:thresholds` and their Gaussian versions ([corollary 21.13](#)).

Proof. (i) By [proposition 21.12\(i\)](#), $\Pi_{<\kappa}$ is ring-invisible below $\theta = \kappa^{<\kappa}$, an infinite cardinal, and $|\Pi_{<\kappa}| = |R_{<\kappa}| = \theta$. Division by an ordinary integer keeps supports, so $2\Pi_{<\kappa} = m\Pi_{<\kappa} = \Pi_{<\kappa}$, and $R_{<\kappa} = D \oplus \Pi_{<\kappa}$ is split ([lemma 21.1\(i\)](#)). [Theorem 43.3](#) and [corollary 43.4](#) with κ replaced by θ , which needs no regularity, give the group statements on the kernels. A homomorphism on $E_\Phi(R_{<\kappa})$ not killing g restricts to the kernel, and conversely $E_\Phi(R_{<\kappa}) \leq G_\Phi(R_{<\kappa}) \leq \text{GL}_N(R_{<\kappa})$ has at most θ elements; the split decomposition $E_\Phi(R_{<\kappa}) = P_\Phi(R_{<\kappa}) \rtimes E_\Phi(D)$ gives the factorization; the Steinberg group has at most θ elements. The Lie statement is [theorem 43.5](#) with $I = \Pi_{<\kappa}$. At $\aleph_1$ use [proposition 21.12\(iv\)](#). (ii) The proof of [corollary 21.13](#) shows that every nonzero ring homomorphism from $\Pi(\mathbb{F})$ has image of cardinality at least $|A_0|$ and that $|\Pi(\mathbb{F})| = |A_0|$. As $\mathbb{F}$ is a field of characteristic zero and division by an ordinary integer keeps supports, $\Pi(\mathbb{F})$ is 2- and m -divisible. Argue as in (i). $\square$

This extends [proposition 21.12](#) and [corollary 21.13](#) from $n \geq 3$ to every root system without A_1 component and to perfect current Lie rings, and bears on [questions 34.4](#) and [45.5](#). It rests on the statements of the sibling report used there (`osq:prop:card11`, `osq:lem:collision`, and for (ii) the proof of `osq:thm:thresholds`).

44 Boundaries and explicit rank-two matrices

44.1 Finite support

On the finite-support subring $A^{\text{fin}} = D + \Pi_k^{\text{fin}}$ of [section 12](#), the coefficient sum $\omega^\eta \mapsto 1$ is a ring homomorphism to k , nonzero on Π_k^{fin} ([proposition 12.2](#), 06 and 11); it induces homomorphisms from $E_\Phi(A^{\text{fin}})$ to the set-sized group $E_\Phi(k)$ that detect positive root parameters, for every Φ . So positivity of exponents alone does not give the theorem: infinite supports, and room to fit a large field into a positive-support multiple, are essential. The map is not extended to arbitrary Hahn series by any summation, which would contradict the ring theorem, and a finitely supported computational model must not replace Π_k in the universal statements (12).

Countable supports, however, suffice. For an infinite cardinal λ let $\Pi_k^{<\lambda} = \{f \in \Pi_k : |\text{supp } f| < \lambda\}$ and $\mathcal{A}_{D,k}^{<\lambda} = D + \Pi_k^{<\lambda}$, the bounded-support rings of the sibling report's support threshold `osq:thm:support`.

Corollary 44.1 (Bounded supports [merge]). *Let λ be an uncountable cardinal, $D \subseteq k$ unital, and Φ without A_1 component. Every homomorphism from $\ker(E_\Phi(\mathcal{A}_{D,k}^{<\lambda}) \rightarrow E_\Phi(D))$, or from the corresponding Steinberg kernel, to a set-sized group is trivial, so ct is a universal set-sized quotient of $E_\Phi(\mathcal{A}_{D,k}^{<\lambda})$ and of $\text{St}_\Phi(\mathcal{A}_{D,k}^{<\lambda})$; and if $\mathbb{Q}$ is perfect, $\mathfrak{l}(\mathcal{A}_{D,k}^{<\lambda}) \rightarrow \mathfrak{l}(D)$ is a universal set-sized Lie-ring quotient. For $\lambda = \aleph_0$ all these fail (the last one for $\mathfrak{l} \neq 0$).*

Proof. $I = \Pi_k^{<\lambda}$ is a ring without unit with $2I = mI = I$, and $\mathcal{A}_{\mathbb{Z},k}^{<\lambda} = \mathbb{Z} \oplus I$ is its Dorroh unitization. If $f : I \rightarrow S$ were a nonzero ring homomorphism into a set-sized ring, $(n, a) \mapsto$

$(n, f(a))$ would be a unital homomorphism $\mathcal{A}_{\mathbb{Z},k}^{\leq \lambda} \rightarrow \mathbb{Z} \ltimes S$ not factoring through ct , against `osq:thm:support(i)` with $D = \mathbb{Z}$. So I is set-invisible, and the proofs of [theorems 38.2, 40.1](#) and [42.3](#) and [corollary 38.3](#), which use only this, $2I = I$ (or $mI = I$) and the split retraction $\mathcal{A}_{D,k}^{\leq \lambda} = D \oplus I$, apply. For $\lambda = \aleph_0$ the augmentation above detects $x_\alpha(\omega)$, and its Lie analogue $\mathbf{b}_i \omega \mapsto \mathbf{b}_i$ is a nonzero homomorphism from $\mathfrak{l}(\Pi_k^{\leq \aleph_0})$ to the set-sized Lie ring $\mathfrak{l}(k)$. $\square$

The support-bounded omnific integers O_κ of the report on first κ coefficients (batch 32), the omnific integers with fewer than κ terms, are $\mathcal{A}_{\mathbb{Z},\mathbb{R}}^{\leq \kappa}$; for uncountable κ the corollary makes $E_\Phi(\mathbb{Z})$ the universal set-sized quotient of $E_\Phi(O_\kappa)$ for every Φ without A_1 component.

44.2 Subgroups can have many small images

A single root subgroup $x_\alpha(\Pi_k) \cong (\Pi_k, +)$ has nonzero coefficient maps to k and, after rational functionals, to $\mathbb{Q}$. Set-invisibility of $P_\Phi(A)$ does not pass to subgroups: it is the interaction of root directions through commutators that forces multiplicativity. A subgroup resembling K_2 need not inherit invisibility, a unipotent direction has nontrivial small representations, and an abelian subgroup of a perfect class group can have many set-sized quotients (11.16, 12).

44.3 The constant term and the Gaussian constant ring

$\text{ct} : \mathbf{Oz} \rightarrow \mathbb{Z}$ is not the standard part: $3 + \sqrt{2}\omega^{1/2}$ has constant term 3 and is not finite, and its whole root-group contribution is invisible to set-sized groups. In $\mathbf{Oz}[i]$ the coefficients at positive exponents are arbitrary complex numbers while the constant must lie in $\mathbb{Z}[i]$; the arithmetic quotient is Gaussian although the purely infinite part has the full complex field (12).

44.4 Elementary versus full groups

The theorems concern $E_\Phi(A)$ and its constant-term kernel. A matrix of determinant one over a non-Euclidean ring need not be elementary ([theorem 18.2](#)), so $E_\Phi(A)$ is not replaced by $G_\Phi(A)$ or even by $\text{SL}_n(A)$; the quotient of full by elementary points is where algebraic K -theoretic information would enter ([questions 34.1](#) and [45.1](#)).

44.5 Explicit rank-two matrices

Source 12 fixes signs by exact matrices. For a nilpotent integral matrix $\mathbf{N}$ with integral divided powers, $\exp(t\mathbf{N}) = \sum_j t^j \mathbf{N}^j / j!$ is a polynomial matrix over every commutative ring; the verifier's rational substitutions check identities and do not assume $1/2 \in A$. In dimension three, $[e_{12}(r), e_{23}(s)] = e_{13}(rs)$ (([7.1](#))). In the standard symplectic representation of dimension four put

$$\mathbf{N}_1 = E_{12} - E_{43}, \quad \mathbf{N}_2 = E_{14} + E_{23}, \quad \mathbf{N}_3 = E_{13}, \quad \mathbf{N}_4 = E_{24};$$

then

$$[\exp(r\mathbf{N}_1), \exp(s\mathbf{N}_2)] = \exp(2rs\mathbf{N}_3), \tag{44.1}$$

$$[\exp(r\mathbf{N}_4), \exp(s\mathbf{N}_1)] = \exp(-rs\mathbf{N}_2) \exp(-rs^2\mathbf{N}_3), \tag{44.2}$$

the two steps (37.2), (37.3) used in [theorem 37.2](#), with one long correction in the second. In dimension seven put

$$\begin{aligned} G_1 &= E_{12} + 2E_{34} + E_{45} + E_{67}, & G_2 &= E_{23} + E_{56}, \\ G_3 &= [G_1, G_2], & G_4 &= \tfrac{1}{2}[G_1, G_3], & G_5 &= \tfrac{1}{3}[G_1, G_4], & G_6 &= [G_2, G_5], \end{aligned}$$

integral matrices, explicitly $G_3 = E_{13} - 2E_{24} + E_{46} - E_{57}$, $G_4 = -2E_{14} + E_{25} + E_{36} - E_{47}$, $G_5 = E_{15} - E_{37}$ and $G_6 = -E_{16} - E_{27}$. With G_1 a simple short and G_2 a simple long root, G_1, G_3 is the short-short pair of (37.4), and

$$[\exp(rG_1), \exp(sG_3)] = \exp(2rsG_4) \exp(3r^2sG_5) \exp(-3rs^2G_6), \quad (44.3)$$

$$[\exp(rG_2), \exp(sG_5)] = \exp(rsG_6), \quad (44.4)$$

so the two error factors of (44.3) are long and the 3 is never inverted. With $G_1^- = E_{21} + E_{43} + 2E_{54} + E_{76}$ and $G_2^- = E_{32} + E_{65}$, $H_1 = [G_1, G_1^-] = \text{diag}(1, -1, 2, 0, -2, 1, -1)$ and $H_2 = [G_2, G_2^-] = \text{diag}(0, 1, -1, 0, 1, -1, 0)$ satisfy $[H_1, G_1] = 2G_1$, $[H_1, G_2] = -3G_2$, $[H_2, G_1] = -G_1$, $[H_2, G_2] = 2G_2$; the cross brackets $[G_1, G_2^-]$, $[G_2, G_1^-]$ and $(\text{ad } G_1)^4 G_2$, $(\text{ad } G_2)^2 G_1$ vanish. These finite realizations check signs; the general identities keep their classical proof (12).

45 Questions of source 12

Source 12 proposes these as continuations of its proofs, not as named open problems of the literature. Each is followed by its status in this report.

Question 45.1 (Full Chevalley groups [12]). For which split semisimple G is $\text{ct} : G(\mathbf{Oz}) \rightarrow G(\mathbb{Z})$ a universal set-sized quotient? Determine the contribution of the non-elementary quotient rather than assuming $E_G(\mathbf{Oz}) = G(\mathbf{Oz})$; a first step would be a stable range in which every set-sized image of the non-elementary part is forced to be arithmetic.

Status: open. For $G = \text{SL}_n$, $n \geq 3$, this is [question 34.1](#).

Question 45.2 (Full rank-one points [12]). What are the set-sized images of $\text{SL}_2(\mathbf{Oz})$ and $\text{SL}_2(\mathbf{Oz}[i])$ outside their elementary subgroups? The countable detectors are defined on E_2 and need not extend.

Status: open. $\text{SL}_2(\mathcal{A}_{D,k}) \neq E_2(\mathcal{A}_{D,k})$ by [theorem 18.2](#); the second part of [question 34.1](#) asks the same.

Question 45.3 (Relative K_2 and central extensions [12]). Determine the kernel of $\text{St}_\Phi(\mathbf{Oz}) \rightarrow E_\Phi(\mathbf{Oz})$ and the behaviour of its constant-term map. Which of its elements have set-sized detectors? Restriction of [theorem 40.1](#) does not answer this.

Status: open ([remarks 20.3](#) and [40.2](#)).

Question 45.4 (Twisted and relative root systems [12]). Extend the reconstruction to unitary, twisted and more general isotropic groups. Is there a relative-root criterion for the transfer of scalar invisibility? Nonreduced systems and relative rank one are informative: their extra root parameters may imitate higher-rank identities without an absolute A_2 subsystem.

Status: open.

Question 45.5 (Exact set-sized thresholds [12]). Classify the Hahn fragments satisfying the collision criterion [proposition 43.2](#), and determine exact detecting cardinalities for all Chevalley types without unproved closure or regularity assumptions.

Status: partly answered [merge]. For the rings $R_{<\kappa}$ and the sibling report's five set-sized models the exact values are determined for every root system without A_1 component ([corollary 43.7](#)); other exponent groups and supports are open, as in [question 34.4](#).

Question 45.6 (Normal subgroup structure [12]). Describe the normal subgroups of $P_\Phi(\mathbf{Oz})$ by ideals, levels or admissible pairs of levels for the two root lengths. A classification must account for proper-class congruence phenomena, not only finite quotients.

Status: open. Known: nonsimplicity ([proposition 39.5](#)), no nontrivial set-sized normal subgroup ([corollary 39.4](#)), and for $\Phi = A_{n-1}$ the separating tower and the chain at $1 + \omega$ ([theorems 10.1](#) and [19.5](#)).

Question 45.7 (Global commutator width [12]). Does $P_\Phi(\mathbf{Oz})$ have finite commutator width, perhaps depending only on Φ ?

Status: open. The rootwise bounds of [proposition 39.1](#) do not answer it; bounded relative root generation or a class-valued obstruction would be a step.

Question 45.8 (Recovering coefficients from abstract groups [12]). Can an abstract isomorphism between omnific elementary groups recover the coefficient ring, its constant-term ideal, or the root datum?

Status: open. The tuple construction recovers a ring quotient from a homomorphism, not the source ring; recovery of coefficient algebra from root commutators has substantial precedent [\[18\]](#).

Question 45.9 (Arithmetic completions [12]). For each D and Φ , determine the profinite completion of $E_\Phi(D)$, which is that of $E_\Phi(\mathcal{A}_{D,k})$ ([remark 39.6](#)); separate the cases with a congruence subgroup theorem from those with a nontrivial congruence kernel, particularly over $\mathbb{Z}[i]$ and in small rank.

Status: open in general; for $\Phi = A_{n-1}$, $n \geq 3$, and $D = \mathbb{Z}$ the completion is $\mathrm{SL}_n(\widehat{\mathbb{Z}})$ ([corollary 19.7](#), with Bass–Milnor–Serre imported).

Question 45.10 (Homology and the Lie–group contrast [12]). Compare the homology and central extensions of $P_\Phi(\mathbf{Oz})$ with those of $\mathfrak{l}(\Pi_{\mathbb{R}})$. How does the rank-one group obstruction coexist with the perfect $\mathfrak{sl}_2$ Lie-ring quotient? A functorial comparison must specify its topology or formal completion.

Status: open.

Question 45.11 (Optimal divisibility hypotheses [12]). The group reconstruction uses $2I = I$ and the Lie reconstruction $mI = I$. Determine optimal root-type-specific weakenings, and what replaces the theorems in positive characteristic and for ideals with residual 2-torsion.

Status: open. In simply laced types the ideal lemma uses no divisibility ([remark 37.3](#), and [theorem 9.2](#) for type A).

Question 45.12 (Formalization [12]). Formalize the finite root-kernel ideal lemma, the image-carrier ring and the bracket-lattice transfer before the class-sized support arguments; a set-sized form of the collision criterion is a ZFC-compatible target.

Status: open. Nothing in this report is formalized; source 07 proposed [proposition 9.7](#) as a first target.

46 Dependencies and scope of source 12

The Chevalley results follow from three chains:

normal forms and set cuts $\implies$ scaled-field collision
 $\implies$ scalar set-invisibility;
 rank-two Chevalley identities $\implies$ the common root kernel is an ideal
 $\implies$ a small image-carrier ring;
 split retraction, finite-word normal generation
 $\implies$ intrinsic kernel invisibility $\implies$ the arithmetic quotient.

The first chain is the sibling report's and [section 8](#); the second is source 12's extension to all root types ([section 37](#)); the third is the normal-closure mechanism of [lemmas 7.1](#) and [9.4](#), applied with the new reconstruction. The Lie classification uses the first chain with the integral relation ([42.1](#)) in place of root geometry, and its converse uses an integral functional on the rational abelianization, residual additive detection and descent. The rank-one converse uses the amalgam and finite simultaneous additive detection.

Source 12's classical inputs are normal forms [[5](#), [7](#)], Chevalley groups and their integral root identities [[6](#), [8](#)], and amalgams and two-row normal forms [[4](#), [17](#)]. Only the specific rank-two identities are taken from [[8](#)]; its general relative commutator theorems, whose hypotheses exclude residue fields of order two, are not invoked, since $\mathbf{Oz}$ has the quotient $\mathbb{Z}/2\mathbb{Z}$ and 2 is not a unit of $\mathbf{Oz}$ (12). An encyclopedia page was used by source 12 for orientation only. Source 12's repository review covered the root README, the catalogue, selected report READMEs and source 11's proof audit, not this article; its novelty claim is limited to an all-root-type and Lie-theoretic development beyond the matrix claims it inspected. Its limitations are listed in [Appendix B](#) (items 12.1–12.28), its finite checks in [Appendix C](#).

A Provenance

A.1 Section by section

Table 2: Provenance of each part of the report.

Part of this report	Sources and choices
Section 2	06 §2, 07 §2, 10 §2, printed once; also in the omnific reports.
Sections 3 and 4	06 §§3–4. The answer to <code>odg:q:affine</code> and the comparison with the omnific Diophantine report are the merge's (remark 4.6).
Section 5	06 §5. $O_n(\mathbf{Oz})$ printed by citation of <code>odg:cor:orthogonal</code> ; the existence of the anisotropic family printed by citation of <code>odg:cor:unipotent</code> ; 06's rational anisotropy argument and explicit matrices kept.
Section 6	06 §6.
Section 7	07 §2.3 (lemma 7.1); 06 §8.1 (its Propositions 8.1, 8.2: propositions 7.2 and 7.3).

Part of this report	Sources and choices
Section 8	06 §7 (absorption); 07 §3 (compression); both identified with the sibling report's lemmas.
Section 9	The $n \geq 3$ theorem: 06's Theorem C with its Theorem 8.3 and Corollaries 8.4, 8.5, and 07's Theorem A with its Theorem 4.1, Corollary 4.2 and Proposition 4.3, printed once in 06's generality for D ; both proofs kept. 11's Theorem A (its Theorem 5.2, Lemma 5.3) is the same theorem and is not reprinted; its proof is kept as the third route, with its §4 (Theorem 4.1, Corollaries 4.3, 4.4, Proposition 4.5, Lemma 4.6: theorem 9.2 , corollary 9.3 , and lemma 9.4); its §3 (the ring obstruction) is cited as <code>osq:thm:universal</code> and lemma 8.1 . 11's Theorem 7.1 and Corollary 7.2 are credited at corollary 9.5 . Remark 9.6 is the merge's, with 11's converse. Proposition 9.7 is 07 §11.1.
Section 10	06 §9.
Section 11	The merge (proposition 11.1), from an observation of the placement dossier; the comparison with the rotation groups is the merge's.
Section 12	06 §10, 07 §§11.2–11.3.
Sections 13 to 18	07 §§5–10 (07's Theorems B, C, D). The Mirzaii–Torres and Cohn citations were checked for this report. 11's Theorem 9.1 is credited at theorem 16.4 ; its Proposition 11.3 at proposition 12.2 .
Section 19	11 §6 (Theorems 6.2, 6.3, 6.4, Proposition 6.5, Corollary 6.6; its Lemma 6.1 is lemma 2.1(ii)), 11 Proposition 7.4, 11 Corollary 9.2 (with the Bass–Milnor–Serre input). 11's Theorem B, printed as these statements about P_n .
Section 20	11 §8 (Lemma 8.1, Theorem 8.2 and its remark).
Section 21	11 §10 (Lemma 10.1, Theorem 10.2 as lemma 21.2 , Theorem 10.3, Corollary 10.4, Propositions 10.5–10.7, Corollaries 10.8, 10.9, the remark on $\aleph_1$, Proposition 10.11); 11's Theorem C. The ring is cited from the sibling report. Proposition 21.12 and corollary 21.13 are the merge's.
Part III	10, all sections. The lattice-projection equivalence, the sufficiency half of the closest-point criterion, the real parity bound and the Pell example are printed by citation of the Hahn–Tate report (subsection 27.1); 10's proofs are kept for its converse, the non-closedness, the Gaussian bounds, arbitrary targets and all omnific statements.
Sections 33 and 34	The merge, with the sources' questions (06 §10.3, 07 §12.4, 10 §14.1, 11 §11); the batch-32 status notes on questions 34.3 and 34.4 are the merge's, from source 12.

Part of this report	Sources and choices
Part IV	12, all sections. Section 35: 12 §1 (Theorems A–D) and the merge’s conventions; 12 §§3–4 (Propositions 3.1, 3.2, Example 3.3, Lemmas 4.1, 4.2, Theorem 4.3, Remark 4.4) are credited to lemmas 2.1 and 8.3 to 8.5, proposition 2.4, and section 12 and <code>osq:thm:universal</code>, not reprinted. Section 36: 12 §2 (Definition 2.1, Lemmas 2.2, 2.3). Section 37: 12 §§5–6 (Lemma 5.1, Theorems 6.1, 6.3, Remark 6.2). Section 38: 12 §7 (Lemma 7.1, Theorem 7.2, Corollary 7.3, Example 7.4, Theorem 7.5). Section 39: 12 §8 (Propositions 8.1, 8.3, 8.5, Remark 8.2, Corollary 8.4, §8.1). Section 40: 12 §9 (Theorem 9.1, Remark 9.2, Proposition 9.3). Section 41: 12 §10; its Theorem 10.1 kept as a second proof of theorem 13.2, its Corollary 10.2, Lemma 10.3 and Theorem 10.4 credited to theorems 14.1 and 15.3, lemma 15.1, and corollary 15.5, its Theorem 10.5 printed (part (ii) the merge’s). Section 42: 12 §11 (Lemmas 11.1, 11.5, Theorems 11.2, 11.3, 11.6, Corollary 11.4, the $\mathfrak{sl}_2$ example); remark 42.8 is the merge’s. Section 43: 12 §12 (Definition 12.1, Proposition 12.2, Theorems 12.3, 12.5, Corollary 12.4); corollary 43.7 is the merge’s. Section 44: 12 §13 and Appendix A; corollary 44.1 is the merge’s. Section 45: 12 §14. Section 46: 12 §15. 12’s Appendices B and C are in Appendix C and in the non-claims.

Section numbers of the sources refer to their own manuscripts, which are not shipped. Where the sources overlap, the statement with the weakest hypotheses is printed. No source proof needed a hypothesis that the printed statement lacks: 06’s proof of the $n \geq 3$ theorem uses only single-product divisibility, gaps and the geometric inverse, and 07’s and 11’s use D only through the split inclusion, so all three apply to every unital $D \subseteq k$. 11’s structural, Steinberg and fragment results are printed for $(D, k) \in \{(\mathbb{Z}, \mathbb{R}), (\mathbb{Z}[i], \mathbb{C})\}$, as 11 states them.

A.2 Renamed symbols

Table 3: Symbols renamed from the sources.

Source	Source symbol	Here, and why
06	$\mathcal{I}_k$	Π_k , the collection’s letter.
06	$\mathcal{B}_k, \mathcal{A}_{D,k}$	$\mathcal{B}_k, \mathcal{A}_{D,k}$ (unchanged meaning).
06	$P_n(k)$	P_n .
06	N_a (congruence subgroups)	$P_n^{(a)}$, since 07’s kernel was N_n .
06	F_r (filtration), U_n	$\text{Fil}_r, U_n = \text{UT}_n(\Pi_k)$; F is the ambient field.
06	$\mathcal{I}_k(\Gamma)$	$\Pi_k(\Gamma)$.
07	$\mathcal{J}_k, c_0$	Π_k , ct.

Source	Source symbol	Here, and why
07	R_D	$\mathcal{A}_{D,k}$.
07	N_n	P_n .
07	$K = \text{Frac } D$	K_D , since K_j are 10's kernels.
07	$G = \text{SL}_2(D)$	G_D , since G is an algebraic group in Part I.
07	T (monomial $\omega^{2\delta}$ or ω^{2a})	τ , or written out.
07	T (transversal)	$\mathcal{T}$.
07	X, Y (finite and infinite points)	$P_{\text{fin}}, P_{\infty}$; X is a scheme in Part I.
07	$L(x)$ (lower root)	$e_{21}(x)$; L is the compressed field.
07	P (coefficient map), A, B (commutator factors), P (index set)	$\mathfrak{p}, X, Y, \mathcal{P}$.
10	$\mathcal{J}, \mathcal{B}, \mathcal{A}, \mathcal{F}$	$\Pi_k, \mathcal{B}_k, \mathcal{A}, F$.
10	$E, \mathcal{E}$	$\mathcal{E}$ (energy), $\mathcal{V}$ (its values), since E_n is the elementary group.
10	$\mathcal{O}, \mathfrak{m}$	$\mathcal{O}_{\text{fin}}, \mathfrak{m}$, as in the omnific Diophantine report.
10	primitive, $U(\mathcal{A}^n)$	unimodular, $\text{Um}(\mathcal{A}^n)$; <i>primitive</i> keeps its meaning of <code>odg:thm:realray</code> , and $U(x)$ is $e_{12}(x)$.
10	V_j	Low_j ; V_j is the Hahn–Tate radical flag.
10	$H = W \cap K_{r+1}$	W' ; H is a target group in Part II.
10	D_i (Gram–Schmidt)	Δ_i ; D is the constant ring.
10	$B(u, v)$ (Gauss)	$\beta(u, v)$; B is a factor matrix.
11	$\Pi_k, A_D, \deg_{\omega}$	$\Pi_k, \mathcal{A}_{D,k}, \deg_{\omega}$ (unchanged meaning).
11	$N_n(A), U_n(I)$	P_n ; $\langle e_{ij}(I) \rangle$, equal to P_n for $I = \Pi_k$ (proposition 7.3); $U(x)$ is $e_{12}(x)$.
11	$K_n(A)$ (Steinberg kernel)	P_n^{St} ; K_D is $\text{Frac } D$ and K_j are prefix kernels.
11	$q = 1 + \omega, N^{(m)}$	written out; $P_n \langle (1 + \omega)^m \rangle$, since q_j are Part III's forms and $P_n^{(a)}$ is the tower.
11	G, K (root-generated group, normal closure)	Σ, N' ; G is an algebraic group, K_D a field.
11	$T(a), \rho_{\varphi}$ (root tuple)	$\vartheta_{\varphi}(a)$; $\mathcal{T}$ is a transversal and ρ_{ℓ} a specialization.
11	F (finite subgroup)	Θ ; F is the ambient field.
11	H (exponent bound in its Proposition 6.5)	h_0 ; H is a target group.
11	$B = D \oplus I$ (split ring)	B in lemma 20.1 , $A_0 = D_0 \oplus I$ in proposition 21.11 .
11	$A_{\kappa,D}, \Pi_{\kappa,k}, N_{n,\kappa,D}$	$R_{<\kappa}, \Pi_{<\kappa}, P_{n,<\kappa}$; the sibling report's $R_{\kappa}^{11}, \Pi_{\kappa}^{11}$.
11	$\omega^{\iota(\gamma)}$ (outer monomials), $q_{\kappa} = 1 + t_1$	X^{γ} , as in the sibling report; $1 + t_1$ written out.

Source	Source symbol	Here, and why
12	I (for the purely infinite ideal), $A_{D,k}$, $\deg$	Π_k , $\mathcal{A}_{D,k}$, $\deg_\omega$; I is an abstract ideal here.
12	T_δ (exponent compression)	Φ_δ of lemma 8.3; Φ is a root system in Part IV, where the rescaling is not used.
12	F (set field of the collision data; detection field $\mathbb{Q}$ or $\mathbb{Q}(i)$)	L ; K_D . F is the ambient field.
12	u, v, μ (collision data, cardinal)	w, v, ν ; u is a unit and μ a minimum value elsewhere.
12	$U, h, J, T(a), R_h, \tau$ (reconstruction)	$\Sigma, \varphi, J_\varphi, \vartheta_\varphi(a), R_\varphi, \vartheta_\varphi$, as in theorem 9.2; τ is the monomial $\omega^{2\delta}$ of lemma 8.5.
12	N (congruence subgroup at $1 + \omega$)	$P_\Phi \langle 1 + \omega \rangle$, extending $P_n \langle a \rangle$.
12	$G_0, B_0, B(A), e, c_j, \deg$	$G_D, C_D, B_D, E_{12}, (g_j)_{21}, \deg_\omega$.
12	$\Lambda, \mathfrak{g}, L(A)$	$\mathfrak{l}, \mathfrak{l}_\mathbb{Q}, \mathfrak{l}(A)$; Λ is an ordinal or a Euclidean lattice, L a field, L_j integral points.
12	b_i, n_{lij}, M, λ	$\mathbf{b}_i, z_{lij}, \mathfrak{H}, \varpi$; n is a matrix size, M a module or bound, λ a cardinal.
12	e, f, h (basis of $\mathfrak{sl}_2$)	$\mathbf{e}, \mathbf{f}, \mathbf{h}$; e_{ij} are elementary matrices, h an index.
12	$N, M, L, K; A, \dots, F, A_-, B_-, H_A, H_B$ (appendix matrices), $X_N(t)$	$N_1, \dots, N_4; G_1, \dots, G_6, G_1^-, G_2^-, H_1, H_2; \exp(tN)$. These letters name rings, groups and fields here.
12	$\tilde{x}_\alpha, P_\Phi, P_\Phi^{\text{St}}, E_\Phi$	unchanged; $P_{A_{n-1}} = P_n$ and $P_{A_{n-1}}^{\text{St}} = P_n^{\text{St}}$.

A.3 Stale repository statements

Sources 06, 07 and 10 pin commits (220784a, fb5c4b5) at which the omnific directories held only the placed base manuscripts of the two omnific reports, with bare labels. The merged reports were written afterwards (bbdd536, 7494d53, 0240140). The following statements are corrected here; the pins are kept as provenance.

- 06 cites “the geometric collision construction in Section 3” of the quotient article, and 07 its “scaled-field collision”: at the pins these were Section 3 of the placed base manuscript; the collection now prints them as `osq:lem:division` and `osq:lem:collision`.
- 06 read only the opening lines of the Diophantine article and the main statements of the quotient article at its pin. Its accounts are accurate for the material it read; the collection now contains the merged reports, in which, in particular, the question `odg:q:affine` answered by Part I appears.
- 06 describes the euclidean-three-space theorem from the catalogue only, as a universal set-sized quotient of $\text{SO}(3, \mathbf{No})$; the report also proves it for $\text{SO}(n, \mathbf{No})$, $n \geq 3$, and $\text{SU}(n, \mathbf{No}[i])$, $n \geq 2$ (`e3:cut:thm:allranks`). 06’s statement that the catalogue describes 51 reports was still what `docs/README.md` said when this report was first written; the catalogue has since grown, and lists 56 reports at the time of the addition of source 11.

4. 06 and 07 leave rank two and SL_n open; within this report rank two is settled by 07 and $\mathrm{SL}_2(\mathbf{Oz}) \neq E_2(\mathbf{Oz})$ ([questions 34.1](#) and [34.2](#)).
5. 10 reports that a repository search for “shortest vector” returned no hit and cites the Lean module `LatticeEnergyCertificate.lean` but not the report it formalizes. The search result is literally true at the pin and today, but the Hahn–Tate report’s Part II, present at the pin, already contains the lattice-projection lemma, the rational-flag minimum criterion, the 2^n parity bound and the same Pell example ([subsection 27.1](#)).
6. 07’s remark that the formalization ledger should not be read as covering its results remains true: `docs/FORMALIZATION.md` maps no label of this report and no omnific statement to Lean.

Source 11 pins `9693b28`, at which this report had already been written (`1e54d5a`); its label `ogl:el:thm:invisible` occurs 16 times in the article at that pin. The following statements of 11 are corrected here, the pin being kept as provenance.

7. 11 says that its searches “did not locate these elementary-group or Steinberg-group statements in the inspected material”. For the elementary-group statements this is not so: its main theorem, the finite residual, the finite-support counterexample and its rank-two question were all in this report at the pin. The report was then missing from the catalogue, which 11 inspected: `docs/README.md` at the pin describes 51 reports while the tree had more report directories, and has no row for this one. For the Steinberg statements the sentence is accurate; no Steinberg group appears elsewhere in the collection.
8. 11’s question on rank two (its Question 11.2) was answered within this report at its pin ([question 34.2](#)).
9. 11 deliberately does not identify the root-generated subgroup with the kernel or assert that it is normal; [propositions 7.2](#) and [7.3](#) (06), present at the pin, prove both. 11’s proofs never depend on the identification, so nothing in them changes.
10. 11 says that its fragments are “related to the set-sized support constructions already studied” in the sibling report. They are the same ring with the same exponent group (`osq:prop:lex`, present at the pin); the embedding in $\mathbf{Oz}$ that 11 gives was added to that report after the pin (`osq:rel:lem:embedding`).
11. 11 says that the sibling report’s exact ring-threshold constructions “motivate” its exact-threshold criterion. One of them, `osq:prop:card11`, present at the pin, satisfies its hypothesis after the collision argument of [proposition 21.12](#).
12. 11’s descriptions of the quotient report (an unrefereed AI-assisted merge dated 22–23 September 2026 whose ring theorem factors every set-sized ring image through `ct`) and of the catalogue (51 reports) are accurate for the pin.

Source 12 pins `7af8a30`, at which this report contained sources 06, 07, 10 and 11, with the batch-31 status notes. The following statements of 12 are corrected here, the pin being kept as provenance.

13. 12 says that an indexed search of the repository for “Chevalley” returned no matching result, and calls this a limited search. As a description of the tree it is false: at the pin the word occurs twice in this article, in [question 34.3](#) (“elementary Chevalley-type groups”) and in its status note (source 11’s suggestion for “split Chevalley groups through root-kernel reconstruction”). That question asked for exactly 12’s extension; 12 answers it

(question 34.3). 12's novelty statement, which it restricts to what it inspected, is unaffected: the tree contained no theorem for root systems other than A_{n-1} and no current Lie ring.

14. 12 inspected this report's guide and source 11's proof audit, not the article. Its account of this report's antecedents (type- A universal quotients, the rank-two amalgam and detection, matrix kernels, Steinberg groups, cardinal arguments) is accurate at the pin; they are [theorem 9.1](#) (type A), [theorem 9.2](#) and [corollary 9.3](#) (the type- A reconstruction and transfer), [remark 9.6](#) (the ring-group implication), [theorems 13.2](#) and [15.3](#) and [corollary 15.5](#) (rank two), [theorems 19.1](#) and [19.2](#) (structure), [theorem 20.2](#) (Steinberg) and [proposition 21.11](#) (the exact-threshold criterion). It does not mention [proposition 21.12](#) and [corollary 21.13](#); its statement that the repository gives exact thresholds in the scalar and matrix settings refers to them.
15. 12 describes this report as merged from AI-assisted unrefereed manuscripts and credits the scalar theorem to the quotient report; both are accurate at the pin.

B What this report does not claim

Every limitation, disclaimer and priority caveat stated by a source is kept here, numbered per source.

B.1 Source 06 (23 items)

- 06.1 An AI-assisted, proof-bearing research draft; not peer reviewed and not verified in Lean.
- 06.2 The combined main statements are proposed contributions, not claims of verified historical priority; they were not located in the inspected sources, which is a qualified novelty assessment.
- 06.3 No named longstanding conjecture is claimed settled.
- 06.4 Normal forms, real closedness, torus structure, nilpotent exponentials, Cartan theory and elementary commutator identities are classical.
- 06.5 The constant-term formalism, common monomial divisibility and geometric collision identities already occur in the repository; the needed forms are reproved, not imported as correct from an unreviewed draft.
- 06.6 The unitriangular and Lie-algebra descriptions are specializations of standard machinery, not independent major discoveries.
- 06.7 The geometric-series identity is a formal identity of normal forms, not convergence in the fine topology.
- 06.8 The only transfer is between real closed fields; no elementary equivalence of $\mathbb{Z}$ and $\mathbf{Oz}$, and no transfer in a language with an omnific predicate, is asserted.
- 06.9 Only the characteristic-zero fiber of an integral model is used; smoothness at the primes is not needed, and ordinary integral points depend on the model.
- 06.10 A conjugate-transpose unitary group is a different real group from the complex orthogonal group of the Gaussian example.
- 06.11 The anisotropic example does not contradict the rigidity of nonzero binary norm fibers.

- 06.12 The comparison table concerns constant-term rigidity, not finite generation of arithmetic subgroups or elementary generation of all omnific matrices.
- 06.13 No convergence of an exponential series at an infinite argument is asserted; all displayed matrix functions are finite polynomials.
- 06.14 The kernel theorem is not a statement about all purely infinite matrix groups: $\text{UT}_n(\Pi_k)$ has nontrivial set-sized characters.
- 06.15 $E_n(D) = \text{SL}_n(D)$ is used only for ordinary D ; no Euclidean algorithm in $\mathbf{Oz}$ is assumed.
- 06.16 The theorem does not make P_n trivial; its tautological proper-class representation is faithful.
- 06.17 No simplicity claim: the tower shows P_n is not simple, and no numerical size is assigned to P_n .
- 06.18 Invisibility fails in every nontrivial fixed set-sized Hahn workspace; the local cardinal form is a conditional lower bound, and Archimedean exponent groups supply no suitable scales.
- 06.19 Finite supports admit a visible image onto $\text{SL}_n(k)$.
- 06.20 Nothing is asserted about rank two or about $\text{SL}_n(\mathcal{A})$; no stable-range theorem is used, and the two questions left are not designated open problems in the literature.
- 06.21 Neither an unproved omnific factorization conjecture nor a global analytic theory on $\mathbf{No}[i]$ is used; published factorization work is context, not a hidden hypothesis.
- 06.22 The repository was read at its pin through its hosted interface, not modified, cloned or built; the inspection was targeted, and the literature search cannot exclude an earlier result in other terminology.
- 06.23 The finite checks do not verify the Hahn support lemma, support gaps, the proper-class collision, Cartan's theorem or the universal quantifiers; the integer filtration instances are not representations of arbitrary elements of Π_k .

B.2 Source 07 (20 items)

- 07.1 The results concern E_n (finite products of elementary matrices), not all of SL_n ; the questions for $\text{SL}_n(\mathcal{A}_{D,k})$ beyond the explicit non-elementary matrices are open.
- 07.2 The two-row presentation is a specialized form of Cohn's classical phenomena; the broad distinction between two rows and larger sizes is not new.
- 07.3 $E_2(\mathbf{Oz})^{\text{ab}} \cong \mathbb{Z}/12 \oplus \Pi$ is a classical specialization, not a novelty claim.
- 07.4 The Cohn matrix pattern and the block identity are not claimed as new.
- 07.5 The proposed contribution is the omnific detection package; priority is not established; the literature and repository check was limited; no named historical factorization conjecture is claimed resolved.
- 07.6 No peer review and no Lean verification.
- 07.7 The scaled-field collision is a prior repository ingredient; no unverified repository theorem is imported as an axiom.

- 07.8 There is no single faithful countable representation, only elementwise separation by varying maps; the separating maps are not a set-indexed family with faithful product.
- 07.9 The coefficient functional is an existence construction, not a computability claim.
- 07.10 No congruence subgroup property is asserted or used.
- 07.11 No numerical description of $\mathrm{SL}_2(\mathbb{Z}[i])^{\mathrm{ab}}$ is given.
- 07.12 The stabilization identity itself does not depend on surreals, and one-step stabilization does not classify the remaining obstructions.
- 07.13 In a fixed set-sized Hahn workspace the compressed field need not lie in a preassigned subfield or birthday-bounded fragment; the optimal threshold at a set-sized stage is not determined.
- 07.14 No classification of homomorphisms from $\mathrm{SL}_2(\mathcal{A}_{D,k})$, higher SL_n , or unstable K -theory obstructions.
- 07.15 The class statements concern class homomorphisms existing in the chosen foundational setting, not externally described maps in a larger metatheory.
- 07.16 No analytic assumption: no continuity, fine-topology limits, global complex exponential, analytic uniqueness principle or differential equation.
- 07.17 The tautological representation on the proper-class space F^n remains faithful and is not covered.
- 07.18 The finite checks validate finite identities and examples only, not the normal-form theorem, the foundations, compression or the group arguments; the PDF build checks typesetting only.
- 07.19 The formalization ledger does not cover these results.
- 07.20 Global choice is part of the stated framework but is not used.

B.3 Source 10 (21 items)

- 10.1 The results concern separated positive pencils only, not arbitrary positive-definite surreal matrices; $(x - \omega y)^2 + y^2$ defeats contraction and even has a reduced basis.
- 10.2 The strongest coinitality assertion uses the full surreal class and fails in fixed Hahn fields such as $k((t^{\mathbb{R}}))$.
- 10.3 The ordinary higher-rank Voronoi background (Amini–Nicolussi) is not a discovery; the rational-flag sufficiency is a specialization of it.
- 10.4 A bounded literature comparison, not an exhaustive certification of novelty.
- 10.5 No named factorization conjecture is resolved; the primality of $\omega^{\sqrt{2}} + \omega + 1$ is L’Innocente–Mantova’s Theorem B.
- 10.6 Normal forms and the constant-term retraction are classical, not new.
- 10.7 The converse “ $\mathrm{ct}(x)$ unimodular implies x unimodular” is not asserted.
- 10.8 The simultaneous improvement is not a sequential limit statement.

- 10.9 The parity bound is a standard finite-lattice argument, not a priority claim.
- 10.10 The finite candidate reduction is not an algorithm for arbitrary surreal inputs.
- 10.11 The Pell identities come from the recurrence, not a classification of Pell solutions over $\mathbf{Oz}$; an infinite index in the recurrence justifies no limit.
- 10.12 The hierarchy table is not a classification of arbitrary surreal forms.
- 10.13 The LLL obstruction concerns an ordinary parameter above the critical value; modified reduction notions and infinitesimal gaps are not analyzed.
- 10.14 No surreal volume or Minkowski theory; the module is a proper class, and its generators are a basis over $\mathcal{A}$, not over $\mathbb{Z}$.
- 10.15 Set-sized families are improved, not class-indexed ones; in the unimodular problem even a countable coinital family cannot be improved simultaneously.
- 10.16 Ordinary coefficient forms are essential; general asymptotic expansions need not have positive-semidefinite coefficients.
- 10.17 Infinitesimal target perturbations can change the nearest point.
- 10.18 A research draft, not peer reviewed or proof-assistant verified; no Lean code, build or audit for these theorems; related Lean modules are not a certificate.
- 10.19 The finite checks verify exact identities, not the class-theoretic or minimization theorems; no floating-point number stands in for an infinitesimal.
- 10.20 The repository comparison was targeted; a search without hits is not proof of absence.
- 10.21 The further directions are not named open problems from the literature.

B.4 Source 11 (24 items)

- 11.1 An unrefereed, AI-assisted, proof-bearing manuscript; its proofs have not been independently certified or formalized in Lean, and existing Lean coverage in the repository is not verification of it.
- 11.2 Its principal group-theoretic results are proposed as original extensions of the cited repository work; priority is not certified.
- 11.3 No named longstanding conjecture is claimed solved.
- 11.4 The repository comparison was limited to the catalogue, the introduction and abstract of the quotient report and targeted searches; not all 51 reports or all source manuscripts were audited, the repository was not modified, and the absence of a search result is not a priority certificate.
- 11.5 The small-ring-image theorem, including its nonunital form, belongs to the repository's quotient report; it is reproved, not claimed, and citing it does not certify the rest of that report.
- 11.6 Recovering coefficient algebra from root commutators has substantial precedent (Voronetsky, including the earlier type- A case), and the normal-generation machinery is standard; neither is claimed as new, and the finite-tuple lemma does not import that paper's deeper results.

- 11.7 The Bass–Milnor–Serre theorem is imported, not reproved; only the preceding factorization is proved.
- 11.8 The main theorems concern $E_n(A)$, not $\mathrm{SL}_n(A)$; $E_n(\mathbf{Oz}) = \mathrm{SL}_n(\mathbf{Oz})$ is not established, nor its Gaussian counterpart, and the question on $\mathrm{SL}_n(A)$ is generated by the work, not a verified literature problem.
- 11.9 Rank two is left separate: neither failure nor a solution is asserted, and $n \geq 3$ is a proved range, not a claim that $n = 2$ is a sharp negative boundary.
- 11.10 The Steinberg constant-term kernel is not K_2 ; no K_2 is calculated, no universal central extension in any unstable rank and no excision theorem is claimed, and nothing about algebraic K -theory in general is implied.
- 11.11 The Steinberg kernel is not claimed centerless or torsion-free, and it differs from the kernel of $\mathrm{St}_n(A) \rightarrow E_n(A)$, which is not computed.
- 11.12 $\mathrm{St}_n(D)$ must not be replaced by $\mathrm{SL}_n(D)$ in the Steinberg theorem.
- 11.13 Triviality of all class-sized representations is not claimed: the inclusion into $\mathrm{GL}_n(\mathbf{No})$ or $\mathrm{GL}_n(\mathbf{No}[i])$ is faithful.
- 11.14 “Arithmetic” means only factorization through $\mathrm{SL}_n(D)$; the representations of $\mathrm{SL}_n(D)$ are not classified or asserted to be algebraic.
- 11.15 It is not claimed that every finite subgroup is conjugate to a constant subgroup.
- 11.16 Invisibility does not pass to arbitrary subgroups: a root subgroup has nonzero coefficient maps to $\mathbb{R}$.
- 11.17 Other root systems do not follow by a change of notation; commutator constants and root lengths must be addressed. (Source 12 addresses them in [Part IV](#); the non-claim describes 11.)
- 11.18 Exact image thresholds are not claimed for the explicit fragments, neither κ nor $\kappa^{<\kappa}$; any application of the exact-threshold criterion must verify its ring hypothesis. (The merge supplies this for $R_{<\kappa}$ in [proposition 21.12](#); the non-claim describes 11.)
- 11.19 The fragment theorems are not extended to singular cardinals or to finite supports.
- 11.20 At $\aleph_1$ nothing excludes an action on a countably infinite set or every finite-dimensional complex representation.
- 11.21 No isomorphism of $\widehat{E_n(\mathbf{Oz}[i])}$ with $\widehat{\mathrm{SL}_n(\mathbb{Z}[i])}$ is asserted.
- 11.22 The congruence chain says nothing about intersections over the class of all principal ideals, and no quotient ring A/aA is claimed to be a set.
- 11.23 No compactness theorem for proper classes, no set of all class homomorphisms, no fine-topology limit (strongly summable sums are coefficientwise Hahn sums) and no identification of an arbitrary integer-part matrix group with its elementary subgroup is used; class homomorphisms are handled one at a time, and the hom-set bijections abbreviate factorization principles.
- 11.24 The finite checks test signs, indices, finite matrix identities and finite geometric remainders; they do not verify arbitrary Hahn supports, cardinal statements, class algebra or the universal assertions. Its proof audit is the manuscript’s own review, not an independent referee report, and its build report is a rendering report, not a correctness certificate.

B.5 The merge (15 items)

- M.1 The results marked [merge] (remarks 4.6 and 9.6, propositions 11.1 and 21.12, and corollary 21.13, and in Part IV theorem 41.1(ii), remark 42.8 and corollaries 43.7 and 44.1) are no more refereed than the sources.
- M.2 Remark 9.6 is a logical implication, not an independent proof of the ring theorem, and covers $K \in \{\mathbb{R}, \mathbb{C}\}$ only.
- M.3 Part I answers `odg:q:affine` only for closed subgroup schemes of GL_N , not for general affine varieties or curves.
- M.4 In the nonrigid case of Part I, no description of all omnific points is claimed; only injective families through the integral points are constructed.
- M.5 Neither theorem 9.1 nor proposition 11.1 implies the rotation-group theorems of the euclidean-three-space report, or conversely; nothing is claimed for $n = 2$ on the finite side.
- M.6 Nothing in this report is formalized in Lean; `docs/FORMALIZATION.md` maps none of its labels.
- M.7 The verification of the Mirzaii–Torres and Cohn citations and of the Milne theorem numbers checks citations, not proofs; 06’s reference to real rank and relative root groups in Milne’s *Reductive Groups*, Sections 22–23, was not checked.
- M.8 Proposition 21.12 and corollary 21.13 answer question 34.4 only for $n \geq 3$, only for the rings named there, and only for the least cardinality of a detecting group; they rest on statements of the sibling report (`osq:prop:card11`, `osq:lem:collision` and, for corollary 21.13, the properties established in the proof of `osq:thm:thresholds`), whose review status is that report’s.
- M.9 Source 11’s third proof of theorem 9.1(iii) is not independent of the collision identities of the other two: it imports the ring theorem. The equivalence recorded in remark 9.6 is an equivalence of statements, not a new proof of either.
- M.10 The material added from source 11 was not reviewed beyond this integration: the proofs were re-derived for the write, 11’s finite checks were rerun, and its Bass–Milnor–Serre and Voronetsky citations were not checked against the originals for this report.
- M.11 The answer to question 34.3 is for split elementary groups: the simply connected form, its Steinberg group, and the relative kernels of other split isogeny forms; the exact converse is for $D \in \{\mathbb{Z}, \mathbb{Z}[i]\}$ in the simply connected elementary category. Twisted, nonsplit and relative-rank-one groups, the full groups $G(\mathcal{A}_{D,k})$, and the converse for non-simply-connected forms of the whole group stay open.
- M.12 Corollary 43.7 holds only for root systems without A_1 component and only in the rings named there; it rests on proposition 21.12 and corollary 21.13 and thus on the sibling report’s statements listed in M.8. Nothing is said about A_1 components at set-sized stages.
- M.13 Remark 42.8 compares two statements; neither is derived from the other, and no statement about all unipotent groups is made. Theorem 41.1(ii) is immediate from source 12’s Theorem B and 07’s countable detection.

- M.14 [Corollary 44.1](#) rests on the sibling report's `osq:thm:support(i)`, whose review status is that report's; it is stated for uncountable λ and root systems without A_1 component only. The identification of O_κ with $\mathcal{A}_{\mathbb{Z},\mathbb{R}}^{<\kappa}$ is by definition; nothing is imported from the report on first κ coefficients.
- M.15 The material added from source 12 was not reviewed beyond this integration: the proofs were re-derived for the write, its finite checks were rerun on a copy, and its citations of Geck, Hazrat–Vavilov–Zhang, Cohn and Serre were not checked against the originals for this report.

B.6 Source 12 (28 items)

- 12.1 An AI-assisted, unrefereed research manuscript; its written arguments are conditional only on the stated classical normal-form and Chevalley-group background, and have not been independently refereed.
- 12.2 The all-type transfer and the Lie classification are proposed contributions; priority was not established by an exhaustive literature review or certified, and no claim of first publication is made.
- 12.3 No named published conjecture is claimed solved.
- 12.4 The scalar quotient, the type- A matrix theorem, the rank-one amalgam and countable detection, the type- A Steinberg version and several matrix-kernel and cardinal arguments are repository antecedents, credited and not claimed as new; they are reproved rather than imported from an unreviewed draft, and citing them does not certify the repository.
- 12.5 The Chevalley relations and integral root identities are classical (Geck; Hazrat–Vavilov–Zhang). The general relative commutator theorems of Hazrat–Vavilov–Zhang are not invoked with their small-residue-field hypotheses dropped: $\mathbf{Oz}$ has the quotient $\mathbb{Z}/2\mathbb{Z}$ and $2 \notin \mathbf{Oz}^\times$; the argument divides a free parameter in Π_k , never a killed parameter or a target element.
- 12.6 The main group statements concern elementary, root-generated groups, not every point of the Chevalley group scheme: $E_\Phi(\mathbf{Oz}) = G_\Phi(\mathbf{Oz})$ is not asserted, no statement about $\mathrm{SL}_n(A)$ is made, and elementary generation of $G_\Phi(\mathbf{Oz})$ is not resolved.
- 12.7 No computation of $K_2(\mathbf{Oz})$; the Steinberg constant-term kernel is not identified with K_2 and is not the kernel of $\mathrm{St}_\Phi(A) \rightarrow E_\Phi(A)$.
- 12.8 The Steinberg kernel is not claimed torsion-free or centerless.
- 12.9 No classification of normal subgroups; nonsimplicity is proved by one proper congruence subgroup, without classifying ideals.
- 12.10 No global bounded commutator width or bounded generation: the bound of at most three commutators is per root element.
- 12.11 The rank-one converse is stated in the simply connected elementary category; no rank-one converse is asserted for every central quotient, and the isogeny statement concerns root-generated groups in the stated forms.
- 12.12 No global exponential correspondence between the abstract class groups and Lie rings is assumed; abstract group maps need not differentiate.

- 12.13 Subgroups of set-invisible groups need not be set-invisible: a root subgroup has nonzero coefficient maps to k and $\mathbb{Q}$, so no K_2 or other subgroup statement follows by restriction.
- 12.14 The transfer proofs use no regularity of κ ; regularity, cofinality and support restrictions may be needed to establish collision data in a particular Hahn fragment and must be checked there.
- 12.15 Action thresholds use $|\mathrm{Sym}(X)| = 2^{|X|}$ for infinite X , not $|X|$.
- 12.16 Finite quotients of $E_\Phi(A)$ are those of $E_\Phi(D)$, but no identification of the profinite completion with a congruence completion, and no blanket Gaussian congruence-subgroup claim, is made.
- 12.17 Finite supports destroy the theorem: the finite-support evaluation is not extended to arbitrary Hahn series, and a finitely supported model must not replace Π_k in the universal statements.
- 12.18 The constant term is not the standard part.
- 12.19 The theorems use the full surreal exponent class; they are not statements about an arbitrary fixed set-sized Hahn ring, whose identity map is a counterexample.
- 12.20 The converse of the ring-group equivalence uses only the Dorroh unitization; a nonunital map need not be linear over a larger constant ring.
- 12.21 Checking root systems up to rank eight (A_ℓ, B_ℓ, C_ℓ , $2 \leq \ell \leq 8$; D_ℓ , $4 \leq \ell \leq 8$; the exceptional types) does not prove the coverage lemma for all ranks; that rests on its written proof.
- 12.22 The finite checks verify polynomial matrix identities, integral divided powers, finite root configurations and finite leading-term examples with polynomial parameters; they do not verify the class-theoretic support argument, Hahn support closure, arbitrary homomorphisms or arbitrary supports, and the rational substitutions in them do not assume $1/2 \in A$.
- 12.23 No Lean formalization is supplied and no repository Lean build was run.
- 12.24 The repository review was targeted (the root README, the catalogue, selected report READMEs and source 11's proof audit), not a line-by-line audit; no blanket assertion is made that no other file anticipates a consequence, and a negative search is not proof of absence.
- 12.25 A user-provided encyclopedia page was used for orientation only; classical sources support the imported machinery, not the new omnific conclusions.
- 12.26 Its proof audit is the manuscript's own record, not a referee report or a formal certificate; its build report is a layout and build review, and word counts of extracted text are not semantic verification.
- 12.27 The twelve questions are proposed continuations, not an inventory of named open problems.
- 12.28 Hom displays with a proper-class source abbreviate factorization statements; no set or class whose members are proper classes is formed, the faithful representation over **No** or **No**[i] is class-sized and is not contradicted, global choice, though harmless, is not used, and the exponent compression is an algebraic normal-form map with no asserted compatibility with the surreal exponential.

C Verification suites

Five finite programs accompany the report, as shipped (byte-identical to the deliveries). The first three and source 12's need Python 3 and SymPy (the recorded runs used Python 3.13.5 and SymPy 1.14.0); source 11's needs only the Python standard library (version 3.10 or later). All five were rerun for this report on copies in a scratch directory with Python 3.14.4 (and SymPy 1.14.0).

- `code/06-algebraic-groups-verification.py` (06): 1,108 exact checks in ten families (908 elementary-identity checks in ranks 3–6, 18 for the two orthogonal examples (5.1) and (5.2), 139 unitriangular instances, 43 geometric remainders and finite-support augmentations). It prints to standard output and writes a file only with `-output`; the recorded run is `data/06-algebraic-groups-verification-results.txt`.
- `code/07-matrix-dichotomy-checks.py` (07): 23 checks, including 120 exact reduced alternating words in the countable target, the Steinberg identities, (17.1), the commutator of example 15.4, the determinant and square-zero identities of M_α , and the stabilized commutator. It always writes `checks-results.json` next to itself; the recorded run is `data/07-matrix-dichotomy-checks-results.json` with `data/07-matrix-dichotomy-checks-output.txt`. The rerun reproduced it except for the recorded Python version.
- `code/10-shortest-vectors-verify.py` (10): six groups of checks (the Cholesky identity, the Pell recurrence and 81 norms, determinants and gcds for indices 0–80, the Gram–Schmidt identity, the three-dimensional examples, and the perturbed-target differences of example 27.6); standard output only, recorded in `data/10-shortest-vectors-verification-output.txt`.
- `code/11-matrix-shadows-verify.py` (11): 2,111 deterministic exact checks (seed 20260923) in twelve families, with exact fractions and finite Laurent polynomials over the lexicographic group $\mathbb{Q} \oplus \mathbb{Q}$: root additivity (272), root inverses (272), root commutators (840), constant-term multiplication and nonnegative supports (200 each), word inverses, conjugated commutators, constant terms of conjugated roots and matrix constant-term multiplication (40 each), exact geometric remainders and positivity of the quotient supports (81 each, the identity $(X^{2e_1} - X^{e_1}) \sum_{r=0}^M X^{e_0 - (r+2)e_1} = X^{e_0} - X^{e_0 - (M+1)e_1}$), and the Whitehead diagonal identity over rational units (5). It prints its results and always overwrites `verification_results.json` next to itself (under the shipped name, `code/verification_results.json`); the recorded run is `data/11-matrix-shadows-verification_results.json`, and the rerun reproduced its content exactly.
- `code/12-chevalley-quotients-verify.py` (12): 2,188 deterministic exact checks (SymPy; seed 20260923 for the rank-one words) in twelve families: the A_2 commutator (1), the C_2 identities (44.1), (44.2) and the halved substitution (3), the G_2 identities (44.3), (44.4) and the halved substitution (3), the G_2 Cartan and cross relations (7) and Serre relations (2), integral divided powers of eight root matrices in exponents 1–7 (56), the $\mathfrak{sl}_2$ brackets (42.3) (3), root-reconstruction witnesses (1,868), exceptional root counts (5), and, for 80 reduced rank-one words of length 1–5 with polynomial parameters in one variable, the leading coefficient (41.2), its nonvanishing and determinant one (80 each). The 1,868 witness checks are the roots of A_ℓ, B_ℓ, C_ℓ ($2 \leq \ell \leq 8$), D_ℓ ($4 \leq \ell \leq 8$), G_2, F_4, E_6, E_7, E_8 : for each root the program finds a pair of roots whose positive combinations, lengths and root-string constant match one of the four configurations of lemma 37.1, and it stops with an error if none exists; the count records one success per root. It checks root geometry, not structure constants. The program prints its results and always writes `verification_results.json`

next to itself (under the shipped name, `code/verification_results.json`); the recorded run is `data/12-chevalley-quotients-verification_results.json`, and the rerun reproduced it except for the recorded Python version. The shipped `code/12-chevalley-quotients-Makefile` names the delivery files `article.tex` (source 12's manuscript, not shipped) and `verify.py`.

These programs check finite identities and examples. They do not verify Hahn summability, support gaps, class-size or cardinal arguments, algebraic-group structure, the reduced-word arguments, the coverage lemma in all ranks, or any minimization or coinitiality theorem, which are proved in the text.

References

- [1] O. Amini and N. Nicolussi, *Higher rank inner products, Voronoi tilings and metric degenerations of tori*, arXiv:2310.06523v1, 2023. Cited by 10 for its Example 3.2, Theorems 3.9 and 8.1 and Corollary 8.3.
- [2] H. Bass, J. Milnor, and J.-P. Serre, *Solution of the congruence subgroup problem for SL_n ($n \geq 3$) and Sp_{2n} ($n \geq 2$)*, Publications Mathématiques de l’IHÉS **33** (1967), 59–137, [doi:10.1007/BF02684586](https://doi.org/10.1007/BF02684586). Cited by 11 for the profinite completion of $SL_n(\mathbb{Z})$ and its real-versus-imaginary caution; not rechecked for this report.
- [3] A. Berarducci, *Surreal numbers, exponentiation and derivations*, arXiv:2008.06878, 2020.
- [4] P. M. Cohn, *On the structure of the GL_2 of a ring*, Publications Mathématiques de l’IHÉS **30** (1966), 5–53, [doi:10.1007/BF02684355](https://doi.org/10.1007/BF02684355). The non-elementary matrix over $k[x, y]$ is on p. 26 (end of Section 7); discretely ordered rings are Section 8.
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